Bally

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April 1999

PINBALL 2000™

REVENGE FROM MARS™

OPERATIONS MANUAL

CE c UL us LISTED

MANUAL INCLUDES

Operations & Adjustments • Testing & Problem Diagnosis • Parts Information
Wiring Diagrams & Schematics

Williams Electronics Games, Inc., 3401 N. California Avenue, Chicago, IL 60618

SOLENOID TABLE

Sol No.	Solenoid Function	Fuse	Power to Playfield	Power Wire Color	Drive Transistor	Drive to Playfield	Drive Wire Color	Solenoid Part Number or Flasher Type
01	LEFT MARTIAN	F101	J102-1	RED-BRN	Q59	J110-13	VIO-BRN	AE1-26-1500
02	RIGHT MARTIAN	F101	J102-1	RED-BRN	Q60	J110-14	VIO-RED	AE1-26-1500
03	JET EXIT POST	F101	J102-1	RED-BRN	Q61	J110-15	VIO-ORG	AE1-26-1500
04	RIGHT GATE	F101	J102-1	RED-BRN	Q62	J110-16	VIO-YEL	A-14406
05	LEFT GATE	F102	J102-2	RED-BLK	Q63	J110-17	VIO-GRN	A-14406
06	DROP TARGET DOWN	F102	J102-2	RED-BLK	Q64	J110-18	VIO-BLU	SM1-26-600
07	DROP TARGET UP	F102	J102-2	RED-BLK	Q65	J110-19	VIO-BLK	AE1-26-1200
08	RIGHT POPPER	F102	J102-2	RED-BLK	Q66	J110-20	VIO-GRY	AE1-25-1000
09	TROUGH EJECT	F103	J102-3	RED-ORG	Q51	J112-11	BRN-BLK	AE1-26-1500
10	LEFT SLINGSHOT	F103	J102-3	RED-ORG	Q52	J112-12	BRN-RED	AE1-26-1200
11	RIGHT SLINGSHOT	F103	J102-3	RED-ORG	Q53	J112-13	BRN-ORG	AE1-26-1200
12	LEFT JET BUMPER	F103	J102-3	RED-ORG	Q54	J112-14	BRN-YEL	AE1-26-1200
13	RIGHT JET BUMPER	F100	J102-7	RED-YEL	Q55	J112-15	BRN-GRN	AE1-26-1200
14	BOTTOM JET BUMPER	F100	J102-7	RED-YEL	Q56	J112-16	BRN-BLU	AE1-26-1200
15	AUTO PLUNGER	F100	J102-7	RED-YEL	Q57	J112-17	BRN-VIO	AE1-23-800
16	RIGHT LOCKUP	F100	J102-7	RED-YEL	Q58	J112-18	BRN-GRY	AE1-23-800
17	CENTER ARROW FLASHER	F109	J102-8	RED-WHT	Q43	J110-1	BLU-BRN	#906
18	NOT USED	F109			Q44	J110-2	BLU-RED	
19	NOT USED	F109			Q45	J110-3	BLU-ORG	
20	NOT USED	F109			Q46	J110-4	BLU-YEL	
21	NOT USED	F109			Q47	J110-5	BLU-GRN	
22	RIGHT POPPER FLASHER	F109	J102-8	RED-WHT	Q48	J110-6	BLU-BLK	#906
23	LEFT ARCH FLASHER	F109	J102-8	RED-WHT	Q49	J110-7	BLU-VIO	#89
24	NOT USED	F109			Q50	J110-8	BLU-GRY	
25	RIGHT ARCH FLASHER	F109	J102-8	RED-WHT	Q67	J112-9	BLK-BRN	#89
26	LEFT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q68	J112-10	BLK-RED	#89
27	RIGHT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q69	J112-19	BLK-ORG	#89
28	ATTACK MARS FLASHER	F109	J102-8	RED-WHT	Q70	J112-20	BLK-YEL	#906
	Flipper Circuits							
33	LOW RIGHT FLIPPER PWR				Q35	J112-1	YEL-GRN	
34	LOW RIGHT FLIPPER HOLD	F104	J103-1	RED-GRN	Q36	J112-2	ORG-GRN	FL1-11629
35	LOW LEFT FLIPPER PWR				Q37	J112-3	YEL-BLU	
36	LOW LEFT FLIPPER HOLD	F105	J103-2	RED-BLU	Q38	J112-4	ORG-BLU	FL1-11629
37	LOCK DIVERTER POWER				Q39	J112-5	YEL-VIO	
38	LOCK DIVERTER HOLD	F106	J103-3	RED-VIO	Q40	J112-6	ORG-VIO	FL1-22241
39	UP/DOWN RAMP POWER				Q41	J112-7	YEL-GRY	
40	UP/DOWN RAMP HOLD	F107	J103-4	RED-GRY	Q42	J112-8	ORG-GRY	FL1-11753

DECLARATION OF CONFORMITY

WILLIAMS ELECTRONICS GAMES, INC.

3401 N. CALIFORNIA AVE.
CHICAGO, IL 60618
U.S.A.

WE, HEREBY DECLARE UNDER SOLE RESPONSIBILITY THAT

THE MODEL: ***"REVENGE FROM MARS" 50270,50370,50470,50770,50970,51070, 51170,51270, 51370,51470,51870,52070,52170,52370,57270***

TO WHICH THIS DECLARATION RELATES IS IN CONFORMITY WITH THE FOLLOWING EUROPEAN PRODUCT SAFETY

ELECTROMAGNETIC COMPATABILITY DIRECTIVE
(89/336/EEC AND AMENDMENTS 91/C162/08, 92/31/EEC,93/68/EEC

ELECTRICAL EQUIPMENT DESIGNED FOR USE WITHIN CERTAIN VOLTAGE LIMITS DIRECTIVE
(73/23/EEC AND AMENDMENTS 88/C168/02, 92/C210/01, 93/68/EEC, 94/C199/03, 95/C214/02)

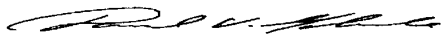
AS IS VERIFIED BY COMPLIANCE WITH THE FOLLOWING STANDARDS

EN 55014:1993 EN55104:1995 EN61000-4-2: 1995
IEC 801-3: 1984 (EN61000-4-3) EN61000-4-4: 1995 EN61000-4-5: 1995
ENV50141: 1993 (EN61000-4-6) EN61000-4-11: 1994 EN60335-1: 1995
IEC 335-2-82 (DRAFT)

FREBRUARY 22, 1999

Date issued:

MANUFACTURE'S SIGNATURE



DAN GALARDE

CORPORATE V.P. OF QUALITY

THESE ARE SOME OF THE NEW FEATURES AVAILABLE WITH PINBALL 2000:

GAME PLAY

The video display allows dynamic interactive features.

The video display allows many playfield devices to occupy the same position at different times in the game.

Video can clearly lead the player through the game, making clear what is happening each time that a target is hit. You see the result at the point of impact with the ball.

The video 'mechanisms' allow for simpler 'real' mechanisms on the playfield. The result is reduced service problems.

ELECTRONICS

The backbox contains a commercial motherboard, housed in a PC style case, and a monitor. The system will become more powerful as new motherboards are introduced.

The Power Driver board is located in the cabinet and is connected to the PC via a standard parallel printer cable. A laptop or a service computer can use this connection for powerful troubleshooting.

The PINBALL 2000 Power Driver board is secured with two screws and has 17 uniquely keyed connectors allowing for easy replacement. The WPC-95 Power Driver board, on the other hand, was secured with nine screws and had 39 connectors.

Each fuse has its own LED indicator.

Blown fuses and burnt out bulbs are shown by built in diagnostics on the video screen.

A PC connector at the coin door allows for advanced diagnostics, (a laptop computer can become a portable test fixture), data transfer and software updates.

The operator may add a low cost modem to the PC for remote data retrieval, an Internet connection and software updates. A revenue sharing system could be implemented on this platform.

SOFTWARE UPDATES

There are no EPROMs in the system. Software updates are done using a PC connected to the front door or via a modem and phone line.

There will be a chip based updated method available for a low-tech customer that is similar to updating a video game hard drive program.

All software resides in the backbox section. The system has been designed such that the software can be updated without uncrating the backbox.

CABINET AND BACKBOX FEATURES

The attendant can gain access to playfield through a key-released front molding. Playfield glass may be removed by location attendant for minor problems) such as trapped balls. Playfield remains locked by passive latch built into coin door.

Computer Case is an easy pull-out drawer in the backbox. This allows for easy access for troubleshooting and service. Entire CPU section may be swapped out in 60 seconds.

Backbox locks provides higher security than previous design and secures CPU case for transit.

CABINET AND BACKBOX FEATURES CONTINUED…

Provision available for CPU case padlock in backbox.

Backglass stores easily over playfield glass and provides reflected image while servicing.

The playfield is at proper playing angle with all leg levelers in their lowest position.

The wiring from the backbox to the cabinet is only a few cables.

Main cable from the cabinet mounted driver board is significantly reduced in length for greater reliability and lower emissions and interference.

The cover for the Power Driver board is hinged to ensure that it is put back in its protective position after service is performed.

DCS II sound system delivers stereo sound directly to the player to enhance game excitement without requiring a high volume level.

CONVERTIBILITY

Operators are able to use the convertibility to rotate their pinball games by moving playfield and a PC board instead of the entire machine. This is possible because of the ease of conversion and the way the devices on the bottom of the playfield are protected by the underside runners.

The playfield has unique runners protecting the parts on the underside of the playfield. These runners provide the following features:

Installation and removal of the playfield is an easy process requiring little physical strength. This ease is enhanced due to the main cable's attachment to the cabinet instead of the playfield.

The devices on the bottom of the playfield are naturally protected in the shipment of these kits.

The playfield may be handled and stored outside of the cabinet without fear of damage to the devices attached to the underside.





Williams Electronics Games, Inc. reserves the rights to make modifications and improvements to its products. The specifications and parts identified in this manual are subject to change without notice.

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SECTION ONE

GAME ASSEMBLY, OPERATION, AND TEST INFORMATION

ROM Summary

IC	TYPE	BOARD	LOCATION	PART NUMBER
Image	ROM	Prism ROM Daughter Card	U100	5341-16205-00
Image	ROM	Prism ROM Daughter Card	U101	5341-16206-00
Image	ROM	Prism ROM Daughter Card	U102	5341-16207-00
Image	ROM	Prism ROM Daughter Card	U103	5341-16208-00
Image	ROM	Prism ROM Daughter Card	U104	5341-16209-00
Image	ROM	Prism ROM Daughter Card	U105	5341-16210-00
Image	ROM	Prism ROM Daughter Card	U106	5341-16211-00
Image	ROM	Prism ROM Daughter Card	U107	5341-16212-00
Sound	ROM	Prism ROM Daughter Card	U109	5341-16213-00
Sound	ROM	Prism ROM Daughter Card	U110	5341-16214-00

PINBALL GAME ASSEMBLY INSTRUCTIONS

It takes two people to assemble a PINBALL 2000 game.
Do not plug in or switch on power to the cabinet until assembly is complete!

Power:	Domestic 120V @ 60Hz Foreign 230V @ 50Hz Japan 100V @ 50HZ	***Dimensions:***	Width: 24" approx. Length: 48" approx. Height: 73" approx.
Temp:	32°F to 100° F, (0°C to 38°C)	***Weight:***	Backbox: 130 LB approx. Cabinet: 230 LB approx.
***Humidity*:**	Not to exceed 95% relative.		

It is not necessary to open the computer case to assemble a PINBALL 2000 game. However, should you ever find it necessary to open the case, be sure to replace the computer case cover. Failure to do so will void FCC, UL and CE compliance, and may cause damage to the PC boards in the case. The foam lining of the cover holds the PC boards securely in place and protects them from vibrations that normally occur in a pinball game.

1. Start by removing the playfield from the cabinet. Set the cabinet on a level surface. Unlock and open the coin door. Pull the yellow lever, (located on the left side of the coin door opening), to the right. The front molding pops up. Lift the molding from the cabinet. Slide the playfield glass down and lift it off of the cabinet. See Figure 1.

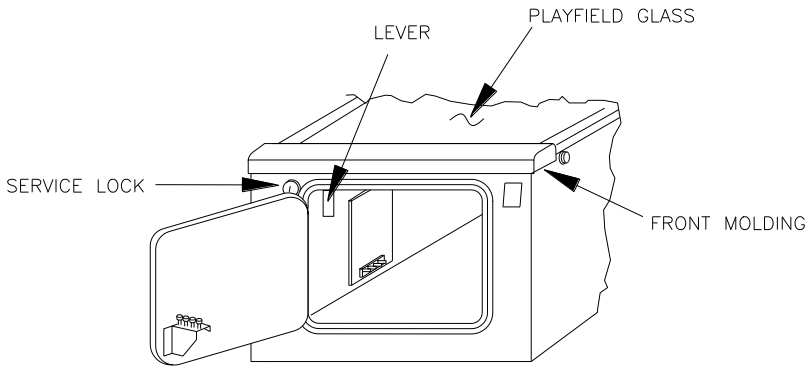


FIGURE 1

2. Lift the playfield by the support bracket under the bottom arch. Rest the playfield on the top of the cabinet as shown in Figure 2. Close the coin door.

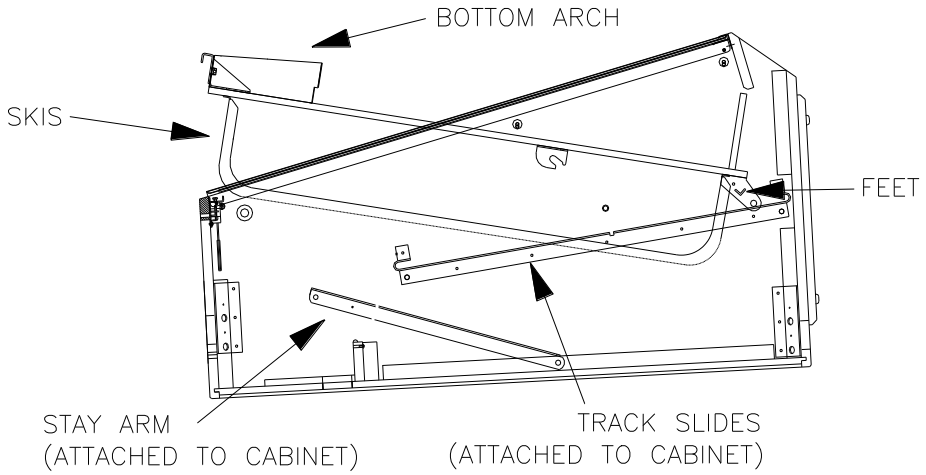


FIGURE 2

3. Firmly grasp the playfield and slide it forward (toward you) along the bottom track slide brackets. Rotate the front of the playfield down and set it on the floor as shown in Figures 3 through 5.

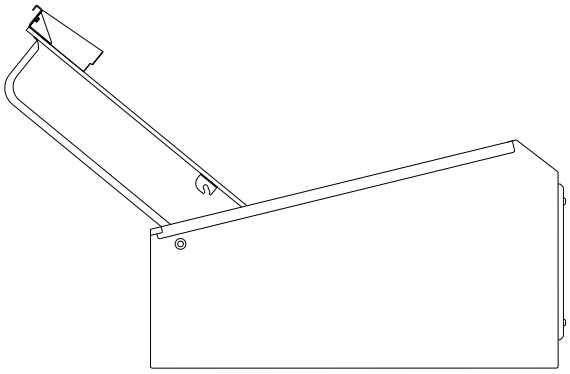


FIGURE 3

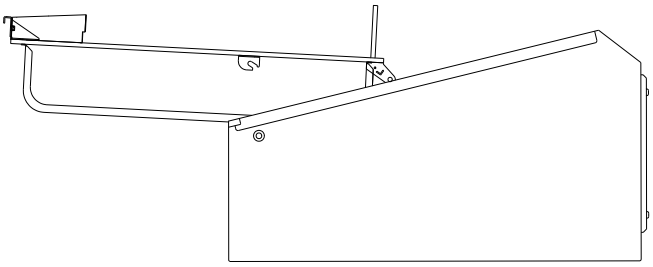


FIGURE 4

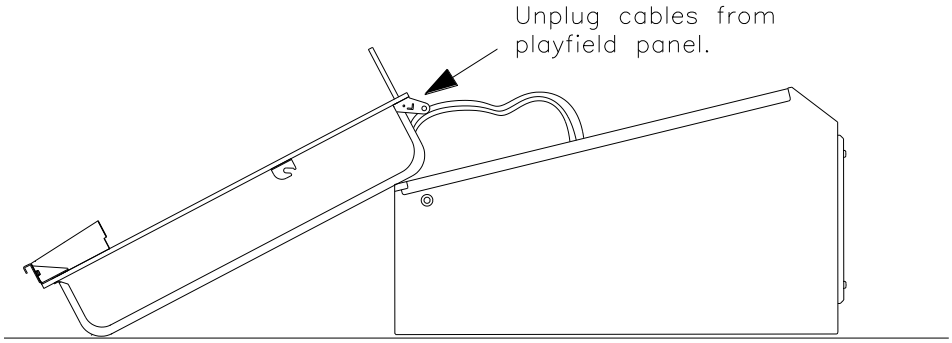


FIGURE 5

4. Unplug the cables from the bracket on the back of the playfield and pull the playfield away. See Figure 5, above.

5. The leg levelers, nuts, and leg bolts are among the parts in the cash box. Install a leg leveler and nut on each of the front and rear legs. Place the cabinet on a support. Using two leg bolts in each leg attach the front and rear legs to the cabinet. Screw the levelers all the way into the legs. Remove the cabinet from the support. See Figure 6, below.

!! DO NOT RAISE THE REAR LEGS. !!

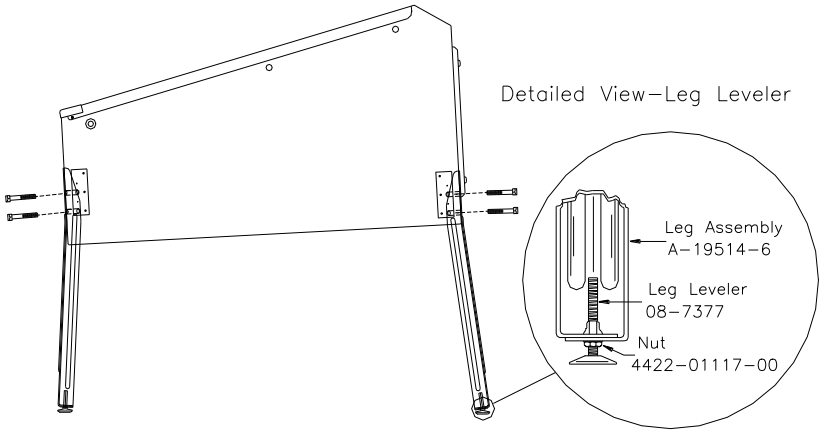


FIGURE 6

6. Next prepare the backbox. Cut the tie-wrap that holds the 9-pin serial cable, the 25-pin parallel cable and the power/speaker cable to the wood shipping brace. The shipping brace is held in place by two screws located on the left and right sides of the backbox. Remove the screws and shipping brace from the backbox. See Figure 7, below.

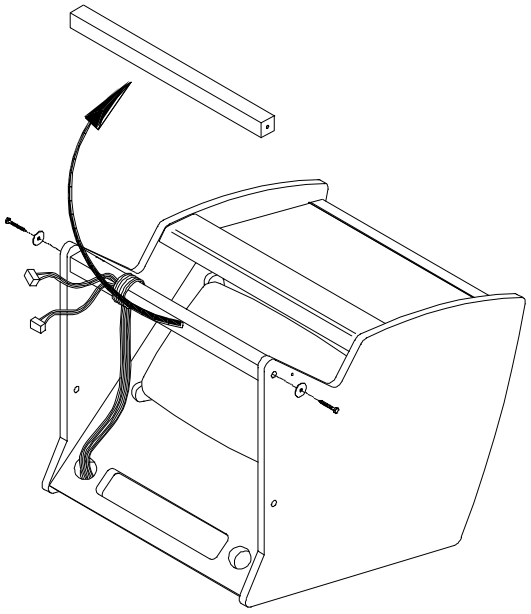


FIGURE 7

7. Stand the backbox upright. Unlock, unscrew and remove the rear door. Pull the three cables out through the back door opening and let them hang down the rear of the backbox. Locate the four mounting bolts, washers and bushings among the parts in the cash box and set them aside. See Figure 8.

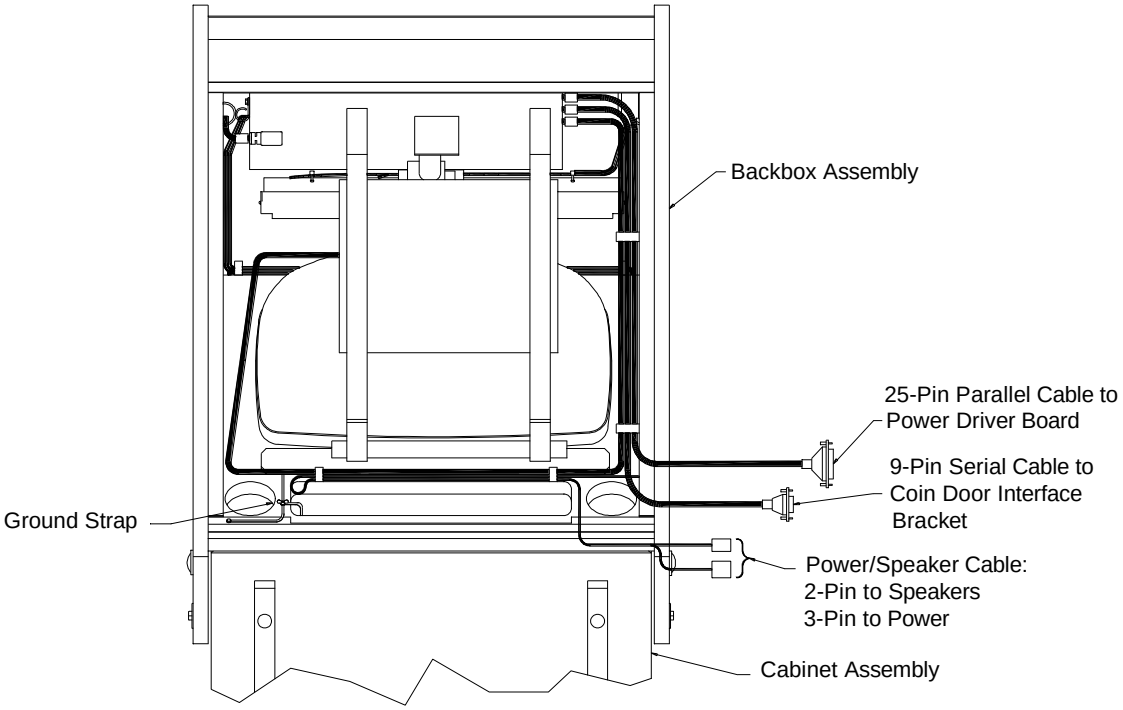


FIGURE 8

8. **WITH THE HELP OF ANOTHER PERSON, carefully lift the backbox and set it on the cabinet. See Figure 9.**

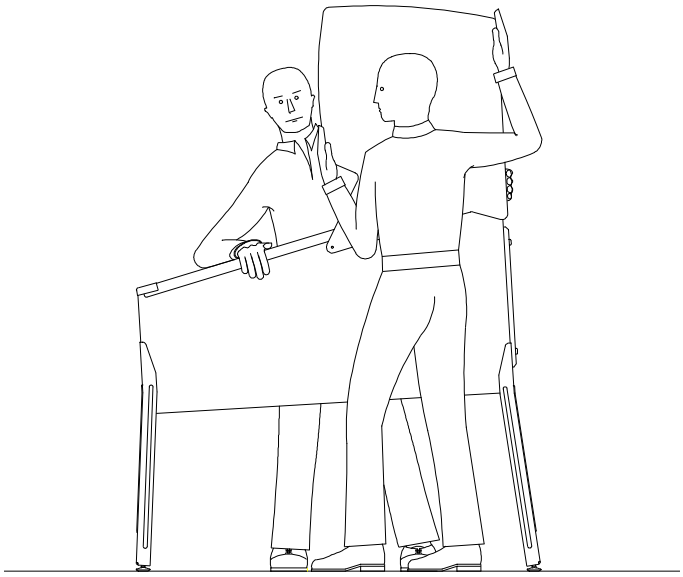


FIGURE 9

9. Line up the four mounting holes in the cabinet with the mounting holes in the backbox. Place a bushing in each of the mounting holes from the inside of the cabinet. Insert a bolt and washer through the outside of the backbox and into the cabinet in each of the four mounting holes. Tighten the bolts. See Figure 10 below.

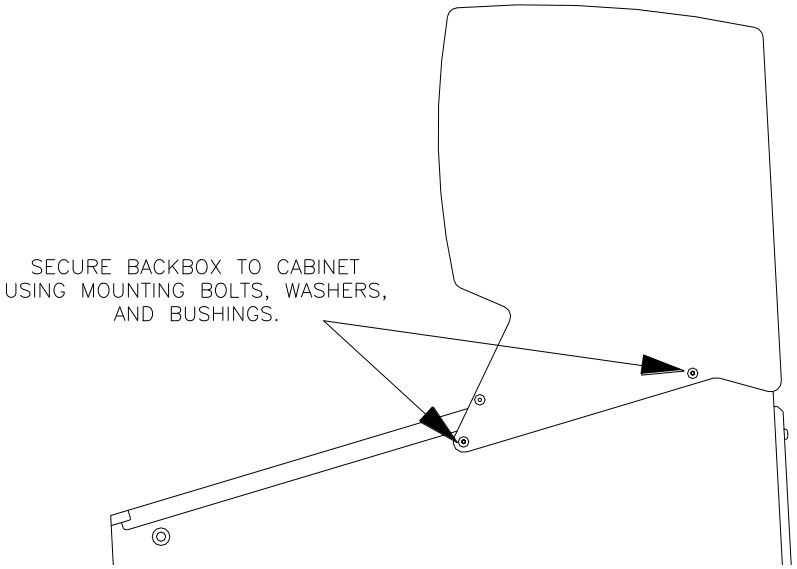


FIGURE 10

! CAUTION

FAILURE TO INSTALL the backbox mounting hardware properly can cause personal injury.

10. Slide the three cables back inside the backbox and connect them. *Do not force cables onto connectors, they should plug in easily*. See Figure 8, on page 1-4 to identify the cables. Route each cable through the unused plastic clamps attached to the inside of the cabinet.

    Plug the 9-pin serial cable into the bracket near the coin door and screw it securely in place (see Figure 11 below). Plug the 25-pin parallel cable into the Power Driver board at J100 and screw it securely in place. The power/speaker cable has two connectors, a 2-pin connector for the speakers and a 3-pin connector for power. Plug the speaker cable connector into the 2-pin connector near the speakers, and plug the power cable connector into the 3-pin isolation tap from the transformer, which is also located near the speakers. Be sure to match the wire colors on the speaker and power cables.

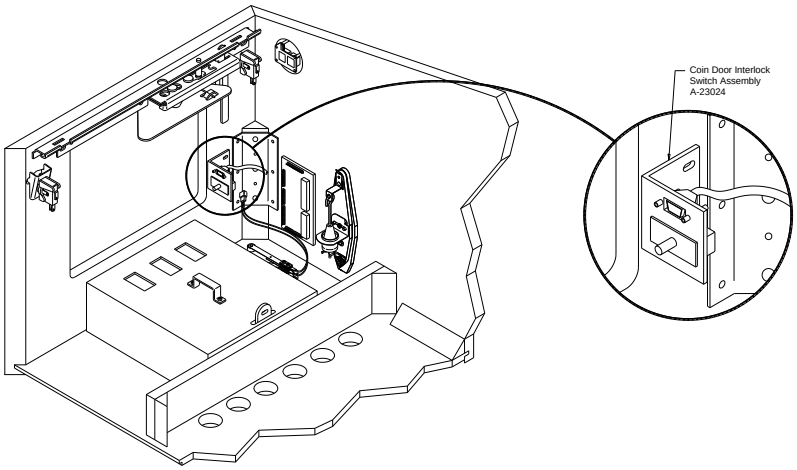


FIGURE 11

    The last cable to attach is the ground strap. The ground strap is located in the rear of the cabinet (see Figure 8 on page 1-4 for ground strap location). Reach through the rectangular hole and pull the ground strap up. Remove the wing nut from the ring lug located to the left of the rectangle hole. Slip the ground strap loop over the ring lug and secure it in place with the wing nut. Replace and lock the backbox rear door. Replacing the screws is optional.

11. Lean the playfield against the cabinet front and plug the cables back into the rear bracket.

12. Carefully, lift the playfield from the front and tip the back of it onto the slides. Slide it back into the cabinet. Be sure that the cables at the back of the playfield are not kinked.

13. Rest the front of the playfield on the top of the cabinet as you did in Step 2. Open the coin door. Lower the playfield all the way into the cabinet. Close and lock the coin door.

14. Be sure the **required number** of balls is installed.

15. Replace the playfield glass. Be sure that the PINBALL 2000 logo is in the lower left corner. See Figure 12 below. Snap the front molding securely into place. Close and lock the coin door.

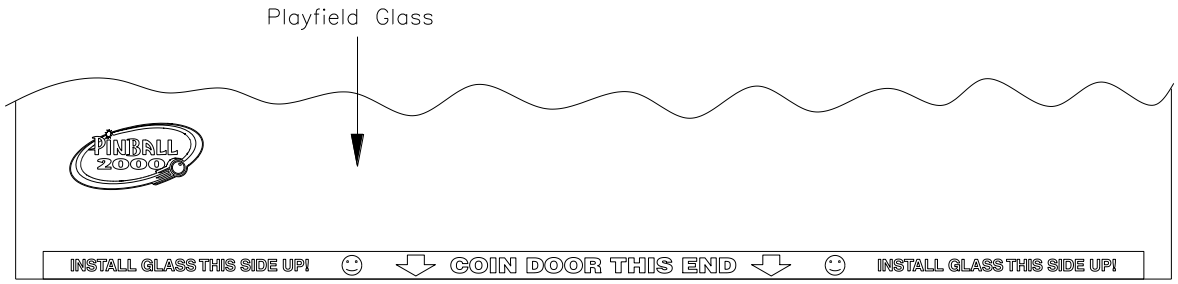


FIGURE 12

16. The level is located on the right shooter rail. This allows the playfield pitch angle to be properly adjusted WITHOUT REMOVING THE GLASS. The first line, (closest to the front of the game), on the level is approximately 6 degrees. Every line thereafter is approximately another 1/2 degree of pitch. The recommended pitch is 6-1/2 degrees. The NOSE of the bubble should be between the first and second line on the level. See Figure 13.

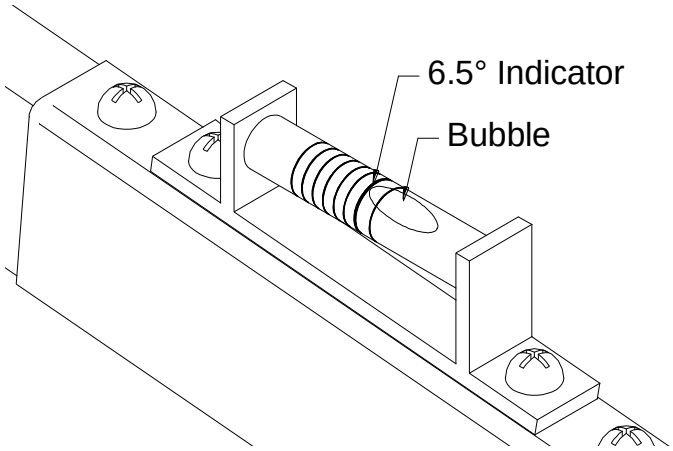


FIGURE 13

17. Move the game into the desired location. Adjust the leg levelers so that the playfield pitch is the recommended 6-1/2 degrees. This places the game in a FIXED position. Begin by screwing the leg levelers all the way into the legs. If the floor is level, the cabinet is designed to position the playfield at exactly at 6-1/2 degrees with the levelers as shown in Figure 1. From this position continue to adjust the levelers to compensate for an uneven or pitched floor. On carpeting, the game must be allowed to settle overnight. Check the level and pitch again the next day. See Figure 14.

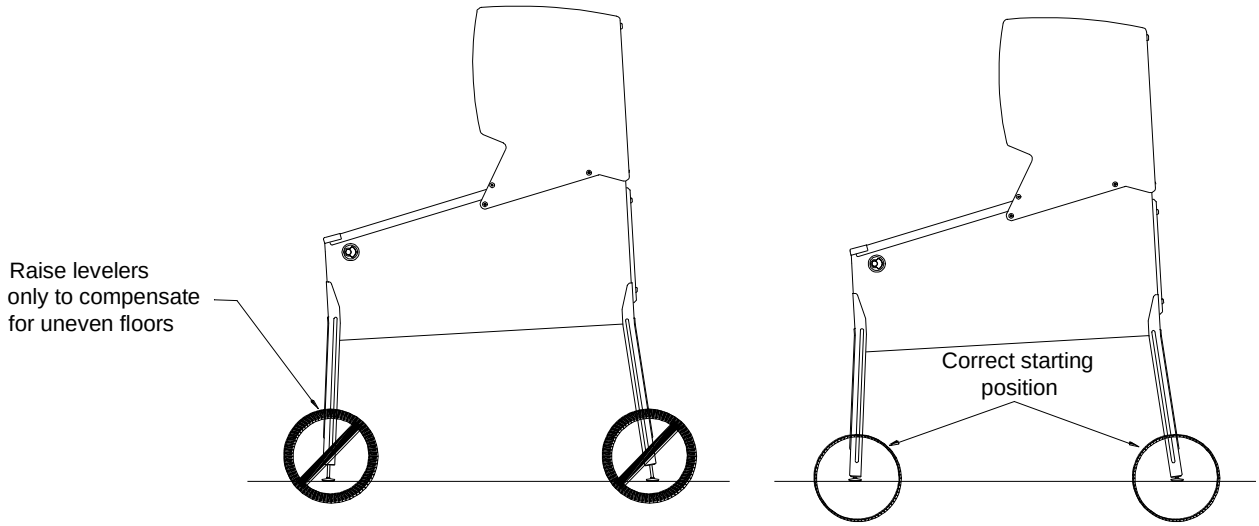


FIGURE 14

IMPORTANT!

Playfield pitch angle can affect the operation of the plumb bob tilt. The plumb bob weight is among the parts in the cash box. After completion of the desired playfield pitch angle setting, the operator should install the weight and adjust this tilt mechanism for proper operation. The unit is factory installed for a 6-1/2 degree angle. If an adjustment is necessary, loosen the screw at the bottom of the unit. Move the pointer, one grove at a time to the left or the right, depending on the degree desired. Hold the pointer in place and tighten screw.

18. Plug the line cord into the game cabinet. Match the prongs on the plug with the holes in the receptacle, and push the line cord securely into place. See Figure 15.

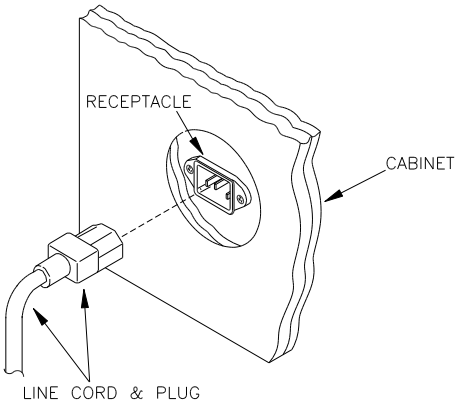


FIGURE 15

19. Move the game into the desired location. Adjust the leg levelers so that the playfield pitch is the recommended 6-1/2 degrees. This places the game in a FIXED position.

20. If a padlock is desired, install the security bar as shown below in Figure 16.

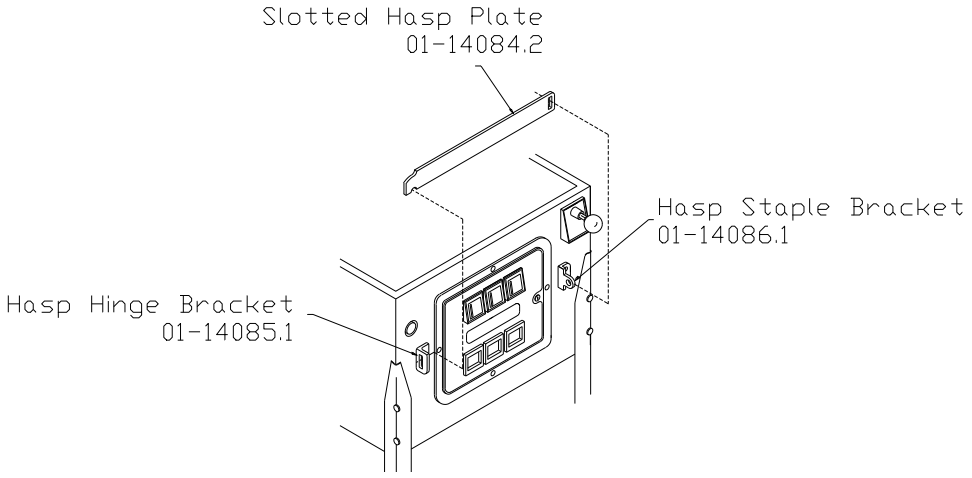


FIGURE 16

21. Plug the game into a properly grounded outlet.

! CAUTION

After assembly and installation at its site location, this game must be plugged into a properly grounded outlet to prevent shock hazard, and to assure proper game operation. DO NOT use a 'cheater' plug to defeat the ground pin. DO NOT cut off the ground pin.

22. POWERING UP. With the coin door closed, plug the game in and switch it on. In normal operation the game performs Start-up Tests. Once the Start-up tests have been successfully completed, the game enters the Attract mode.

23. IMPORTANT: Fill out and return the registration card.

RAISING THE PLAYFIELD

⚠ CAUTION

This game uses ski assemblies to raise and lower the playfield. Be sure the back of the playfield skis slide along the cabinet tracks until the feet hook into the loop at the end of the track.

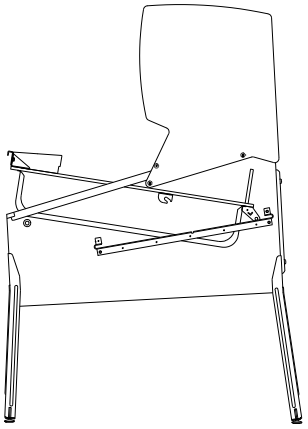
Before Raising the Playfield:

Be sure there are no balls present in any of the ball-holding playfield devices (i.e. poppers). Raising the playfield with balls present in these locations may cause them to come loose and damage the playfield. Use the "Empty Balls Test" to remove all of the balls from these locations.

To Raise the Playfield:

1. Open the coin door. Remove the front molding and the playfield glass. Lift the playfield by the metal bracket under the bottom arch. Rest it on the top of the cabinet.

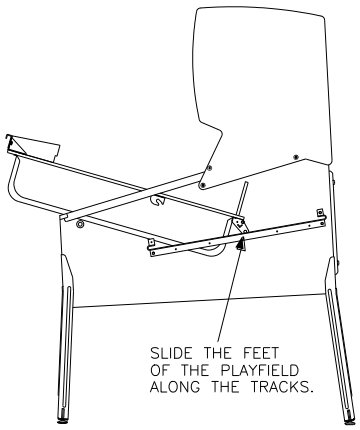
2. Close the coin door. Grasp the playfield by the skis and pull it toward you along the track slides. Rotate the front of the playfield up and back toward the backbox.

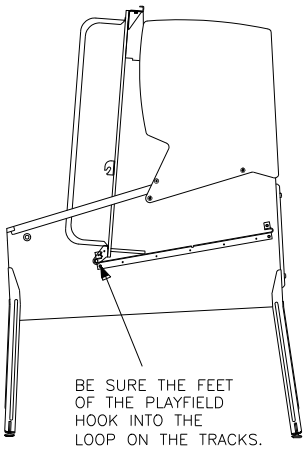


To Lower the Playfield

1. Rotate the front of the playfield down and push it back, away from you, along the track slides. Rest it on top of the cabinet. Open the coin door.

2. Grasp the playfield by the metal bracket under the bottom arch and lower it back into the cabinet. Replace the playfield glass and the front molding. Close and lock the coin door.





GAME CONTROL LOCATIONS

Cabinet Switches

The ON/OFF SWITCH is on the bottom of the cabinet near the right front leg.
The START BUTTON is a push-button to the left of the coin door on the cabinet exterior. Press the Start button to begin a game, or during the diagnostic mode, to ask for HELP.

Service Lock

The service lock allows the operator to gain access to the playfield without allowing access to the cash box. With the coin door closed, turn the key in the service lock and the front molding pops up. Remove the molding and slide the glass down and off of the cabinet. The operator can clean the playfield or remove a trapped ball without being able to lift the playfield and gain access to the cash box.

Coin Door Buttons

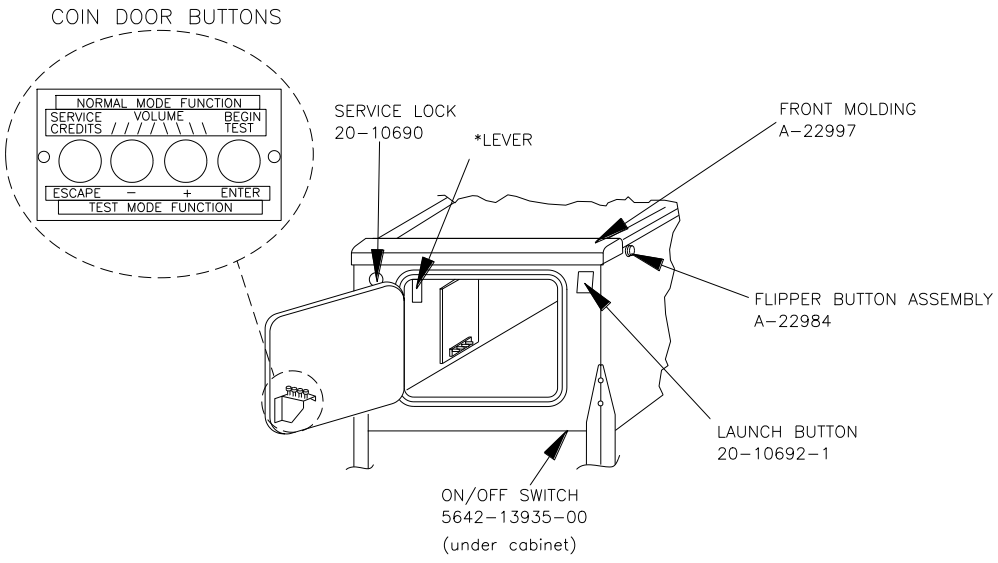
The operator controls all game adjustments, obtains bookkeeping information, and diagnoses problems, using only four push-button switches mounted on the inside of the coin door. The coin door buttons have two modes of operation Normal Function and Test Function.

NORMAL FUNCTION

The SERVICE CREDITS button puts credits on the games that are not included in any of the game audits.
The VOLUME UP (+) button raises the sound level of the game. Press and hold the button until the desired level is reached.
The VOLUME DOWN (-) button lowers the sound level of the game. Press and hold the button until the desired level is reached. See the Adjustment menu to turn the sound down all the way.
The **BEGIN TEST button starts the Menu System operation and changes the coin door buttons from Normal Function to Test Function.

TEST FUNCTION

The ESCAPE button allows you to get out of a menu selection or return to the Attract mode.
The UP (+) button allows you to cycle forward through the menu selections or adjustment choices.
The DOWN (-) button allows you to cycle backward through the menu selections or adjustment choices.
The ENTER button allows you to get into a menu selection or lock in an adjustment choice.



The START BUTTON (p/n 20–9663–16) is not shown on this drawing. It is located below the service lock.

*The LEVER is part of the LOCK BRACKET ASSEMBLY (p/n A–22996).

** To reset the High Scores, press and hold the BEGIN TEST button.

GAME OPERATION

⚠ CAUTION

After assembly and installation at its location, this game must be plugged into a properly grounded outlet to prevent shock hazard, and to assure proper game operation. DO NOT use a 'cheater' plug to defeat the ground pin on the line cord. DO NOT cut off the ground pin.

POWERING UP. With the coin door closed, plug the game in, and switch it on. The game performs Start-up Tests. A test pattern and 'PLEASE WAIT' appears on the screen. The screen changes and 'TESTING' appears on the screen. Once the Start-up tests have been successfully completed the game goes into the Attract mode.

Note: *After the game has been on location for a time, the Start-up tests may contain messages concerning game problems.*

ATTRACT MODE. During the Attract mode, the screen shows a series of messages informing the player of the recent highest scores.

CREDIT POSTING. Insert coin(s). The screen shows the number of credits purchased.

STARTING A GAME. Press the Start button. A ball is ejected into the shooter lane. Press the Ball Launch button on the cabinet to send the ball onto the playfield and begin game play. If credits are posted, additional players may enter the game by pressing the Start button once for each player, before the end of play on the first ball.

TILTS. Actuating the cabinet slam tilt switch inside the cabinet ends the current game and proceeds to the Game-over mode. With the third closure of the plumb bob tilt switch, the player loses the remaining play of that ball, but can complete the game.

END OF A GAME. All earned scores and bonuses are awarded. If a player's final score exceeds the specified value, the player receives a designated award for achieving the current highest score. A random digit set appears in the display. An award may be given when the last two digits of any player's score match the random digits.

GAME-OVER MODE. The **Game-over** screen shows the high scores and the game proceeds to the Attract Mode.

MENU SYSTEM OPERATION

The System Menu allows you to choose from several menus, which in turn, lead to other menus to choose from. To access the System Menu, open the coin door and press the Enter button. The System Menu appears on the screen. Press either the Up or the Down buttons to move the cursor up or down the screen. Notice how the menu options are highlighted in order. A menu option must be highlighted for it to be selected. To open a System Menu item, (Diagnostics menu, Adjustments menu etc.), press the Enter button when that menu is highlighted. To return to the System Menu (from the Diagnostics menu, Adjustments menu, etc.) press the Escape button at any time. The following outline shows the System Menu and the next lower level of menus available from the System Menu.

SYSTEM MENU

DIAGNOSTICS

- Switch Edges
- Single Switches
- DIP Switch
- Solenoid Test
- Single Lamp Test
- Lamp Row/Column
- All Lamps
- Video Tests
- Fuse Check

ADJUSTMENTS

- Audio Adjustments
- System Adjustments
- Feature Adjustments
- Game Pricing
- High Scores
- Communication

BOOKKEEPING

- Main Audits
- Earning Audits
- Standard Audits
- Feature Audits
- Histograms
- Timestamps
- Earnings Charts

UTILITIES

- System Information
- Show Shell Window
- Set Location I.D.
- Update Game Code
- Clear Audits
- Clear Coins
- Reset High Scores
- Set Clock
- Factory Adjustments
- Factory Reset
- Install Presets
- Clear Credits

PRINTOUTS

- Print Audits
- Print Adjustments
- Print Everything

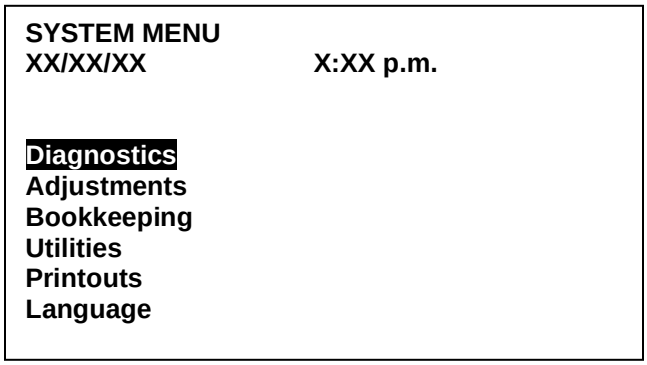
LANGUAGE

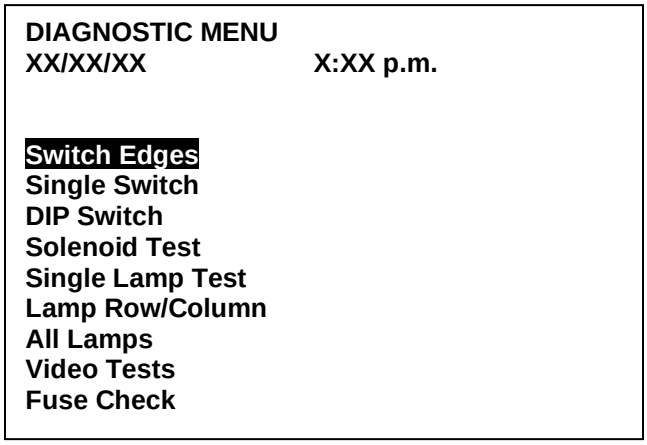
- English
- Deutsch
- Francais
- Espanol

DIAGNOSTICS

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Diagnostic menu. Press the Enter button to open the menu when it is selected. Once the Diagnostic menu is open, use the Up and Down buttons to move the cursor and select a test. Press the Enter button to enable the test. Press the Escape button to return to the Diagnostic menu. Press the Escape button a second time to return to the System Menu.

In order to operate the tests that use the +50V or +20V circuits, pull the interlock switch button out. The interlock switch is located on a bracket in the coin door opening.





SWITCH EDGES TEST

Press each of the switches one at a time. The name and number of the switch is on the screen. If a switch other than the one pressed, or no switch at all is indicated, the system has detected a problem with the switch circuit. To return the Diagnostic menu, press the Escape button.

SINGLE SWITCHES TEST

This test isolates a single switch and shows its state in the display. A mechanical switch is 'made' when the display reads closed. An opto switch is 'made' (opto beam broken) when the display reads open. Use the Up or Down buttons to select the switch to be tested. To return the Diagnostic menu, press the Escape button.

DIP SWITCH TEST

This test is used to show the positions of the DIP Switches. To return to the Diagnostic menu, press the Escape button.

SOLENOID TEST

The Solenoid test has three modes -- Repeat, Stop, and Run. Only one solenoid should pulse at a time. The system has detected a problem if more than one solenoid pulses, a solenoid comes on and stays on, or no solenoids pulse during the Repeat and Run modes.

Repeat: The Repeat mode activates an individual solenoid. Press the Enter button to start this test. The name of the first solenoid shows in the display and the corresponding coil pulses. Press the Up or Down buttons to cycle through the solenoids, one at a time. The same solenoid pulses until you press the Up or Down buttons to advance to the next one. To return the Test menu, press the Escape button. To advance to the next test mode, press the Enter button.

Stop: The Stop mode halts the Solenoid test. No solenoids should be active. To return the Test menu, press the Escape button. To advance to the next test mode, press the Enter button.

Run: The Run mode cycles through the solenoids automatically. The display shows the name and number of the solenoid currently being pulsed. To return the Test menu, press the Escape button. To return to the Repeat mode, press the Enter button.

SINGLE LAMP TEST

The Single Lamp test checks each lamp circuit individually. Press the Up or Down buttons to scroll through this test. A lamp should light for each name and number that is displayed. Any other results indicate the system has detected a problem. To return to the Diagnostic menu, press the Escape button.

LAMP ROW/COLUMN

This test allows individual rows and columns in the lamp matrix to be operated. This is useful for troubleshooting wiring and driver problems.

Press the Up and Down buttons to cycles through the different rows and columns. To return to the Diagnostic menu, press the Escape button.

ALL LAMPS TEST

This test causes all the controlled lamps to flash at the same time. Every controlled lamp should flash. Any other results indicate the system has detected a problem. To return to the Diagnostic menu, press the Escape button.

VIDEO TEST

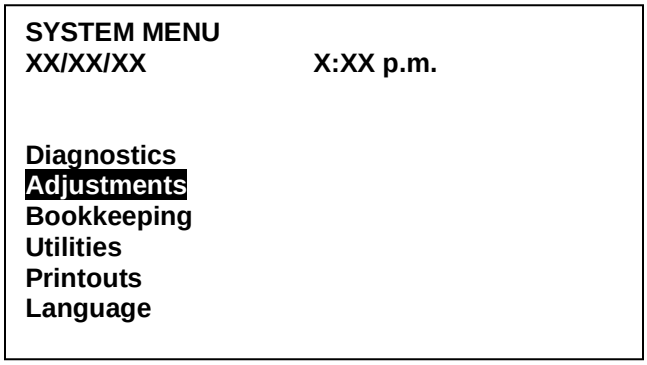
FUSE CHECK

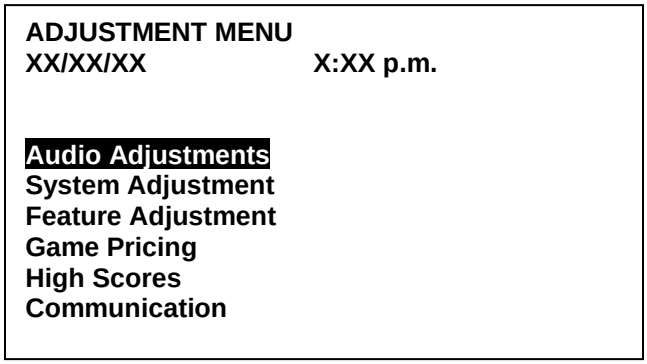
ADJUSTMENTS

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Adjustment menu. Press the Enter button to open the menu when it is selected. Once the Adjustment menu is open, use the Up and Down buttons to move the cursor and select a sub-menu. Press the Enter button to enable the sub-menu and view the next level of functions.

Use the Up and Down buttons to move the cursor and select a sub-menu function. Press the Enter button to enable that function. Use the Up and Down buttons to change the value. Press the Enter button to lock in the new value, or press the Escape button to retain the original value and return to the sub-menu.

Press the Escape button to return to the sub-menu. Press the Escape button a second time to return to the Adjustment menu and a third time to return to the System Menu.





AUDIO ADJUSTMENTS

Current Volume

This adjustment shows the current volume level of the game.
Settings:
Factory Setting: XX

Minimum Volume Override

The volume can be turned off.
Settings: Yes or No
Factory Setting: No

Volume Interlock

Settings: Yes or No
Factory Setting: No

SYSTEM ADJUSTMENTS

Extra Ball

This adjustment leads to a menu of adjustments for the Extra Ball feature.

Match

This adjustment lead to a menu of adjustments for the Match feature.

Replay

This adjustment leads to a menu of adjustments for the Replay feature.

Power Saver

This adjustment leads to a menu of adjustments for the Power Saver feature.

Clock/Format

This adjustment leads to a menu of adjustments for the Clock feature.

Balls Per Game

A "game" is defined by specifying the number of balls to be played.
Settings: 1 to 10
Factory Setting: 3

Game Restart

When you press the Start button during or after the 2nd ball, the game in progress ends and a new game begins. This adjustment has three settings to determine how to handle this.

Settings: **Never** – Don't allow a new game start until the current game is over.
Slow - Restart if the Start button is pressed continuously for over 1/2 second. This helps to prevent the unintended restart of the game in progress.
Instantly - Restart as soon as the Start button is pressed. When you press the Start button during game over, or during the 1st ball (to add a player), it is always handled instantly.

Factory Setting: Slow

Tournament Play

Equalize random game features and global score values during multi-player games.
Settings: Yes or No
Factory Default: No

Maximum Tickets per Player

The amount of tickets each player can earn.
Settings: 00 to 100.
Factory Default: 25

Special Award

The award a player receives when he/she earns a Special.
Settings:
Factory Setting: Credit

Slam Tilt Penalty

Whether of not the player is penalized for causing a Slam Tilt.
Settings: Yes or No
Factory Setting: No

Tilt Warning

The number of total actuation’s of the plumb bob that can occur before the game is "tilted".
Settings: 1 to 10
Factory Setting: 3

System Adjustments Continued…

Allow Chase Ball

The Chase Ball feature will attempt to keep a game operational in the event that a ball becomes stuck on the playfield. After a number of unsuccessful ball searches, the game software will end the player's current ball, give a bonus award, then serve a new ball (or end the game if that was the last ball in the player's game). This ball 'chases' the stuck ball on the playfield and will hopefully knock it loose. Setting this adjustment to NO will revert to the previous behavior of searching endlessly until the stuck ball becomes freed, or the machine's power is turned off and back on.

Settings: Yes or No
Factory Setting: Yes

Midas Enabled

This enables the second serial port to work with the Midas Security Unit.

Settings: Yes or No
Factory Setting: No

FEATURE ADJUSTMENTS

Scene Select Timer

The starting number of seconds used for timing down the scene select screen at the start of each player's first ball. When set to OFF, the select screen will remain displayed until the player presses the launch button.

Settings: 05-10, Off

Scene Select Wrap

Allows action button presses for the scene select screen to "wrap" the selected beam (group) to the opposite side.

Settings: Yes, No

Paris Saucer Hits

The number of hits required on the saucer to qualify the "Paris In Peril" scene to be completed. The higher the value, the more difficult it is to complete the "Paris In Peril" scene.

Settings: 06-10

Big-O-Beam Barn Hits

The number of hits required on the barn to qualify the "Big-O-Beam" scene to be completed. The higher the value, the more difficult it is to complete the "Big-O-Beam" scene.

Settings: 04-10

Mars Kneads Women TVs with Static

The number of TV screens containing "static" at the start of the "Mars Kneads Women" scene. Each screen with static requires an additional shot to complete. The higher the value, the more difficult it is to complete the "Mars Kneads Women" scene.

Settings: 00, 02, 04, 06

Tower Struggle Tower Hits

The number of hits required on the tower to qualify the "Tower Struggle" scene to be completed. The higher the value, the more difficult it is to complete the scene.

Settings: 04-10

Feature Adjustments Continued…

Question Mark Difficulty

Controls the difficulty of the "Question Mark" scenes (Martian Tank, Martian Aerobics, Martian Bowling, and Martian Autopsy). Harder settings require more shots to qualify the completion of the scenes.

Settings: Extra Easy, Easy, Medium, Hard, Extra Hard

Drive-In Demolition Difficulty

Controls the difficulty of the "Drive-In Demolition" scene. Harder settings require more shots to qualify the completion of the scene.

Settings: Extra Easy, Easy, Medium, Hard, Extra Hard

Alien Abduction Difficulty

Controls the difficulty of the "Alien Abduction" scene. Harder settings require more shots to qualify the completion of the scene.

Settings: Extra Easy, Easy, Medium, Hard, Extra Hard

Martian Happy Hour Difficulty

Controls the difficulty of the "Martian Happy Hour" scene. Harder settings require more shots to qualify completion of the scene.

Settings: Extra Easy, Easy, Medium, Hard, Extra Hard

Secret Weapon Difficulty

Controls the difficulty of the "Secret Weapon" scene. Harder settings require more shots to qualify completion of the scene.

Settings: Extra Easy, Easy, Medium, Hard, Extra Hard

Bonus Wave Count

The number of shots required during the "Bonus Wave Multiball" scenes (Fuel, Saucer, Weapons) to achieve the next bonus award (Jackpot, Saucer Light, Extra Ball, etc.) The higher the number, the more difficult it is to obtain bonus awards during the "Bonus Wave Multiball" scenes.

Settings: 04-06

Capture Multiball Difficulty

A difficulty adjustment for Capture Multiball, which affects the number of shots to the center loop "capture zone" to achieve Multiball. The "Easy" setting always requires four shots to this area for Multiball. The "Medium" setting requires four shots for the first Multiball and six shots for each Multiball thereafter. The "Hard" setting always requires six shots for Multiball.

Settings: Easy, Medium, Hard

Martian Attack Timer

The number of seconds the player is given to complete the "Martian Attack" feature.

Settings: 20-40

Hypno Beam Difficulty

This is a difficulty adjustment for Hypno Beam Multiball. Controls the Multiball ball save timer and the number of shots available to add balls for the Multiball.

Feature Adjustments Continued…

Missile Extra Ball #1

The number of ”Missile” shots (Left Loop, Left Ramp, Right Ramp, Right Loop) required to light the first “Missile” extra ball.

Settings: 10-30, Off

Missile Extra Ball #2

The number of “Missile” shots (Left Loop, Left Ramp, Right Ramp, Right Loop) required to light the second “Missile” extra ball.

Settings: 40-60, Off

Timed Plunger

The seconds before automatically launching the ball onto the playfield at ball start when the ball is sitting in the shooter lane.

Settings: 30-90, Off

Disable Rubber Martians

This setting is provided for use when the solenoid driven, “Rubber Martians” are broken and/or intermittent. Setting this adjustment to “Yes” will keep the Martians from being activated during game play and ball search.

Settings: Yes, No

Disable Left Gate

Provided for use when the Left Gate is broken and/or intermittent. Setting this adjustment to “Yes” will keep the Left Gate from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

Disable Right Gate

Provided for use when the Right Gate is broken and/or intermittent. Setting this adjustment to “Yes” will keep the Right Gate from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

Disable Lock Diverter

Provided for use when the Lock Diverter is broken and/or intermittent. Setting this adjustment to “Yes” will keep the Lock Diverter from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

Disable Up/Down Ramp

Provided for use when the Up/Down Ramp is broken and/or intermittent. Setting this adjustment to “Yes” will keep the Up/Down Ramp from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

Feature Adjustments Continued…

Disable Jet Exit Post

Provided for use when the Jet Exit Post is broken and/or intermittent. Setting this adjustment to "Yes" will keep the Jet Exit Post from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

Disable Right Ramp Drop Target

Provided for use when the Right Ramp Drop Target is broken and/or intermittent. Setting this adjustment to "Yes" will keep the Right Ramp Drop Target from being activated during game play and ball search, and provide additional game play software compensation for the broken/intermittent device.

Settings: Yes, No

GAME PRICING

Free Play

This adjustment determines whether or not the player can play the game for free.
Settings: Yes or No
Factory Setting: No

Maximum Credits

This is the maximum amount of credits that can be bought at one time.
Settings:
Factory Setting: 10

Hide Coin Audits

This determines if the coin audits are visible to service personnel.
Settings: Yes or No
Factory Setting: No

HIGH SCORES

Highest Scores

The game maintains a record of the four highest scores achieved to date.
Settings: Yes or No
Factory Setting: Yes

High Score To Date Award

This is the award given for achieving the High Score to Date or the Champion High Score to Date.
Settings: Credit or Ticket.
Factory Setting: Credit

Champion Credit

The number of credits or tickets awarded for a Grand Champion Score.
Settings: 0 to 10
Factory Settings: 1

High Scores Continued…

H.S.T.D. 1 Credits
H.S.T.D. 2 Credits
H.S.T.D. 3 Credits
H.S.T.D. 4 Credits

The number of credits or tickets awarded whenever a player exceeds the four highest scores.
Settings: 00 to 10
Factory Setting: 1

H.S.T.D. Reset Every

The number of games to be played before an automatic reset of the displayed Highest Score occurs. The operator selects the values provided at reset in the Back-up High Scores.
Settings: OFF (disabled), 250 to 20,000.
Factory Settings: 2000 Games

Default Champion

The Back-up Grand Champion Score.
Settings: 00 to 200,000,000
Factory Settings: 50,000,00

Default H.S.T.D. 1	**40,000,000**
Default H.S.T.D. 2	**30,000,000**
Default H.S.T.D. 3	**20,000,000**
Default H.S.T.D. 4	**10,000,000**

The first through fourth Default High Score values. The game automatically restores this value when the "High Score Reset Every" value is reached.
Settings: 00 to 200,000,000
Factory Settings: Default H.S.T.D. 1 = 40,000,000
Default H.S.T.D. 2 = 30,000,000
Default H.S.T.D. 3 = 20,000,000
Default H.S.T.D. 4 = 10,000,000

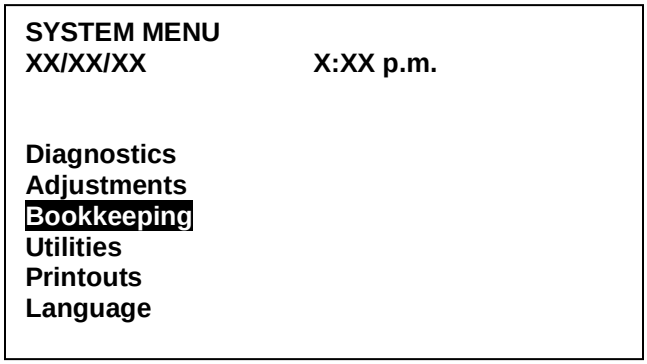
BOOKKEEPING

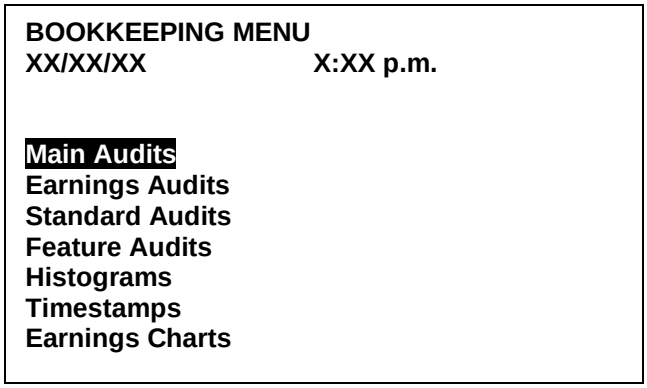
Audits Cannot Be Set, They Can Only Be Cleared.

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Bookkeeping menu. Press the Enter button to open the menu when it is selected. Once the Bookkeeping menu is open, use the Up and Down buttons to move the cursor and select a sub-menu. Press the Enter button to enable the sub-menu and view the next level of functions.

Use the Up and Down buttons to move the cursor and select a sub-menu function. Press the Enter button to enable that function. Use the Up and Down buttons to reset the value to zero. Press the Enter button to lock in the zero value, or press the Escape button to retain the original value and return to the sub-menu.

Press the Escape button to return to the sub-menu. Press the Escape button a second time to return to the Bookkeeping menu and a third time to return to the System Menu.





MAIN AUDITS

No Main Audits at this time.

EARNING AUDITS

Total Earnings	XX	Recent Slot 6	XX
Recent Earnings	XX	Recent Slot 7	XX
Recent Paid Credits	XX	Recent Slot 8	XX
Total Paid Credits	XX	Total Slot 1	XX
Recent Service Credits	XX	Total Slot 2	XX
Total Service Credits	XX	Total Slot 3	XX
Recent Slot 1	XX	Total Slot 4	XX
Recent Slot 2	XX	Total Slot 5	XX
Recent Slot 3	XX	Total Slot 6	XX
Recent Slot 4	XX	Total Slot 7	XX
Recent Slot 5	XX	Total Slot 8	XX

STANDARD AUDITS

Games Started	XX		Right Drains	XX	XX%
Total Plays	XX		Game Tilts	XX	
Average Game Time	XX		Slam Tilts	XX	
Balls Played	XX		Plumb Bob Tilts	XX	
Average Ball Time	XX		Ball Roll Tilts	XX	
Extra Balls	XX	XX%	1 Player Games	XX	
Tickets Awarded	XX	XX%	2 Player Games	XX	
Match Awards	XX	XX%	3 Player Games	XX	
Total Replay Awards	XX	XX%	4 Player Games	XX	
Replay 1 Awards	XX	XX%	Ball Searches	XX	
Replay 2 Awards	XX	XX%	Ball Searches (5)	XX	
Replay 3 Awards	XX	XX%	Chase Balls	XX	
Replay 4 Awards	XX	XX%			
Left Drains	XX	XX%			

FEATURE AUDITS

Alien Abduction Started –

The number of times the “Alien Abduction” scene was started.

Martian Happy Hour Started –

The number of times the “Martian Happy Hour” scene was started.

Secret Weapon Started –

The number of times the “Secret Weapon” scene was started.

Fuel Started –

The number of times the “Fuel” (Bonus Wave Multiball) scene was started.

Tower Struggle Started –

The number of times the “Tower Struggle” scene was started.

Question Mark Started –

The number of times the “Question Mark” (“?”) scene was started.

Drive In Demolition Started –

The number of times the “Drive In Demolition” scene was started.

Saucer Started –

The number of times the “Saucer” (Bonus Wave Multiball) scene was started.

Paris In Peril Started –

The number of times the “Paris In Peril” scene was started.

Big-O-Beam Started –

The number of times the “Big-O-Beam” scene was started.

Mars Kneads Women Started –

The number of times the “Mars Kneads Women” scene was started.

Martian Tank Started –

The number of times the “Martian Tank” Question Mark scene was started.

Martian Aerobics Started –

The number of times the “Martian Aerobics” Question Mark scene was started.

Feature Audits Continued…

Martian Bowling Started –
The number of times the “Martian Bowling” Question Mark scene was started.

Martian Autopsy Started –
The number of times the “Martian Autopsy” Question Mark scene was started.

Ball Capture Lit –
The number of ball captures lit for Capture Multiball.

Balls Captured –
The number of balls captured for Capture Multiball.

Capture Multiball Started –
The number of times the Capture Multiball feature was started.

Capture Multiball Super Jackpot –
The number of times the Super Jackpot was collected during Capture Multiball.

Martian Attack Lit –
The number of times the “Martian Attack” feature was lit by completing the martians standup targets.

Martian Attack Started –
The number of times the “Martian Attack” feature was started.

Martian Multiball Started –
The number of times the “Martian Multiball” feature was started.

Martian Bombs Collected –
The number of times a “Martian Bomb” was collected.

Martian Bombs Used –
The number of times a “Martian Bomb” was used.

Bonus Wave Extra Ball –
The number of extra balls awarded from “Bonus Wave Multiball”.

Mothership Multiball Started –
The number of times “Mothership Multiball” was started.

Missile Extra Ball #1 Lit –
The number of times the first extra ball was lit from the “Missile” shots.

Missile Extra Ball #2 Lit –
The number of times the second extra ball was lit from the “Missile” shots.

Saucer Light #3 –
The number of times the third saucer light was collected.

Saucer Light #6 –
The number of times the sixth saucer light was collected.

All Saucer Lights –
The number of times all of the saucer lights were collected.

Stroke of Luck Lit –
The number of times the Stroke of Luck feature was lit.

Feature Audits Continued…

Stroke of Luck Awards –
The number of times the Stroke of Luck feature was awarded.

Stroke of Luck "500,000" –
The number of times Stroke of Luck awarded "500,000".

Stroke of Luck "1,000,000" –
The number of times Stroke of Luck awarded "1,000,000".

Stroke of Luck "+5 Bonus X" –
The number of times Stroke of Luck awarded "+5 Bonus X".

Stroke of Luck "Hold Bonus X" –
The number of times Stroke of Luck awarded "Hold Bonus X".

Stroke of Luck "Spot 3 Missiles" –
The number of times Stroke of Luck awarded "Spot 3 Missiles".

Stroke of Luck "1 Saucer Light" –
The number of times Stroke of Luck awarded "1 Saucer Light".

Stroke of Luck "Light Capture" –
The number of times Stroke of Luck awarded "Light Capture".

Stroke of Luck "Award Capture" –
The number of times Stroke of Luck awarded "Ball Capture".

Stroke of Luck "Start Capture Multiball" –
The number of times Stroke of Luck awarded "Start Capture Muliball".

Stroke of Luck "Light Martian Attack" –
The number of times Stroke of Luck awarded "Light Martian Attack".

Stroke of Luck "1 Martian Bomb" –
The number of times Stroke of Luck awarded "1 Martian Bomb".

Stroke of Luck "Light Extra Ball" –
The number of times Stroke of Luck awarded "Light Extra Ball".

Stroke of Luck "Hypno Beam Multiball" –
The number of times Stroke of Luck awarded "Hypno Beam Multiball".

Hypno-Beam Extra Balls –
The number of extra balls collected from the Hypno-Beam feature.

Hypno-Beam Jackpots –
The number of jackpots collected from the Hypno-Beam feature.

Attack Mars Lit –
The number of times the "Attack Mars" feature was qualified.

Attack Mars Started –
The number of times the "Attack Mars" feature was started.

Attack Mars Won –
The number of times the "Attack Mars" feature was won.

HISTOGRAMS

Score Histograms
Ball Histograms

TIMESTAMPS

Total Uptime	XX
Current Uptime	XX
System Downtime	XX
Last Game Start	XX
Last Service Credit	XX

EARNINGS CHARTS

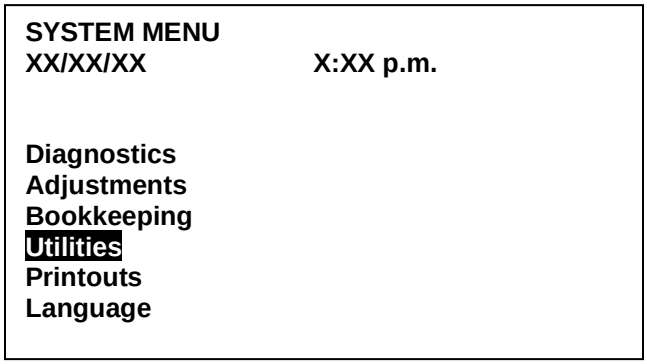
Last 24 Hours
Last 21 Hours
Last 21 Days
Last 12 Hours

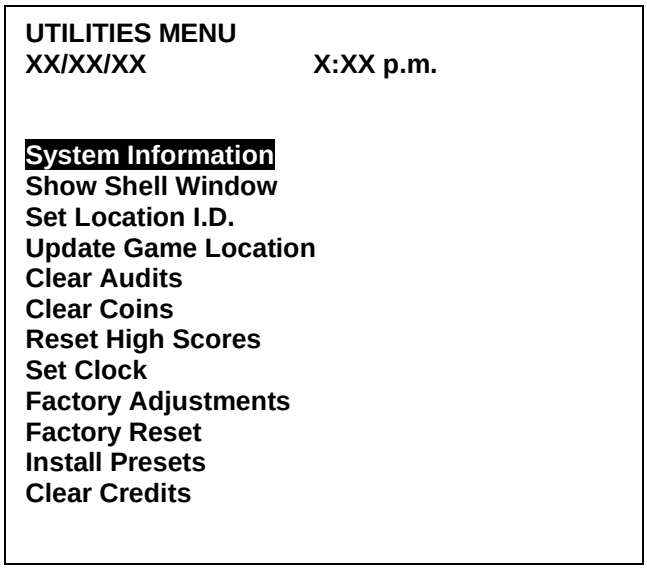
UTILITIES

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Utilities menu. Press the Enter button to open the menu when it is selected. Once the Utilities menu is open, use the Up and Down buttons to move the cursor and select a function. Press the Enter button to enable the function.

Use the Up and Down buttons to change the function setting. Press the Enter button when the desired setting is displayed and a 'countdown' screen appears. Press the Enter button during the 'countdown' to lock in the new setting. Or, wait for the 'countdown' to complete and the new setting is locked in automatically. Press the Escape button, during the 'countdown' to retain the original setting and return to the Utilities Menu.

Press the Escape button to return to the Utilities menu. Press the Escape button a second time to return to the System Menu.





SYSTEM INFORMATION

The System Information utility displays game specifications. It cannot be cleared.

SHOW SHELL WINDOW

This is used to show the system terminal window. It is for diagnostic use.

SET LOCATION I.D.

This utility allows the operator to show the game's location on printouts. Press the Enter button to open the utility. Use the Up and Down buttons to move the cursor and select letters. Press the Enter button to lock in desired letters.

UPDATE GAME CODE CODE (You will typically want to run the latest update).

Use this menu to enable or disable the update game code (stored in the PRISM board FLASH memory).

Enable Updated Code

If the update game code is enabled, the system will run the last version of update game code installed (REFERENCE SOFTWARE UPDATES IN NEW FEATURES OF PINBALL 2000).

Disabled Updated Code

If the update game code is disabled, the system will run the version of game code located in the original PRISM board ROMs.

WARNING TO THE OPERATOR.

In early releases of the Revenge from Mars game code ROMs, DO NOT perform the Disabled Updated Code. Doing so will cause your game to run in Free Only Mode and the UPDATE GAME CODE utility menu will not allow the updated game code to be enabled.

If you accidentally disable the update game code, do one of the following:

1) Plug a keyboard into the keyboard port (located near the computer case in the backbox) or,
2) Plug a computer terminal into the COM 1 port (located near the coin door).

Then, at the command (%) prompt type the following:

% fupdate enable

to re-enable the Updated Game Code.

Note: This operation takes about 15 seconds to complete.

CLEAR AUDITS

Press the Enter button to clear the all of the audits.

CLEAR COINS

Press the Enter button to clear the Earnings Audits.

RESET HIGH SCORES

Press the Enter button to clear the High Score to Date Table.

SET CLOCK

Press the Enter button to activate the clock. Use the Up and Down buttons to change the time, then press the Enter button to lock in the time.

FACTORY ADJUSTMENT

Press the Enter button to restore the adjustments to factory settings.

FACTORY RESET

Press the Enter button to restore the adjustments to their factory setting, clear the Audits, Reset the High Score, and the Location I.D.

INSTALL PRESETS

Press the Enter button to open the utility. Use the Up and Down buttons to move the cursor and select from the available Presets. When the desired Preset is selected, press the Enter button to lock it in. Press the Escape button, during the 'countdown' to retain the original setting and return to the Utilities Menu. See the Preset Table on the next page.

PRESET TABLES

Difficulty Settings

Adjustment	Extra Easy	Easy	Medium	Hard	Extra Hard
Paris Saucer Hits	7	7	8	9	9
Big-O-Beam Barn Hits	6	6	7	8	8
MKW TVs with Static	2	2	2	2	2
Tower Struggle Tower Hits	6	6	7	8	8
Question Mark Difficulty	Extra Easy	Easy	Medium	Hard	Extra Hard
Drive-In Demolition Difficulty	Extra Easy	Easy	Medium	Hard	Extra Hard
Alien Abduction Difficulty	Extra Easy	Easy	Medium	Hard	Extra Hard
Martian Happy Hour Difficulty	Extra Easy	Easy	Medium	Hard	Extra Hard
Secret Weapon Difficulty	Extra Easy	Easy	Medium	Hard	Extra Hard
Capture Multiball Difficulty	Medium	Medium	Hard	Hard	Hard
Missile Extra Ball #1	15	15	20	25	25
Missile Extra Ball #2	40	45	50	55	60
Martian Attack Timer	40	35	30	25	20
Hypno Beam Difficulty	Easy	Medium	Hard	Hard	Hard

3 and 5 Ball Settings

Adjustment	3 Ball	5 Ball
Balls Per Game	3	5
Score Level #1	50,000,000	80,000,000
Score Level #2	250,000,000	300,000,000
Score Level #3	500,000,000	600,000,000
Score Level #4	800,000,000	900,000,000

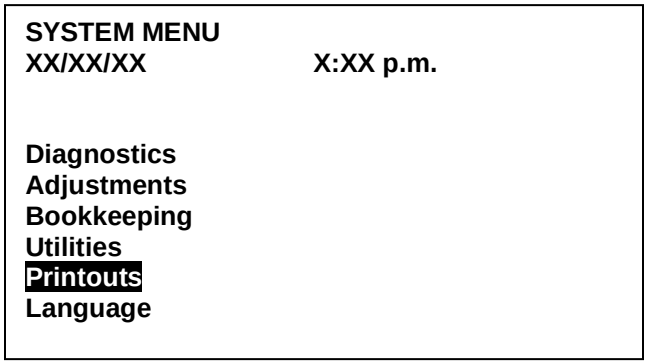
CLEAR CREDITS

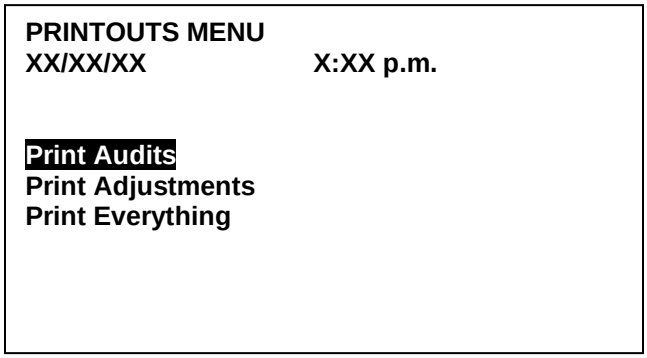
Press the Enter button to clear the game Credits.

PRINTOUTS

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Printouts menu. Press the Enter button to open the menu when it is selected. Once the Printouts menu is open, use the Up and Down buttons to move the cursor and select a function. Press the Enter button to enable the function.

Press the Escape button to return to the Printouts menu. Press the Escape button a second time to return to the System Menu.





PRINT AUDITS

Print all System Audits.

PRINT ADJUSTMENTS

Print a record of all System Adjustments. A "*" signifies a changed adjustment.

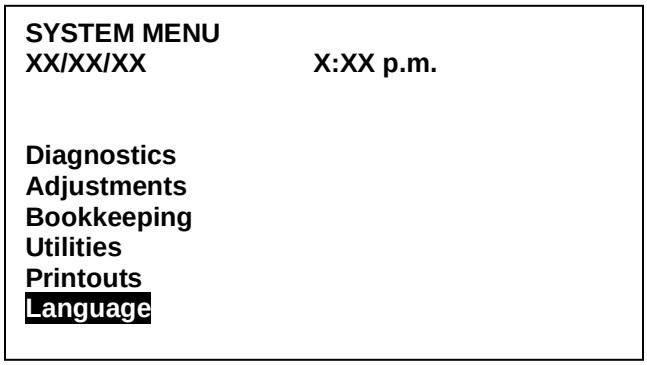
PRINT EVERYTHING

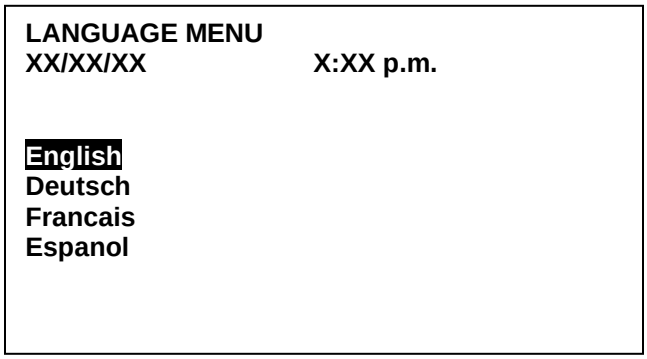
Print both audits and adjustments.

LANGUAGE

Open the coin door, press the Enter button and the System Menu appears on the screen. Use the Up and Down buttons to move the cursor and select the Language menu. Press the Enter button to open the menu when it is selected. Once the Language menu is open, use the Up and Down buttons to move the cursor and select a language. Press the Enter button to enable that language.

Press the Escape button to return to the Language menu. Press the Escape button a second time to return to the System Menu.





ROUTINE MAINTENANCE INFORMATION

- **CLEANING**

  Good game action and extended playfield life are the results of regular playfield cleaning. During each collection stop, the playfield glass should be removed and thoroughly cleaned using only a non-abrasive sprayed liquid glass cleaner and a soft cloth. The playfield should be wiped off with a clean, lint-free cloth. The game steel balls should be cleaned and inspected for any chips, nicks, or pits. Replace any damaged balls to prevent playfield damage.

  Regular, more extensive, playfield cleaning is recommended. However, avoid excessive use of water and caustic or abrasive cleaners because they tend to damage the playfield surface. Playfield wax (or any carnauba based wax), or polish may be used sparingly, to prevent a buildup on the playfield surface. Do not use cleaners containing petroleum distillates on any playfield plastics because they may dissolve the plastic material or damage the artwork.

- **GLASS PARTS**

  Wear protective gloves and use both hands when carrying or moving parts made of glass. Glass pieces can fly eight feet (two meters) or more from the point of impact.

SERVICE INFORMATION

Only trained personal should service the game. Always turn off power and unplug the game before attempting any service!

- **HIGH VOLTAGE CONSIDERATIONS**

  CRTs and their power supplies can retain energy for long periods of time after the power has been turned off. This is especially true when a defective circuit prevents a normal discharge. Connect a very well insulated ground strap to the metal chassis. Slide the free end of the strap under the CRT anode cap until contact is made. Wait two minutes for charge recovery, then discharge the anode a second time.

- **RADIATION**

  The high voltages used in the Video Monitor Assembly are capable of generating X-rays under fault conditions. Do not substitute high voltage components or modify the circuit without factory authorization. Follow the manufacturer's directions for measuring and adjusting the CRT anode voltage.

- **BATTERY**

  Note the position of the battery. Gently lift the contact arm to release tension. Do not bend the arm. Slide the battery out of it holder. To reinstall the battery, orient the cell near the holder and slide it under the contact arm. Do not force the battery into the holder.

! CAUTION

Danger of explosion if battery is incorrectly replaced. Replace with only the same or equivalent type recommended by manufacturer. Do not attempt to recharge these batteries. Avoid direct shorts across terminals or from terminals to ground. Dispose of used batteries according to manufacturer's instructions.

♦ **MEMORY**

The ROM chips contain the computer operating instructions. Memory devices are very sensitive to static charges. Use grounding precautions when handling these parts.

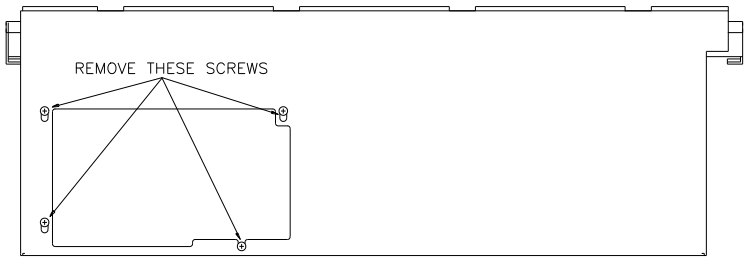
To remove a ROM chip from its socket, carefully note its position and then lift the chip using a chip extraction tool. To reinstall a ROM, place the chip over its socket so that the notch in the chip lines up with the silkscreen on the board, and press down firmly to seat. Do not force the chip into the socket. Do not bend the pins under the chip.

! CAUTION

Use wrist straps, conductive mats or other methods of safely discharging static electricity. Hold all integrated circuits by their packages and not by the metal pins. Extractor and insertion tools are recommended to avoid bent or broken pins caused by excessive force.

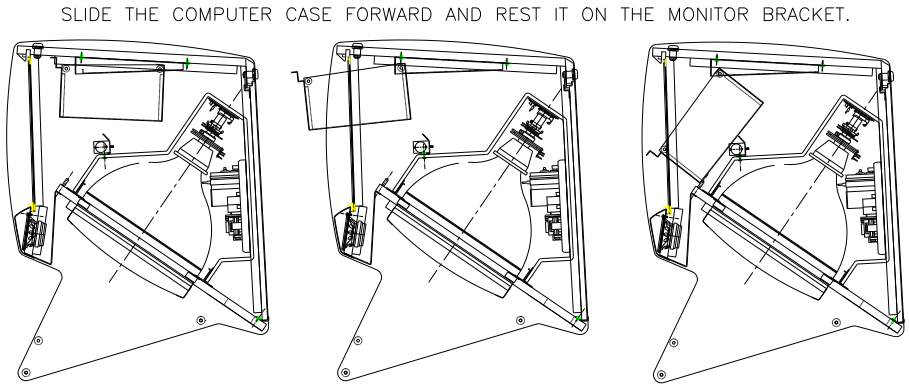
♦ **POWER SUPPLY**

Unlock and remove the backbox rear door. Unplug the IEC cord and remove the four outer screws surrounding the power supply that secure it to the case. See the drawing below.



Rear view of the computer case

Unlock and remove the backglass and the translite. Unscrew the two screws on either side of the power switch. Grasp the computer case from the front and slide it toward you. Let it hang in front of the backbox, see the drawing below.



Side View of Backbox

Open the top of the computer case. Unplug the cable harnesses leading from the power supply; one plug is going to one connector on the power amp and two plugs are going to one connector on the CPU board. Lift the power supply out of the computer case.

NOTICE

Always replace the computer case cover. Failure to do so will void FCC, UL and CE compliance, and may cause damage to the PC boards in the case. The foam lining of the cover holds the PC boards securely in place and protects them from vibrations that normally occur in a pinball game.

- **GLASS PARTS**
  Wear protective gloves and use both hands when carrying or moving parts made of glass. Fluorescent tubes and CRTs will usually implode if broken. Glass pieces can fly eight feet (two meters) or more from the point of impact.

- **FLUORESCENT TUBE, STARTER, OR FLUORESCENT LIGHT ASSEMBLY**
  Remove the backglass and translite. Remove both plastic lamp locks. Grasp the bulb at each end and give it a quarter turn. Gently pull the bulb straight out to remove it from its socket. The starter also requires a quarter turn for removal. Do not force the bulb or starter during reinstallation. Clean the bulb to remove fingerprints and dust, then reinstall the translite and backglass.

  To remove the entire light fixture, remove the lamp locks and the bulb as described above. Disconnect the fluorescent light assembly connector from its power cable. Remove the screws that hold the assembly to the cabinet, then lift out the assembly.

  If you drop a fluorescent tube or a CRT and it breaks, it will implode! Use care in handling.

- **CIRCUIT PROTECTION**
  To avoid equipment damage, any replacement fuse must match the original in fuse type, voltage rating, and current rating. Do not use fuses with different time delay characteristics.

- **MONITOR**
  Unlock and remove the backbox rear door. Disconnect all of the cables going to the monitor. Remove the four flange nuts securing the monitor's mounting brackets to its mounting panel. CAREFULLY, Pull the monitor from the backbox. Clean the face of the monitor before reinstalling it.

  ! CAUTION
  THE VIDEO MONITOR IS HEAVY, WITH MOST OF THE WEIGHT TOWARD THE FRONT OF THE ASSEMBLY. Be sure it is firmly supported as it is removed from the cabinet. The monitor does require isolation from the A.C. line voltage in normal operation. When operating outside the backbox or servicing the monitor on a test bench, YOU MUST ISOLATE THE MONITOR FROM LINE VOLTAGE WITH A TRANSFORMER. THE MONITOR MUST BE POWERED AT 110V.

- **LUBRICATION**
  The two main lubrication points of the Ball Release mechanism are the pivots for the arm. The mechanisms of other playfield devices are somewhat similar to the Ball Release, and have the same lubrication requirements. A medium viscosity oil (switch target grease) is satisfactory for these devices.

  Because of the functional design (arm-actuated via solenoid plunger operation), the pivot points of the Left and Right Kickers ("Slingshots") all require lubrication as a regular servicing procedure. Lubrication to ensure proper operation also applies to the target blades of the Drop Targets. MBI Instrument Grease, also known as Drop Target Switch Lubricant, with a Williams' part number of EI165, is a recommended lubricant.

- **SWITCH CONTACTS**
  Playfield Switches. For proper game operation, switch contacts should be free of dust, dirt, contamination, and corrosion. Blade switch contacts are plated to resist corrosion. Cleaning blade switch contacts requires gentle closing of the contacts on a clean business card or piece of paper, and then pulling the paper about 2 inches, which should restore the clean contact surface. Adjust the switch contacts to a 1/16-inch gap.

Switch Contacts Continued…

Flipper Switches. This game uses the Fliptronic II Electronic Flipper System. The End-of-Stroke switches are NORMALLY OPEN. The switch should close when the flipper is energized. All E.O.S. switches are gold flashed computer grade leaf switches. Only low computer current is carried through these switches. DO NOT FILE or abrasively clean these switches! DO NOT REPLACE these switches with the tungsten high current switches, as intermittent operation could occur.

NOTE: *Unlike the old style of flipper, an E.O.S. switch failure does not harm the flipper. The game notifies the operator that the switch is misadjusted in the test report, but continues to play. The E.O.S. switches are a means by which the new electronic flippers feel and play with all of the subtleties of the old flippers.*

EPILEPSY WARNING

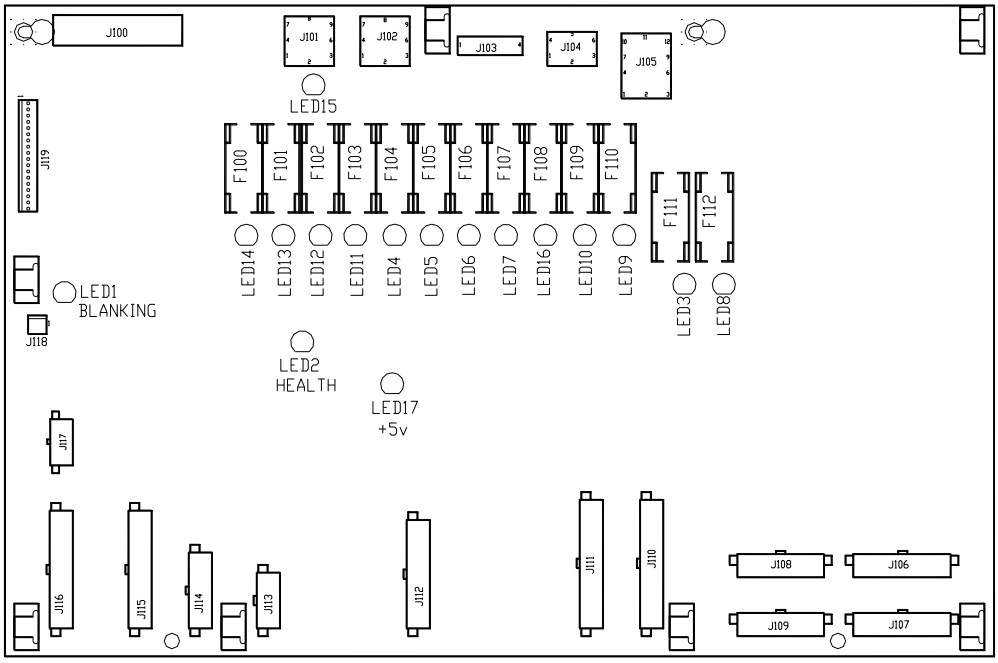
A very small portion of the population has a condition which may cause them to experience epileptic seizures or have momentary loss of consciousness when viewing certain kinds of flashing lights that are present in our daily environment. These persons may experience seizures while watching some kinds of television pictures or playing certain video games. People who have not had any previous seizures may nonetheless have an undetected epileptic condition.

If you or anyone in your family has experienced symptoms linked to an epileptic condition (e.g., seizures or loss of awareness), immediately consult your physician before using any video games.

We recommend that parents observe their children while they play video games. If you or your child experience the following symptoms: dizziness, altered vision, eye or muscle twitching, involuntary movements, loss of awareness, disorientation, or convulsions, DISCONTINUE USE IMMEDIATELY and consult your physician.

LED AND FUSE LIST

POWER DRIVER BOARD



LED LIST

LED 1	Watchdog	**LED 10**	+20V Flashlamps
LED 2	Health	**LED 11**	Solenoid Power 1
LED 3	+18V Lamp Matrix A	**LED 12**	Solenoid Power 2
LED 4	+50V Lower Right Flipper	**LED 13**	Solenoid Power 3
LED 5	+50V Lower Left Flipper	**LED 14**	Solenoid Power 4
LED 6	+50V Upper Right Flipper	**LED 15**	+20V
LED 7	+50V Upper Left Flipper	**LED 16**	+12V
LED 8	+18V Lamp Matrix B	**LED 17**	+5V
LED 9	+50V Solenoids		

FUSE LIST

Loc.	Value		Part Number	Description
F100	T4.0A	250V	5371-14530-00	Solenoid Power 4
F101	T4.0A	250V	5371-14530-00	Solenoid Power 1
F102	T4.0A	250V	5371-14530-00	Solenoid Power 2
F103	T4.0A	250V	5371-14530-00	Solenoid Power 3
F104	T4.0A	250V	5371-14530-00	Lower Right Flipper Power
F105	T4.0A	250V	5371-14530-00	Lower Left Flipper Power
F106	T4.0A	250V	5371-14530-00	Upper Right Flipper Power
F107	T4.0A	250V	5371-14530-00	Upper Left Flipper Power
F108	T4.0A	250V	5371-14530-00	12VAC Unregulated
F109	T4.0A	250V	5371-14530-00	20VAC Flashlamps
F110	T6.3A	250V	5731-14529-00	50VAC Solenoids
F111	T5.0A	250V	5731-14046-00	18VAC Lamp Matrix A
F112	T5.0A	250V	5731-14046-00	18VAC Lamp Matrix B

LINE FUSE

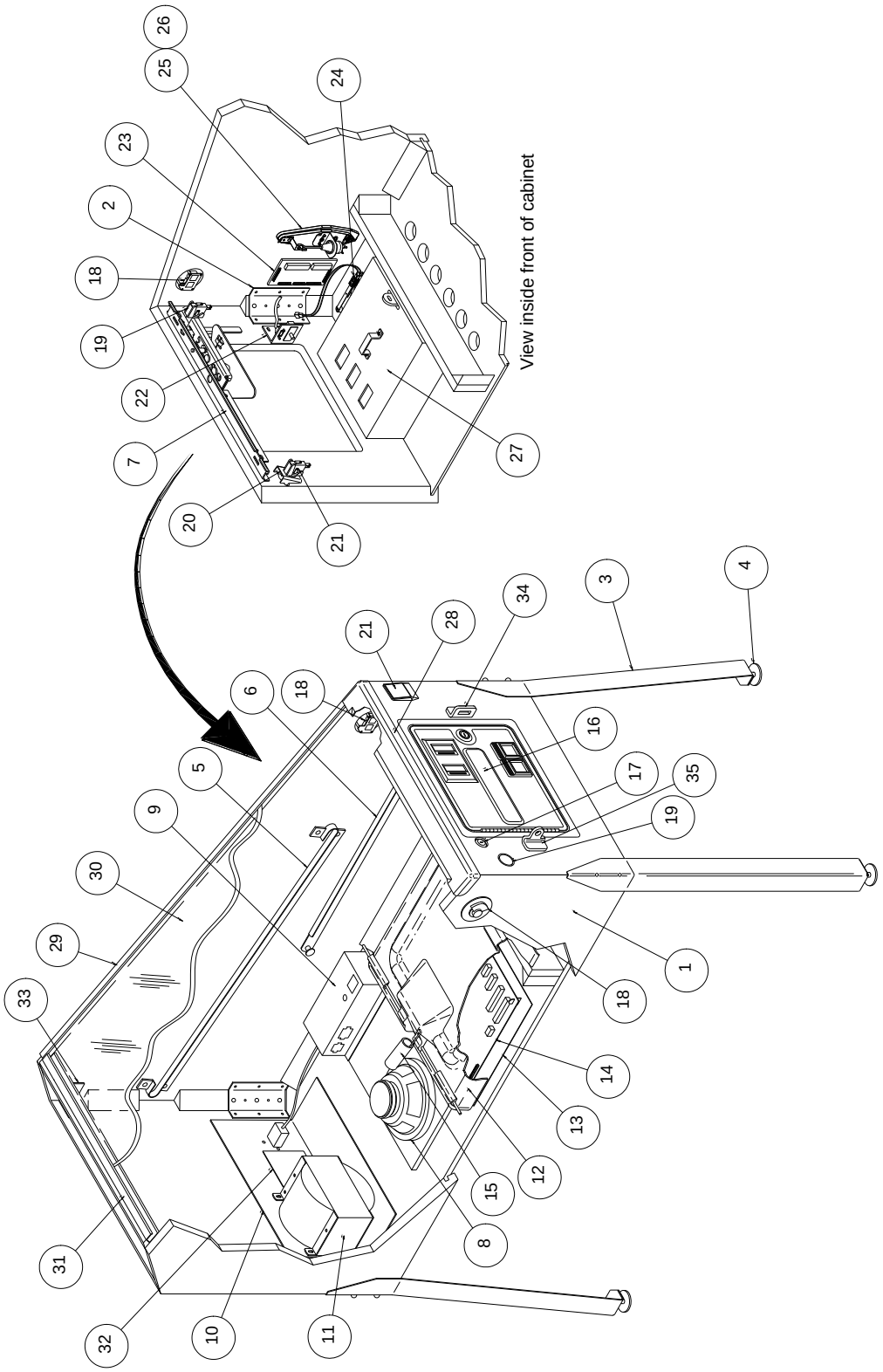
Location	Value		Part Number
Foreign	T4.0A	250V	5731-14530-00
Domestic	T5.0A	250V	5731-14046-00

SECTION TWO

PARTS INFORMATION

Items listed as Associated Parts are not sold as part of the main assembly. Associated Parts are only sold separate.

50070-CAB
Cabinet Assembly



50070-CAB
Cabinet Assembly Parts List

Item	Part Number	Description
1	**04-12480.5**	Cabinet
2	**01-11400-1**	Leg Mounting Bracket (x4)
3	**A-19514-6**	Leg Assembly – Black Ridge (x4)
4	**08-7377**	Leg Leveler/Adjuster (x4)
5	**01-15132-1**	Playfield Slide Track – Right
6	**04-12610**	Stay Arm
7	**A-22996**	Lockdown Bracket Assembly
8	**5555-16059-00**	Speaker, 6.5" Woofer, 4Ω
9	**A-22966**	Power Control Box Assembly
10	**04-12596.1**	Drip Plate
11	**5610-15930-00**	Transformer Assembly
12	**A-22998**	Driver Board Cover Assembly
13	**A-23299**	Driver Board Back Plate Assembly
14	**04-12329**	Power Driver Board PCB Assembly
15	**03-10011**	Cover Stop Tube
16	**09-96032-33**	Coin Door – USA
17	**20-10690**	Hand Molding Lock
18	**A-22984**	Flipper Button Assembly (x2)
19	**20-10696**	Push Button w/Switch – *START* (Yellow)
20	**01-15184**	Launch Button Bracket
21	**20-10692-1**	Push Button w/Switch – *LAUNCH* (Red)
22	**A-23024**	Coin Door Interlock Switch Assembly
23	**A-22964**	Coin door Interface PCB Assembly & Spacer
24	**A-17195-2**	Tilt Switch Assembly w/Cable
25	**04-10346.1**	Tilt Mechanism Assembly
26	**20-6502-A**	Plumb Bob
27	**A-23139**	4-Ball Cash Box Assembly
28	**A-22997**	Hand Molding Assembly
29	**A-22976-1**	Side Molding Assembly (x2)
30	**04-12739.1**	Tinted Playfield Glass Assembly
31	**03-8091**	Rear Molding
32	**01-15214**	Hole Cover
33	**01-15212**	Cabinet Corner Bracket
34	**01-14085.1**	Hasp Hinge Bracket
35	**01-14086.1**	Hasp Staple Bracket

Miscellaneous Parts: (Not Shown)

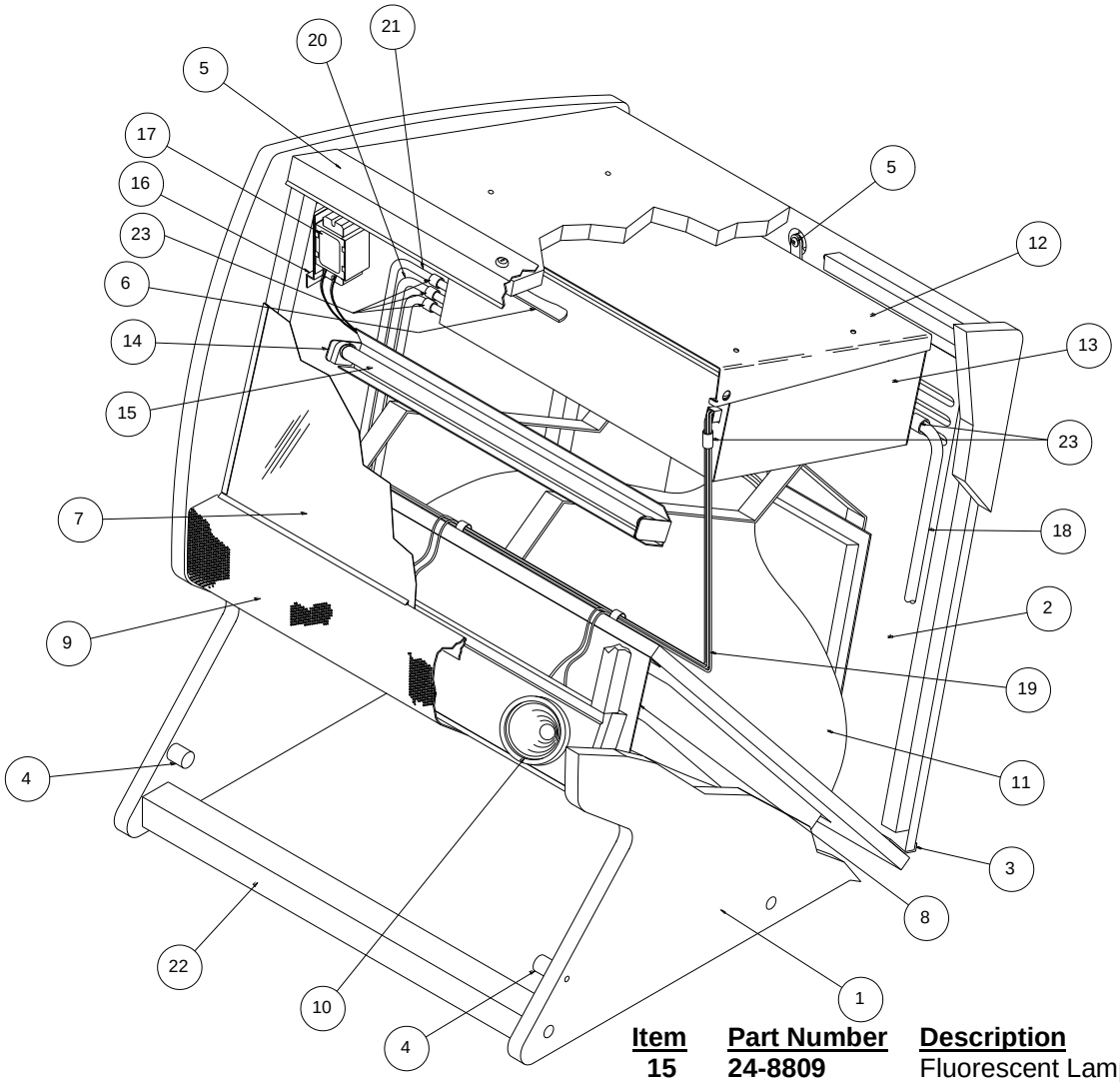
Part Number	Description
01-15132-2	Playfield Slide Track – Left
04-12628.2	Safety Pin Bracket
01-15136-1-2	Lock Plate
*****	Cordset
20-6500	Steel Ball, ∅1-1/16" (x4)
01-6389-1	Cash Box lock Bracket
01-12352	Clip Bracket
4322-1125-40	3/8 x 16 Hex Acorn Head Leg Bolt
01-14084.2	Slotted Hasp Plate

Cabinet Cables:

Part Number	Description
H-22740-2.1	Cabinet Cable
H-22740-1.2	Cabinet Cable
H-22736	Secondary power Cable
H-22737-2	Cabinet Power/Speaker Cable
H-22955	Cabinet Switch/Lamp Cable

***** *See Power Interface/Cordset Application Chart on page 2-31*

50070-BB
Back Box Assembly



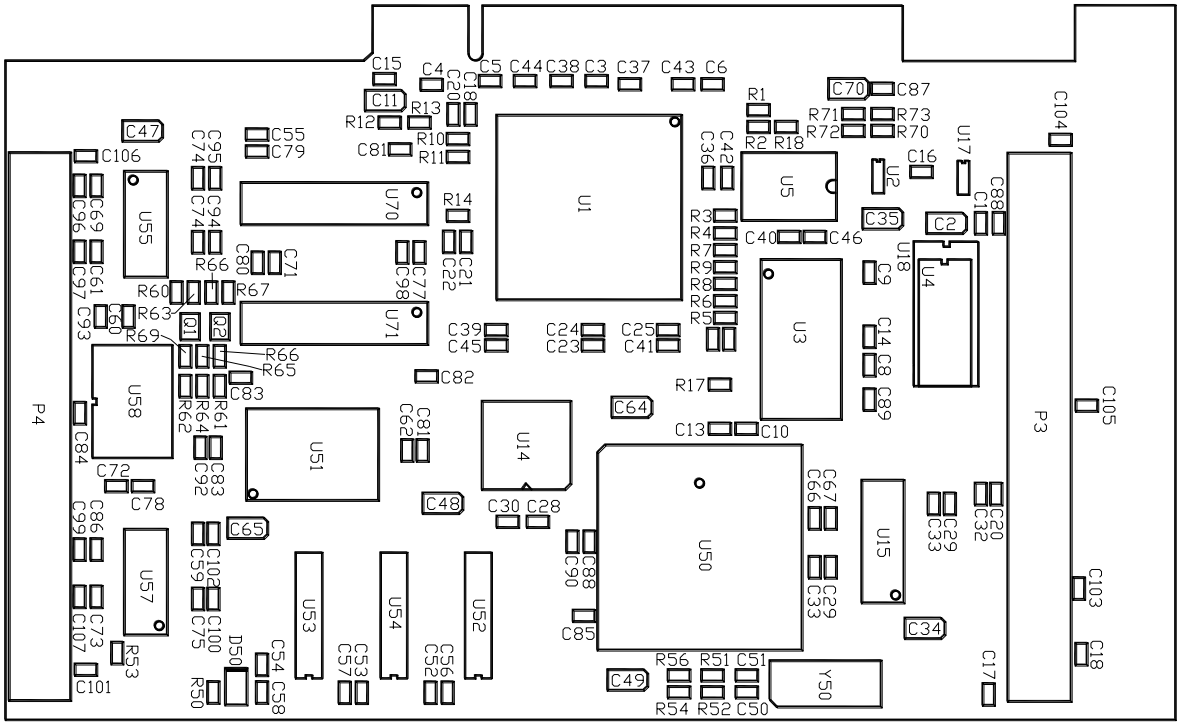
Item	Part Number	Description
1	04-12489.4	Back Box
2	A-22982	Back Boor Assembly
3	01-15131	Door Channel
4	03-9992	Knob
5	A-22981	Lock & Plate Assembly
6	01-15134	Front Lock Cam
7	A-22975-50070	Marquee Assembly
8	04-12607.1	Speaker Grill Bracket
9	04-12606.1	Speaker Grill
10	5555-16223-00	Speaker - 4Ω, 25W
11	5675-16058-02	Monitor – 19”
12	01-15130.2	Computer Case Mounting Bracket
13	A-22724	Computer Case Assembly
14	A-22980	Fluorescent Lamp Assembly
15	24-8809	Fluorescent Lamp 15W, 18”
16	04-12752	Ballast Mounting Bracket
17	A-23155	Ballast Assembly
18	A-23160	CPU/AC Power Cable
19	H-22737-1.1	AC/Speaker Cable
20	20-10583	Extension Cable db09 – 10ft
21	20-10584	Extension Cable db25 – 10ft

Miscellaneous Parts:

Item	Part Number	Description
22	11-1464	Back Box Shipping Brace
23	5556-15272-00	Ferrite
*24	02-4352-2	Pivot Bushing
*25	4322-10228-12	CS 3/8-16 x ¾ Button Head – Nylon Patch
*26	4700-00111-00	FW .390 x .875 x .062

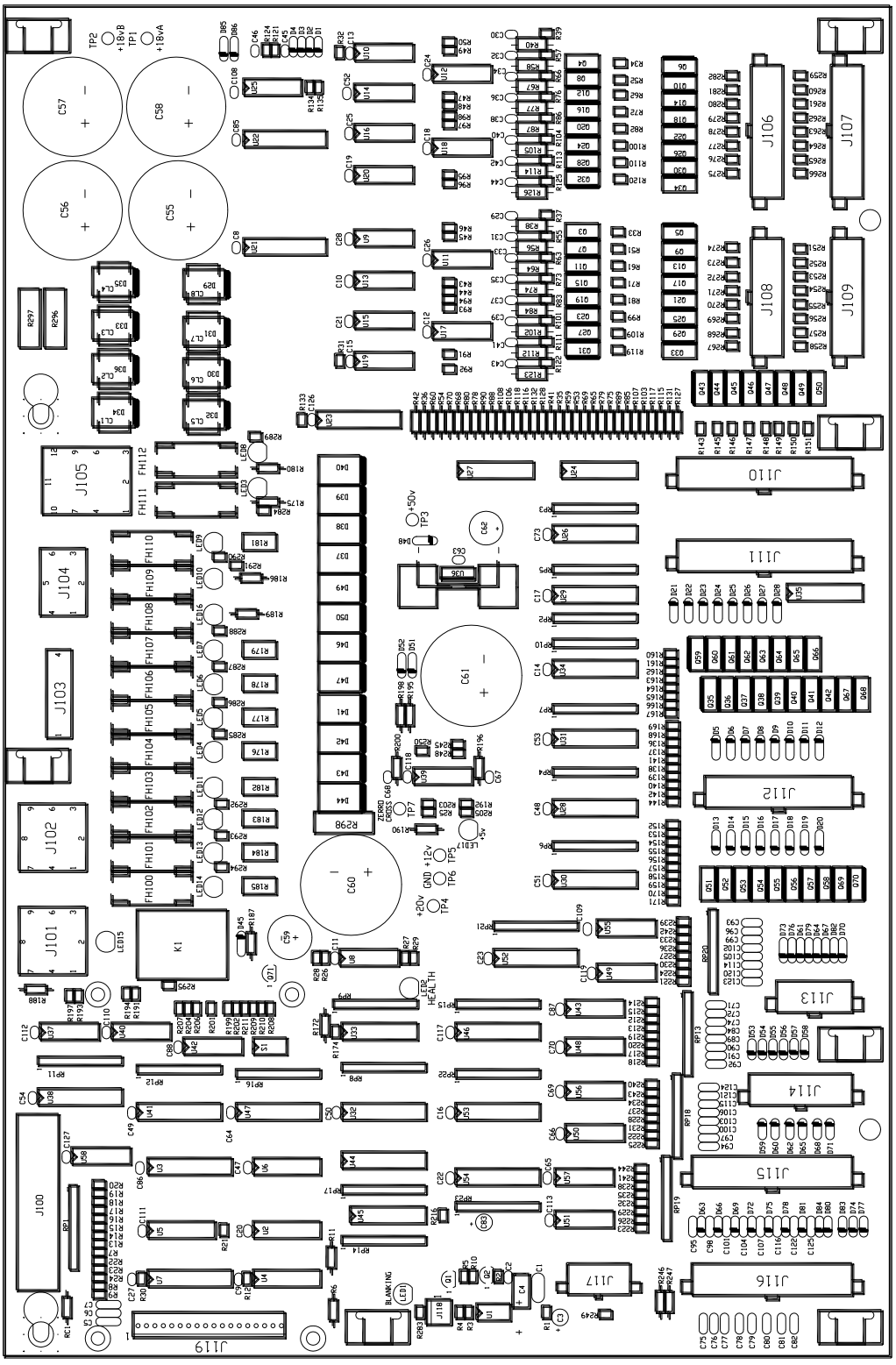
***** *Item not shown*

04-12602
Prism PCB Assembly



Designator	Part Number	Description
R1-R14, R17, R18, R52-R54, R56, R65, R66, R69-R73	**5020-14156-00**	Resistor, 10KΩ, 5%, 1/10W, R0805
R50	**5020-14160-00**	Resistor, 330Ω, 5%, 1/10W, R0805
R62, R63, R64, R67, R68	**5020-14152-00**	Resistor, 100Ω, 5%, 1/10W, R0805
C2, C11, C34, C35, C47, C48 C49, C64, C65, C50, C51	**5051-14151-00**	Capacitor, 10μf, 16V ±20%, SMD TANT
C1, C7-C10, C17, C19, C21 C24-C29, C36-C40, C52-C55, C67, C90-C100	**5052-14149-00**	Capacitor, 22ρf, 100V 10%, SMD
C3-C6, C12-C16, C20, C22, C23, C41-C46, C101-C107	**5052-14164-00**	Capacitor, .01μf, 50V 20%, SMD
C18, C30-C33, C56-C63, C66, C68, C69, C78-C89	**5052-14165-00**	Capacitor, .1μf, 50V 20%, SMD Z5U
Q1, Q2	**5052-14368-00**	Capacitor, 22ρf, 100V 10%, SMD
U15	**5160-16056-00**	Transistor, NPN 2N3904 SC70
U55, U57	**5283-16051-00**	IC, Bus Transceiver, 74FCT162543T
U4	**5285-14601-00**	IC, 74FCT162245
U2	**5340-16204-00**	IC, SRAM, 32K x 8, 70ns, SOP28
U58	**5345-16049-00**	IC, EE Serial PROM
U3	**5345-16053-00**	IC, Flash 8Mbit TSOP48
U5	**5345-16196-00**	IC, Flash EEPROM, 28F320J5 SSOP
U17	**5521-16050-00**	OSC, 30MHz SMT
U50	**5432-16052-00**	IC, NVRAM Controller SOIC8
U51	**5400-15450-00**	ADSP-2104KP-40
U52, U53, U54	**5410-14590-00**	IC, SDRC ASIC Video
U1	**5349-14606-00**	IC, SRAM, 32K x 8 – 15, 28PSOJ
U14	**5430-16048-00**	IC, PCI Bus Interface, PCI9502
Y50	**A-23290**	IC, CPLD U14 Assembly
D50	**5520-16054-00**	OSC, 16MHz Crystal SMT
P3, P4	**5671-14653-00**	LED, Yellow SMD
	5792-16057-00	Connector, Dual Row 2 x 40 SMT

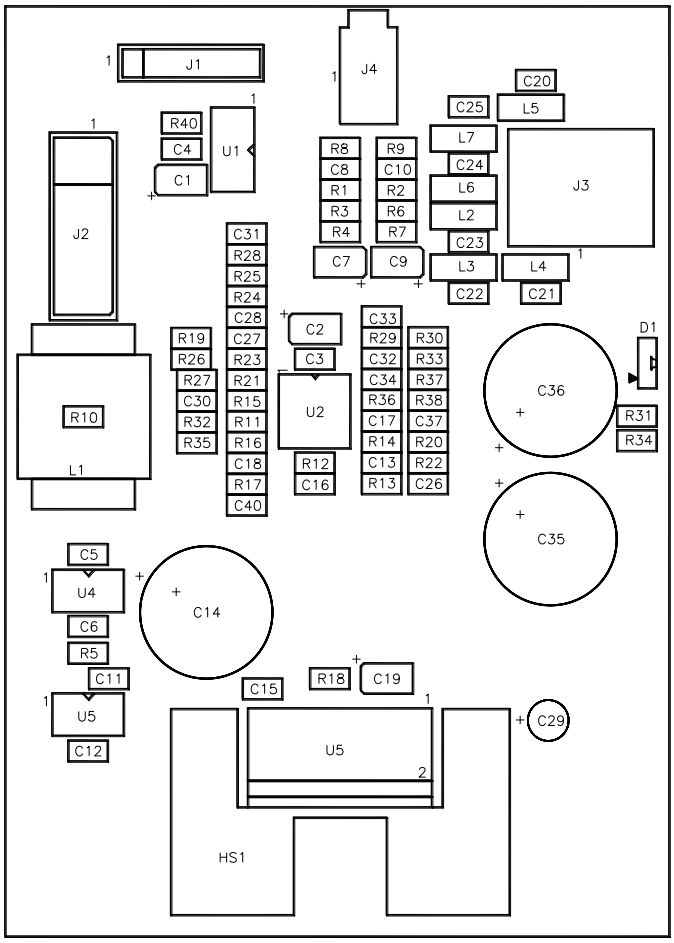
04-12329
Power Driver Board



04-12329
Power Driver Board Parts List

Part Number	Description	Designator
5043-08996-00	Capacitor, 0.1µf, 50V ±20% Axial	C1, C2, C8-C54, C63-C70, C73, C85-C88, C108-C113, C117-C119, C126, C127
5040-13098-00	Capacitor, 4.7µf, 35V ±20%	C3, C83
5040-09365-00	Capacitor, 1µf, 63V +50%-10% Axial	C4
5048-11029-00	Capacitor, 100ρf 50V ±5% Axial	C5-C7
5040-15413-00	Capacitor, 10000µf, 35V, 25mm	C55-C58, C60, C61
5040-09537-00	Capacitor, 100µf, 100V ±20% Radial	C59
5040-13102-00	Capacitor, 470µf, 35V ±20% Radial	C62
5048-11030-00	Capacitor, 470ρf, 50V Axial	C71, C72, C74-C82, C84, C89, C90-C107, C114-C116, C120-C125
5070-08919-00	Diode, 1N4148, 150mA	D1-D4, D53-D84
5070-09054-00	Diode, 1N4004, 1.0A	D5-D28, D45, D48, D51, D52
5070-14526-00	Diode, P600G, 6A, 400 PIV	D29-D44, D46, D47, D49, D50
5733-14528-00	Fuse Holder, 5x20mm, 10A	FH100-FH112
5731-14530-00	Fuse, 5x20mm, T4A, 250V	F100, F101, F102, F103, F104, F105, F106, F107, F108, F109
5731-14529-00	Fuse, 5X20mm, T6.3A, 250V	F110
5731-14046-00	Fuse, 5x20mm, SB T5A	F111, F112
5705-14562-00	Heatsink, 10-220 Wave Sol 287	HS1
5791-16065-00	25 Pin D-SUB Connector, R/A PCB Mount	J100
5792-13223-09	9 Pin Female, .084 MLX	J102, J101
5792-13223-04	4 Pin Female, .084 MLX	J103
5792-13223-06	6 Pin Female, .084 MLX	J104
5792-13223-12	12 Pin Female, .084 MLX	J105
5791-13356-18	Header-PCB Mount, 0.165 center 18P	J106, J107
5791-13356-16	Header-PCB Mount, 0.165 center 16P	J108, J109
5791-13356-24	Header-PCB Mount, 0.165 center 24p	J110, J111
5791-13356-20	Header-PCB Mount, 0.165 center 20p	J112
5791-13356-10	Header-PCB Mount, 0.165 center 10P	J113
5791-13356-14	Header-PCB Mount, 0.165 center 14P	J114
5791-13356-22	Header-PCB Mount, 0.165 center 22P	J115, J116
5791-13356-08	Header-PCB Mount, 0.165 center 8P(N)	J117
5791-13830-03	3H Straight Square Pin, .100 Solid Tab	J118
5792-15928-18	Connector, 18 Pin, 22-02-7183 Top EN	J119
5580-12782-00	Relay DPDT 12VDC	K1
5671-13732-00	LED, Display Red, T1¾	LED1-LED15, LED17
5190-10270-00	Transistor, 2N3906 PNP	Q1, Q2
5162-12635-00	Transistor, Tip 102	Q3, Q4, Q7, Q8, Q11, Q12, Q15, Q16, Q19, Q20, Q23, Q24, Q27, Q28, Q31, Q32
5192-12428-00	Transistor, Tip 107	Q5, Q6, Q9, Q10, Q13, Q14, Q17, Q18, Q21, Q22, Q25, Q26, Q29, Q30, Q33, Q34
5220-15710-00	Transistor-FET, STP20N10L, N-Channel	Q35-Q70
5160-10269-00	Transistor, 2N3904, NPN	Q71
5019-09669-00	SIP Resistor, 1KΩ, 9R 10, 5%	RP14
5019-16328-00	SIP Resistor, 2.2KΩ, 9R 10P, 5%	RP19
5019-10661-00	SIP Resistor, 10KΩ, 9R 10, 5%	RP1-RP13, RP15-RP18, RP20-RP23
5010-13517-00	Resistor, 15Ω, 1/4W,	R247
5010-13594-00	Resistor, 1kΩ, 1/8W, 5% Film	R1, R22, R23, R31-R34, R51, R52, R61, R62, R71, R72, R81, R82, R99, R100, R109, R110, R119, R120, R191, R193, R194, R197, R201, R204, R205-R215, R217-R244
5010-13365-00	Resistor, 4.7kΩ, 1/8W, 5%	R2
5010-13366-00	Resistor, 470Ω, 1/8W, 5%	R3, R4, R7, R8, R9, R136-R171
5010-13368-00	Resistor, 3.9kΩ, 1/8W, 5%	R5
5010-09160-00	Resistor, 220Ω, 1/4W, 5%	R6, R11, R172, R190, R195, R196, R198, R200
5010-14395-00	Resistor, 6.8kΩ, 1/8W, 5%	R10, R284, R289, R291
5010-13593-00	Resistor, 10kΩ, 1/8W, 5% Film	R12, R21, R25-R28, R29, R43-R50, R91-R98,R121, R124, R134, R135, R192, R199, R245, R248-R282
5010-13364-00	Resistor, 33Ω, 1/8W, 5%	R13-R20, R24, R30, R133, R174, R216
5010-09358-00	Resistor, 1kΩ, 1/4W, 5%	R35, R36, R53, R54, R65, R68, R75, R78, R85, R88, R103, R106, R115, R116, R127, R128
5010-13372-00	Resistor, 220Ω, 1/8W, 5%	R37, R39, R55, R57, R63, R66, R73, R76, R83, R86, R101, R104, R111, R113, R122, R125, R283
5010-12427-00	Resistor,.22Ω, 1W, 5%	R38, R40, R56, R58, R64, R67, R74, R77, R84, R87, R102, R105, R112, R114, R123, R126
5010-08992-00	Resistor, 560Ω, 1/4W, 5%	R41, R42, R59, R60, R69, R70, R79, R80, R89, R90, R107, R108, R117, R118, R131, R132
5010-14093-00	Resistor, 0Ω, 1/8W, 5%	R246
5010-09999-00	Resistor, 2KΩ, 1/4W, 5%	R175, R180, R186-R189
5010-14711-00	Resistor, 10kΩ, 1W, 5%	R176-R179, R181-R185
5010-13595-00	Resistor, 2.7kΩ, 1/8W, 5% Film	R202
5010-13843-00	Resistor, 22kΩ, 1/8W, 5% Film	R203
5010-13841-00	Resistor, 47kΩ, 1/8W, 5%	R285-R290, R292-R295
5011-13292-00	Resistor, 330Ω, 2W, 5%	R296-R298
5645-12554-00	Switch DIP, 4 Position	S1
5824-09248-00	Test Point #1502-1	TP5, TP6
5431-10449-00	IC, 555 Timer	U1
5315-12812-00	IC, 74HCT138, 3 to 8 Decoder	U2, U4, U5, U6
5315-12813-00	IC, 74HCT245, Octal Transceiver	U3
5315-13079-00	IC, 74HCT574, Octal D-Latch	U7, U26, U28-U32, U34, U44
5281-09487-00	IC, 74LS74, Dual D F/F	U8-U10, U13-U16, U19, U20
5370-12272-00	IC, LM339, Quad Comp	U11, U12, U17, U18, U25, U37, U39, U40, U42, U43, U48, U49, U50, U51, U55, U56, U57
5315-12821-00	IC, 74HCT240, Octal Buffer	U21-U23, U38, U41, U46, U47, U52-U54
5162-12422-00	Trans, ULN 2803 Oc-Drl	U24, U27, U33, U35, U45
5250-09157-00	Reg, 7805, 1.0A, 5V	U36
5311-12538-00	IC, 74HC14, Hex S-T-	U58
5671-12993-00	LED, Green, 5mm, T1¾ Diffused	LED16
5070-09266-00	Diode, 1N5817, 1.0A, Schottky	D85, D86
DNP	Do Not Populate	TP1-TP4, TP7, RC1, CL1-CL8
MRD	Missing Reference Designators	R173, RP24-RP27, C126, R129, R130

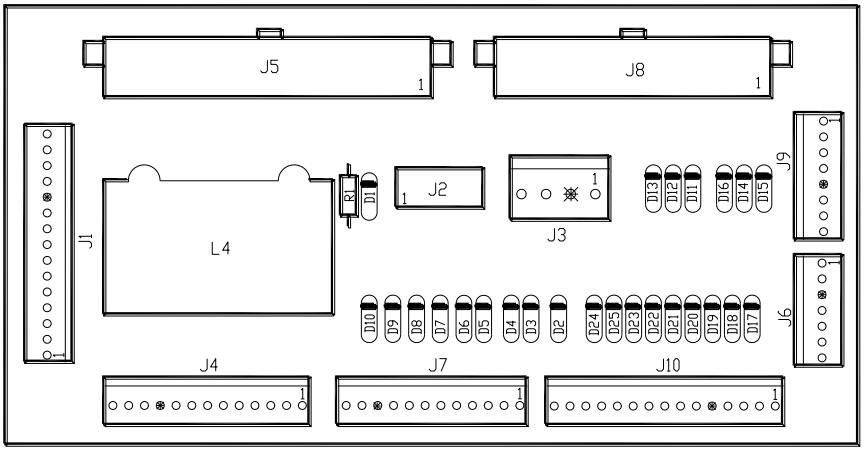
04-12621
P2000 Audio Amplifier PCB Assembly



Designator	Part Number	Description
R18	**5020-14156-00**	Resistor, 10KΩ, 1/10W 5%R0805
R19, R29	**5020-16081-00**	Resistor, 18KΩ, 1/10W 5% 0805
R24, R36	**5020-16082-00**	Resistor, 68KΩ, 1/10W 5% 0805
R14	**5020-16080-00**	Resistor, 82KΩ, 1/10W 5% 0805
R1, R2, R4 R7	**5020-16079-00**	Resistor, 56KΩ, 1/10W 5% 0805
R13	**5020-14383-00**	Resistor, 9.1KΩ, 1/10W 1% SMD-8085
R10, R12, R20 R21, R26, R30 R32	**5020-14349-00**	Resistor, 0KΩ, 1/10W SMD-0805
R11, R15, R16 R17	**5020-16222-00**	Resistor, 100KΩ, 1/10W 5% SMD-0805
R3, R6	**5020-15029-00**	Resistor, 2.4KΩ, 1/10W 5% SMD-0805
R25, R28, R31 R37, R38	**5020-14157-00**	Resistor, 1KΩ, 1/10W 5% SMD-0805
C1, C2, C7, C9	**5051-14151-00**	Capacitor, 10µf, 16V ±20%, SMD
C19		TANT
C3, C4, C15, C31 C37	**5052-14165-00**	Capacitor, .1µf, 50V ±20%, SMD Z5U
C10, C8	**5052-14166-00**	Capacitor, 1000ρf, 50V ±5%, SMD
C17	**5052-14170-00**	Capacitor, 6800ρf, 50V ±5%, SMD
C20, C21, C22, C23, C24, C25	**5052-14367-00**	Capacitor, 150ρf, 50V ±5%

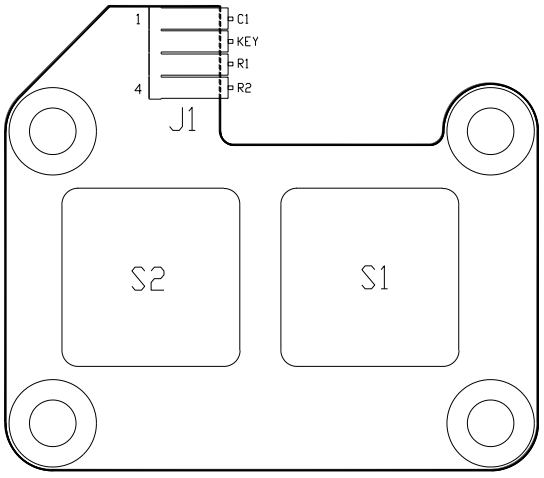
Designator	Part Number	Description
C5, C6, C11, C12 C30, C32, C40	**5052-15041-00**	Capacitor, .22µf, 25V Y5V, 0805
C13, C27, C28, C33, C34	**5052-16078-00**	Capacitor, .068µf, 25V ±10%, 7R 0805
C14, C35, C36	**5040-14630-00**	Capacitor, 2200µf, 25V Radial
C29	**5040-11036-00**	Capacitor, 47µf, 16V Radial ±20%, 105C
C26	**5052-16221-00**	Capacitor, 68µf, 16V Y5V +80% -20%, 0805
D1	**5671-16083-00**	LED, Red Side View, SMT
U4	**5250-16069-00**	IC, Regulator 78L05 SOIC8
U5	**5250-16070-00**	IC, Regulator 78L08 SOIC8
U2	**5370-16072-00**	IC, LM324A Quad OpAmp SOIC14
U3	**5370-16073-00**	IC, Audio Amp TDA7375
U1	**5371-16074-00**	IC, DAC Stereo, 16Bit CS4333, SOP8
L2 - L7	**5553-16075-00**	Ferrite BD, 60Ω, 6A, 1806 SMD
J1	**5791-16076-08**	Connector, JST PH, Vertical 8-Pin
J3	**5791-16071-06**	Header, PCB Mount, R/A MiniFit 6-Pin
J2	**5796-14955-00**	Connector, 4 CKT, Vertical Header 10A/250V
HS1	**5705-16067-00**	Heat Sink, Multiwatt 15, 2 Inch

04-12492
Coin Door Interface PCB Assembly



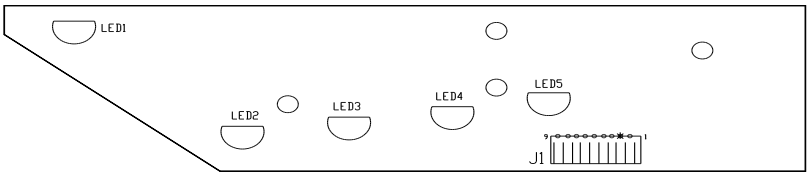
Designator	Part Number	Description
D1-D25	**5070-09054-00**	Diode, 1N4004, 1.0A
J5	**5791-13356-24**	Header, PCB Mount, 0.165 Center, 24 Pin
J8	**5791-13356-20**	Header, PCB Mount, 0.165 Center, 20 Pin
J9	**5791-13830-08**	8H, Straight Square Pin, .100 Solid Tab
J6	**5791-13830-07**	7H, Straight Square Pin, .100 Solid Tab
J1, J10	**5791-13830-15**	15H, Straight Square Pin, .100 Solid Tab
J7	**5791-13830-12**	12H, Straight Square Pin, .100 Solid Tab
J4	**5791-13830-13**	13H, Straight Square Pin, .100 Solid Tab
J3	**5791-13830-04**	4H, Straight Square Pin, .156
J2	**5791-11000-10**	10 Square Pin Straight Header, .100 w/Shroud
R1	**5010-13517-00**	Resistor, 15Ω, ¼W

04-12612
2 Switch Flipper PCB Assembly



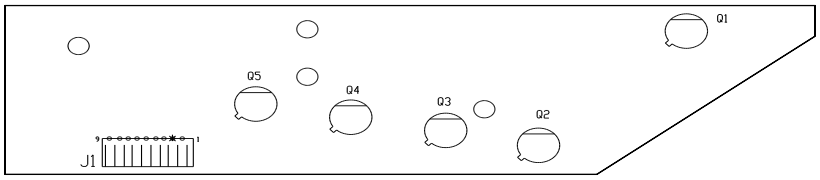
Designator	Part Number	Description
J1	**5791-12622-04**	R/A Square Pin Header, .100 Solid Tab
S1, S2	**5641-16194-00**	Keyswitch w/Jumper

A-18617-1
Trough IR LED PCB Assembly



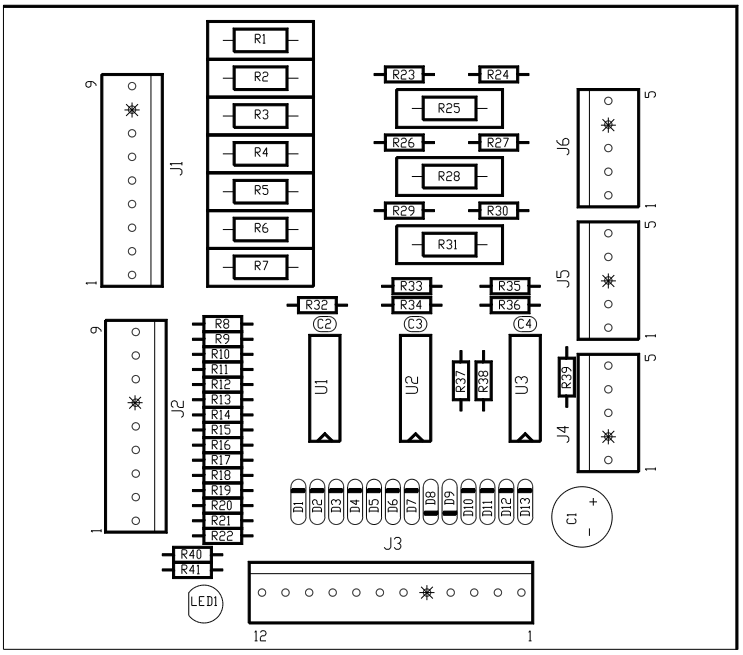
Designator	Part Number	Description
LED1 – LED5	**5671-12731-00**	Infra Red Diode
J1	**5791-12622-09**	Connector, 9-Pin Header Sq.

A-18618-1
Trough IR Photo Transistor PCB Assembly



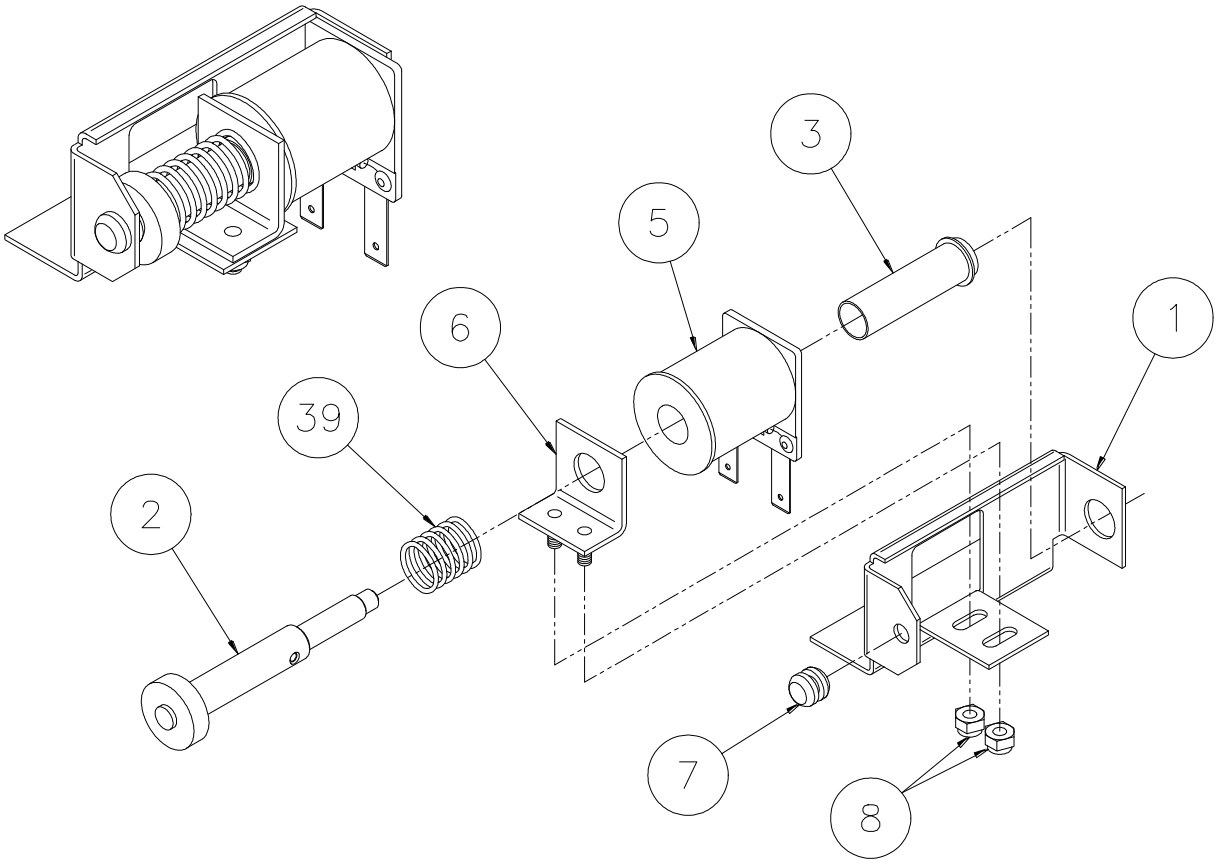
Designator	Part Number	Description
Q1 – Q5	**5163-14114-00**	Infra Red Photo Transistor
J1	**5791-12622-09**	Connector, 9-Pin Header Sq.

A-20246
10-Opto PCB Assembly w/Bracket



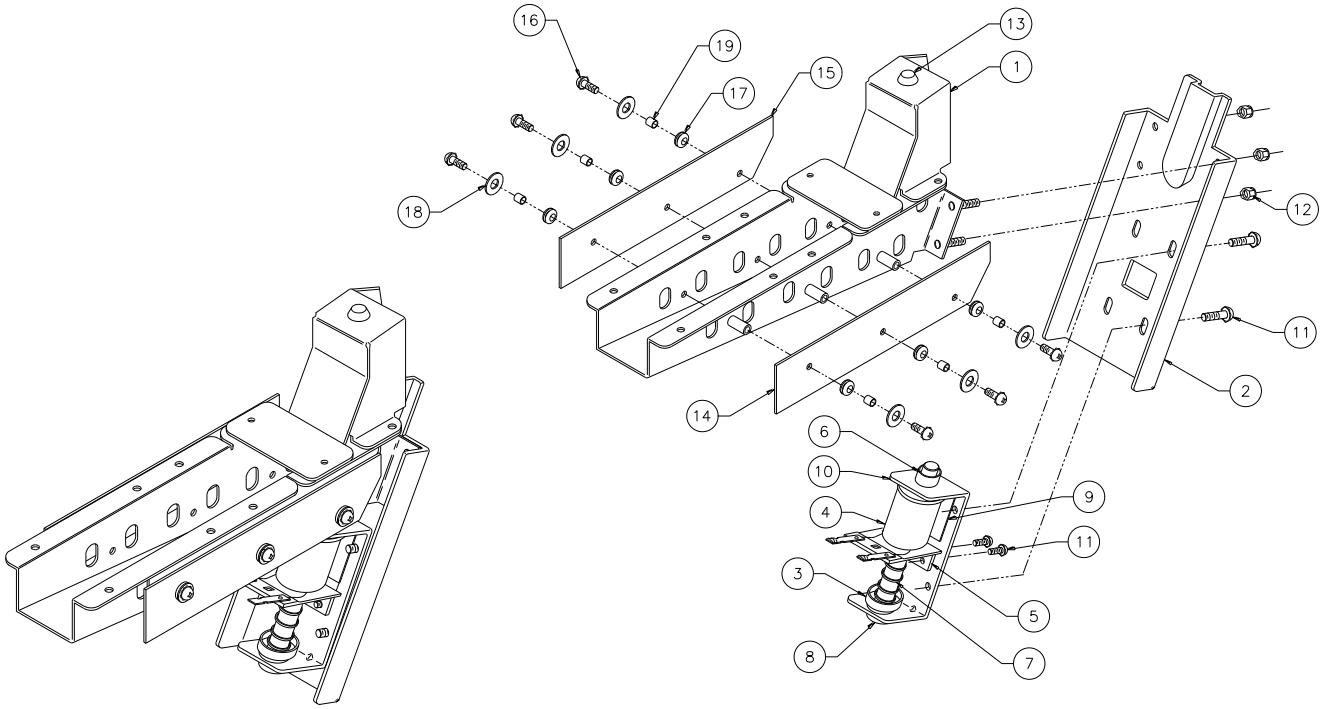
Designator	Part Number	Description
-	**A-18159.1**	10-Opto PCB Assembly
C1	**5040-10974-00**	Capacitor, 100M, 35V radial
C2–C4	**5043-08996-00**	Capacitor, 0.1M, 50V ±20% Ax.
D1-D13	**5070-09054-00**	Diode 1N4004, 1.0A
J1, J2	**5791-10862-09**	Connector, 9-Pin Header
J3	**5791-10862-12**	Connector, 12-Pin Header
J4- J6	**5791-10862-05**	Connector, 5-Pin Header
LED1	**5671-13732-00**	LED, Display Red T-1 ¾
R1-R7, R28, R31, R50	**5010-12928-00**	Resistor, 270Ω, 2W, 5%
R8-R24, R26, R27, R29, R30	**5010-09999-00**	Resistor, 2KΩ, ¼W, 5%
R32, R35, R39-R41	**5010-09162-00**	Resistor, 100KΩ, ¼W, 5%
R34, R36-R38,	**5010-08774-00**	Resistor, 22KΩ, ¼W, 5%
R33	**5010-08776-00**	Resistor, 68KΩ, ¼W, 5%
U1-U3	**5370-12272-00**	IC, LM339 Quad Comp
-	**01-10756**	PCB Mounting Bracket
-	**07-6688-18N**	Rivet, 1/8” x 3/16”

A-22429-4
Auto-Fire Assembly



Item	Part Number	Description
1	01-14618.1	Auto-Fire Bracket
2	A-6306-2	Plunger
3	03-7067	Coil Tubing
4	04-10910-1	Coil Centering Bracket
5	AE1-23-800	Coil Assembly
6	10-135	Spring
7	23-6420	Rubber Grommet
8	4408-01119-01	Nut 8-32 ESN/NTM

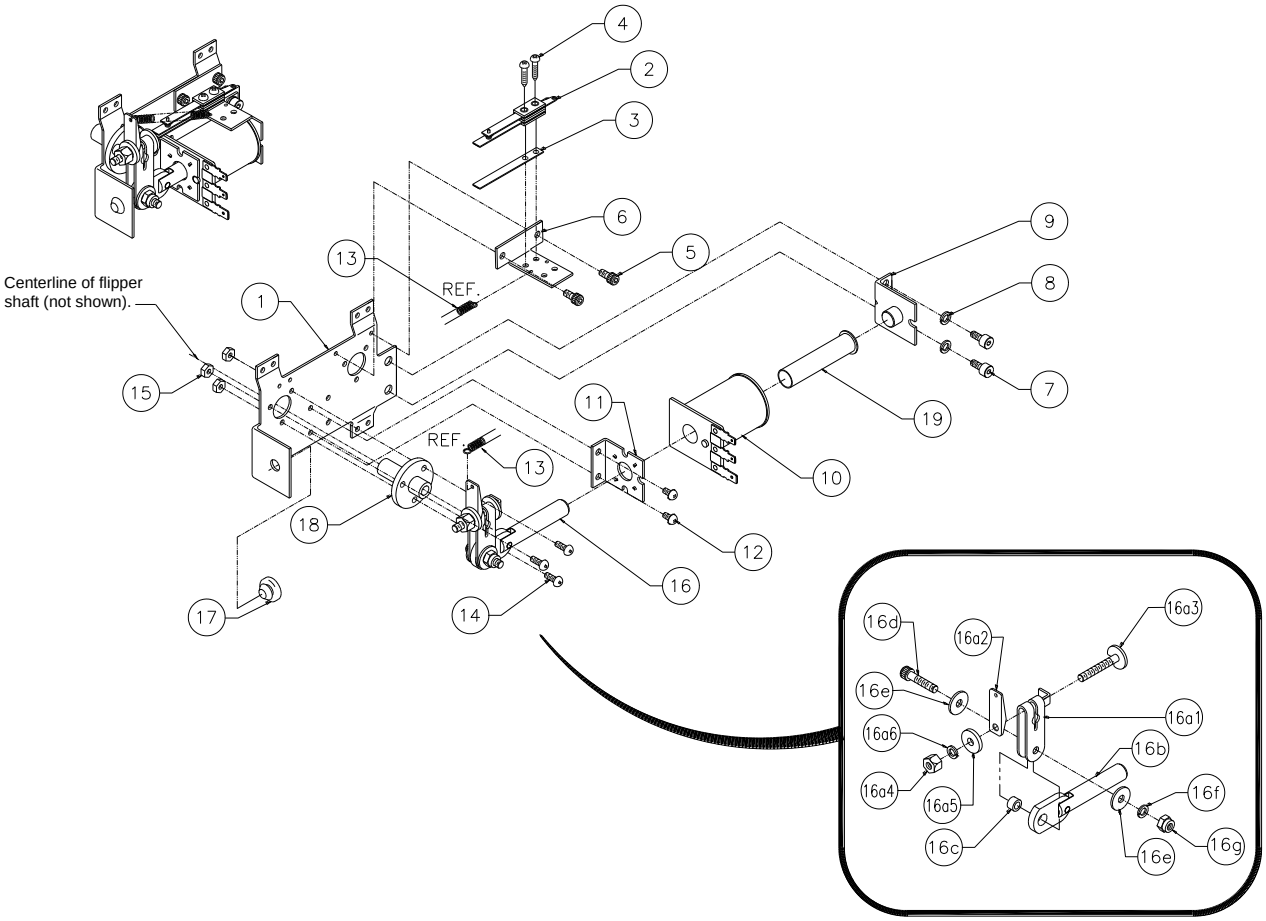
A-19963-4
Ball Trough Assembly



Item	Part Number	Description
1	**A-16809-2**	Ball Trough Welded Assembly
2	**01-11587**	Ball Trough Front
3	**A-6306-2**	Bell Armature Assembly
4	**AE1-26-1500**	Coil
5	**01-8-508-T**	Coil Retainer Bracket
6	**03-7067-5**	Coil Tubing
7	**10-135**	Spring
8	**23-6420**	Rubber Grommet
9	**03-8523**	Insulator
10	**01-11586**	Coil Mounting Bracket
11	**4008-01227-05**	MS 8-32 x 5/16” Pin-Hd-S

Item	Part Number	Description
12	**4408-01119-00**	Nut 8-32 ESN
13	**23-6702**	Bumper Plug
14	**A-18617-1**	Trough IRED LED PCB Assembly
15	**A-18618-1**	Trough IRED Transistor PCB Assembly
16	**4006-01003-10**	MS 6-32 x 5/8” P-PH-S
17	**23-6626**	Rubber Grommet
18	**4700-00004-00**	FW .146 x .375 x .032
19	**02-4975**	Bushing

A-23004-R-2 Flipper Assembly



Item	Part Number	Description
1	**A-14877.1-R**	Flipper Bracket Assembly, Right
2	**SW-1A-194**	Switch Assembly
3	**03-9878**	Shock Deflector
4	**4006-01003-08**	MS 6-32 x ½" P-PH-S
5	**4008-01227-05**	MS 8-32 x 5/16" Pin Hd-SEMS
6	**01-14998**	Switch Bracket
7	**4010-01066-06**	CS-10-31 x 3/8" SH
8	**4701-00004-00**	LW #10 Split
9	**A-12390**	Flipper Stop Bracket Assembly
10	**FL1-11629**	Flipper Coil (Blue)
11	**01-7695-1**	Solenoid Bracket
12	**4006-01003-04**	MS 6-32 x ¼" P-PH-S
13	**10-364**	Spring
14	**4006-01005-06**	MS 6-32 x 3/8" P-PH
15	**4406-01117-00**	Nut 6-32 Hex
16	**A-23000**	Flipper Crank Link Assembly
16a)	**A-22999**	Flipper Crank Assembly
16a)1.	**01-11764-R**	Flipper Crank, Right

Item	Part Number	Description
16a)2.	**01-9376**	Bracket Retaining Spring
16a)3.	**04-11222**	Locking Stud
16a)4.	**4410-01127-00**	Nut 10-32 Hex Heavy Duty
16a)5.	**4700-00107-00**	FW .203 x .625 x .104
16a)6.	**4701-00004-00**	LW #10 Split
16b)	**A-15847**	Flipper Link Assembly
16c)	**02-4676**	Bushing – Link Spacer
16d)	**4010-01086-14**	CS 10-32 x 7/8" SH
16e)	**4700-00023-00**	FW .203 x .625 x .059
16f)	**4701-00004-00**	LW #10 Split
16g)	**4410-01132-00**	Nut 10-32 ESNA
17	**23-6577**	Bumper Plug
18	**03-7568**	Flipper Bushing
19	**03-7066-5**	Coil Tubing

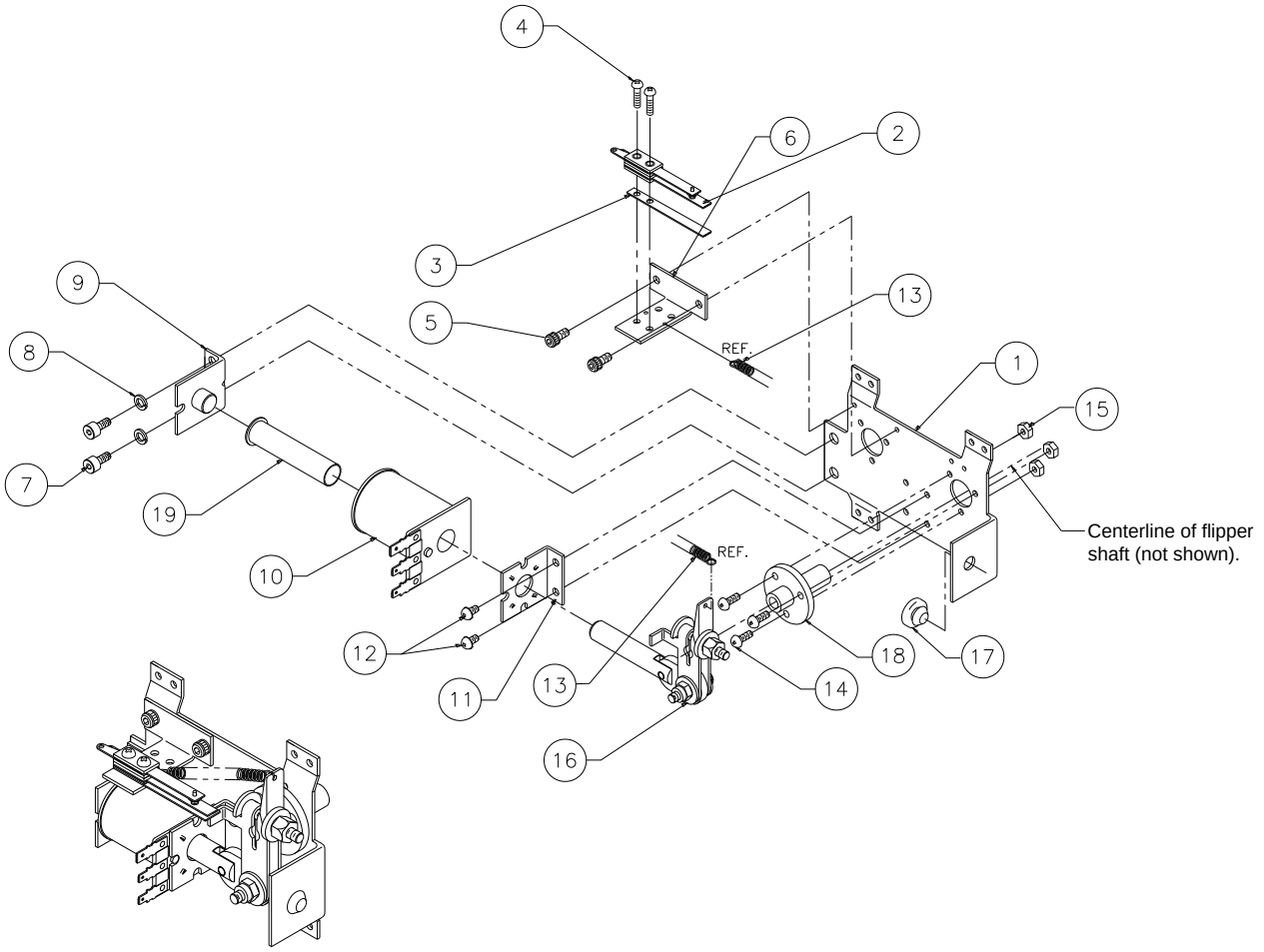
Associated Parts: (Not Shown)

23-6695	Flipper Ring
20-10110-5	Flipper Bat w/Shaft

Notes…

1. *Each Flipper Assembly is mounted beneath the playfield, in conjunction with the Plastic Flipper & Shaft, and Flipper Rubber on the upper side of the playfield.*
2. *With the flipper, in the non-activated position, the E.O.S. Switch contacts must have a gap of .062 (±015) inch. When flipper is activated switch must close.*
3. *Any adjustment of the E.O.S. switch must be made at a minimum distance of 0.25 inch from the switch body.*
4. *Longer blade of E.O.S. switch must be made straight. Gap adjustment is done by adjusting shorter blade.*
5. *All moving elements of the assembly must operate freely without any evidence of binding.*
6. *Apply Loctite™245 when reattaching screws to the Flipper Stop Assembly, the Solenoid Bracket, and the Flipper Bushing.*

A-23001-L-2 Flipper Assembly



Item	Part Number	Description
1	**B-13104.1-L**	Flipper Bracket Assembly, Right
2	**SW-1A-194**	Switch Assembly
3	**03-9878**	Shock Deflector
4	**4006-01003-08**	MS 6-32 x ½" P-PH-S
5	**4008-01227-05**	MS 8-32 x 5/16" Pin Hd-SEMS
6	**01-14998**	Switch Bracket
7	**4010-01066-06**	CS-10-31 x 3/8" SH
8	**4701-00004-00**	LW #10 Split
9	**A-12390**	Flipper Stop Bracket Assembly
10	**FL1-11629**	Flipper Coil (Blue)
11	**01-7695-1**	Solenoid Bracket
12	**4006-01003-04**	MS 6-32 x ¼" P-PH-S
13	**10-364**	Spring
14	**4006-01005-06**	MS 6-32 x 3/8" P-PH
15	**4406-01117-00**	Nut 6-32 Hex
*16	**A-23000**	Flipper Crank Link Assembly
16a)	**A-22999**	Flipper Crank Assembly
16a)1.	**01-11764-R**	Flipper Crank, Right
16a)2.	**01-9376**	Bracket Retaining Spring
16a)3.	**04-11222**	Locking Stud
16a)4.	**4410-01127-00**	Nut 10-32 Hex Heavy Duty
16a)5.	**4700-00107-00**	FW .203 x .625 x .104
16a)6.	**4701-00004-00**	LW #10 Split
16b)	**A-15847**	Flipper Link Assembly
16c)	**02-4676**	Bushing – Link Spacer
16d)	**4010-01086-14**	CS 10-32 x 7/8" SH
16e)	**4700-00023-00**	FW .203 x .625 x .059
16f)	**4701-00004-00**	LW #10 Split
16g)	**4410-01132-00**	Nut 10-32 ESNA
17	**23-6577**	Bumper Plug
18	**03-7568**	Flipper Bushing
19	**03-7066-5**	Coil Tubing

Associated Parts: (Not Shown)

Part Number	Description
23-6695	Flipper Ring
20-10110-5	Flipper Bat w/Shaft

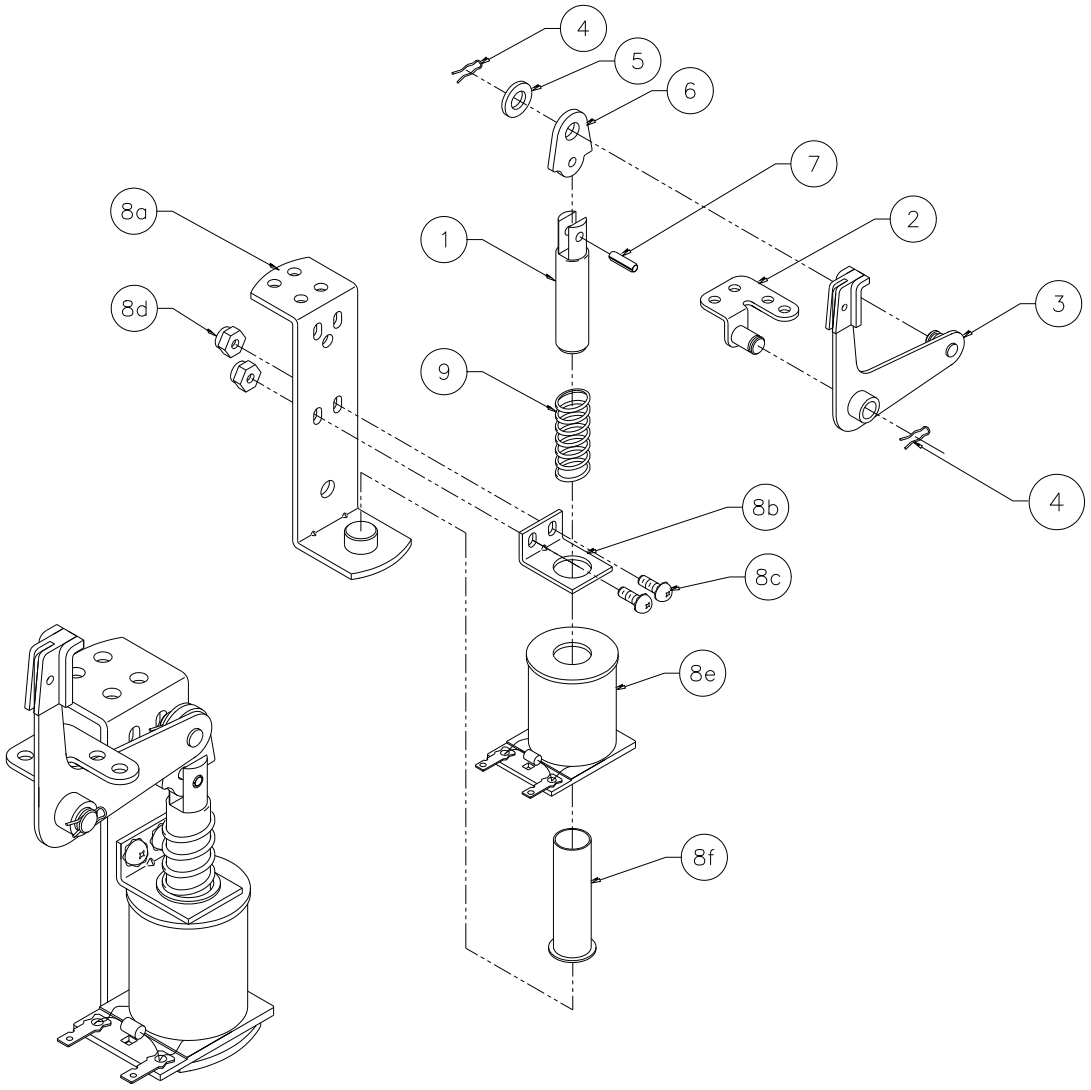
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*See A-23004-R, Flipper Assembly, for assembly detail drawing.

Note: ***Switch Assembly to be attached to the Switch Bracket using the outermost threaded holes on the bracket***

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A-17811 Kicker Assembly

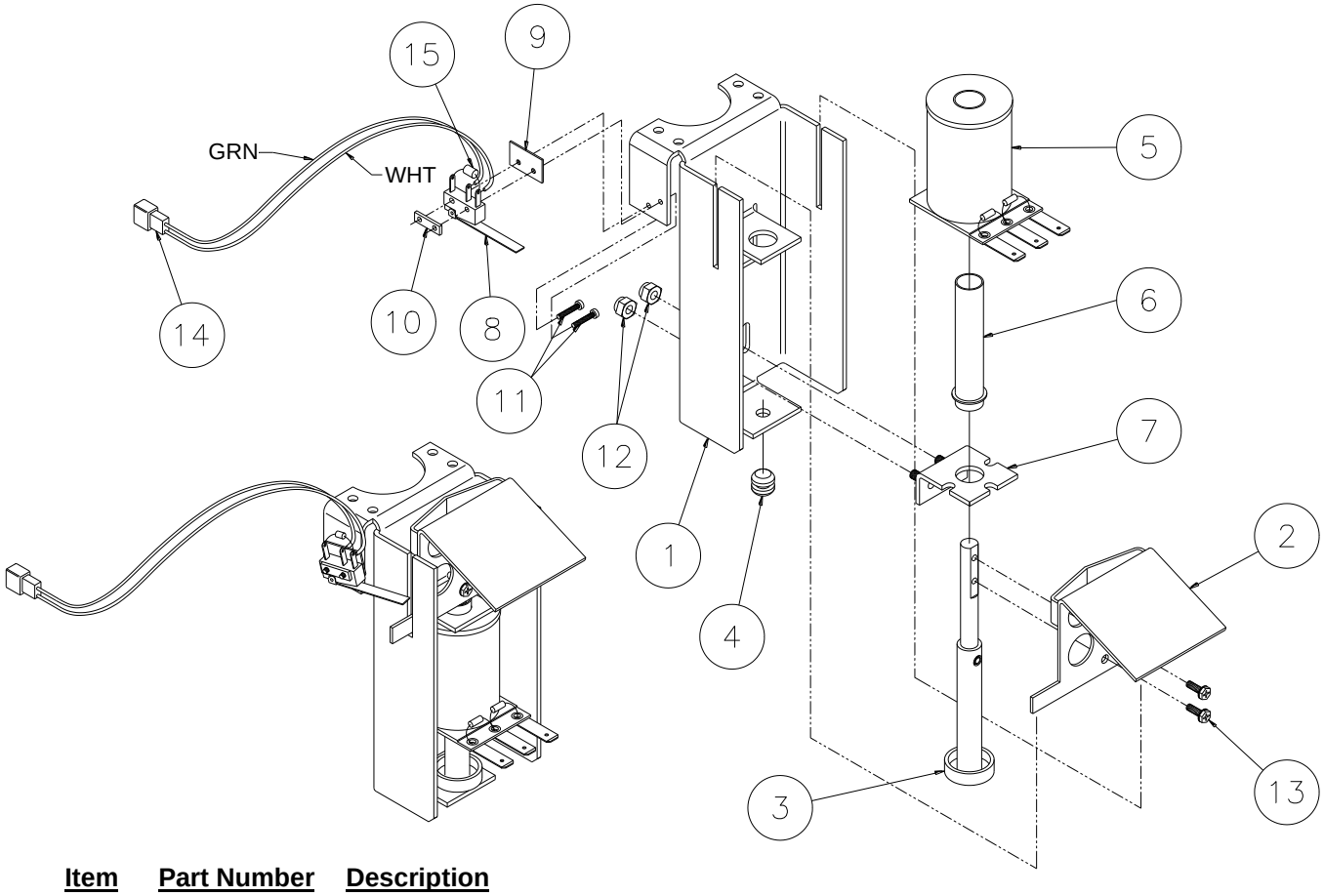


Item	Part number	Description
1	**02-2364**	Coil Plunger
2	**A-17810**	Kicker Mounting Bracket Assembly
3	**A-12664**	Crank Assembly
4	**12-6227**	Hair Pin Clip
5	**4700-00030-00**	**F**W .265 x .500 x .067
6	**03-8085**	Armature Link
7	**20-8716-5**	Roll Pin 1/8” x 7/16”

Associated Parts:

Item	Part number	Description
8	**A-22207-6**	Coil & Bracket Assembly-Left
8a)	**A-17808**	Bracket & Stop Assembly
8b)	**01-8-508-S**	Coil Retainer Bracket
8c)	**4006-01017-06**	MS 6-32 x 3/8” P-RH-S
8d)	**4406-01119-00**	Nut 6-32 ESN
8e)	**AE1-26-1200**	Coil Assembly
8f)	**03-7066**	Coil Tubing
9	**10-128**	Kicker Spring

A-22989 Skip Ramp Assembly

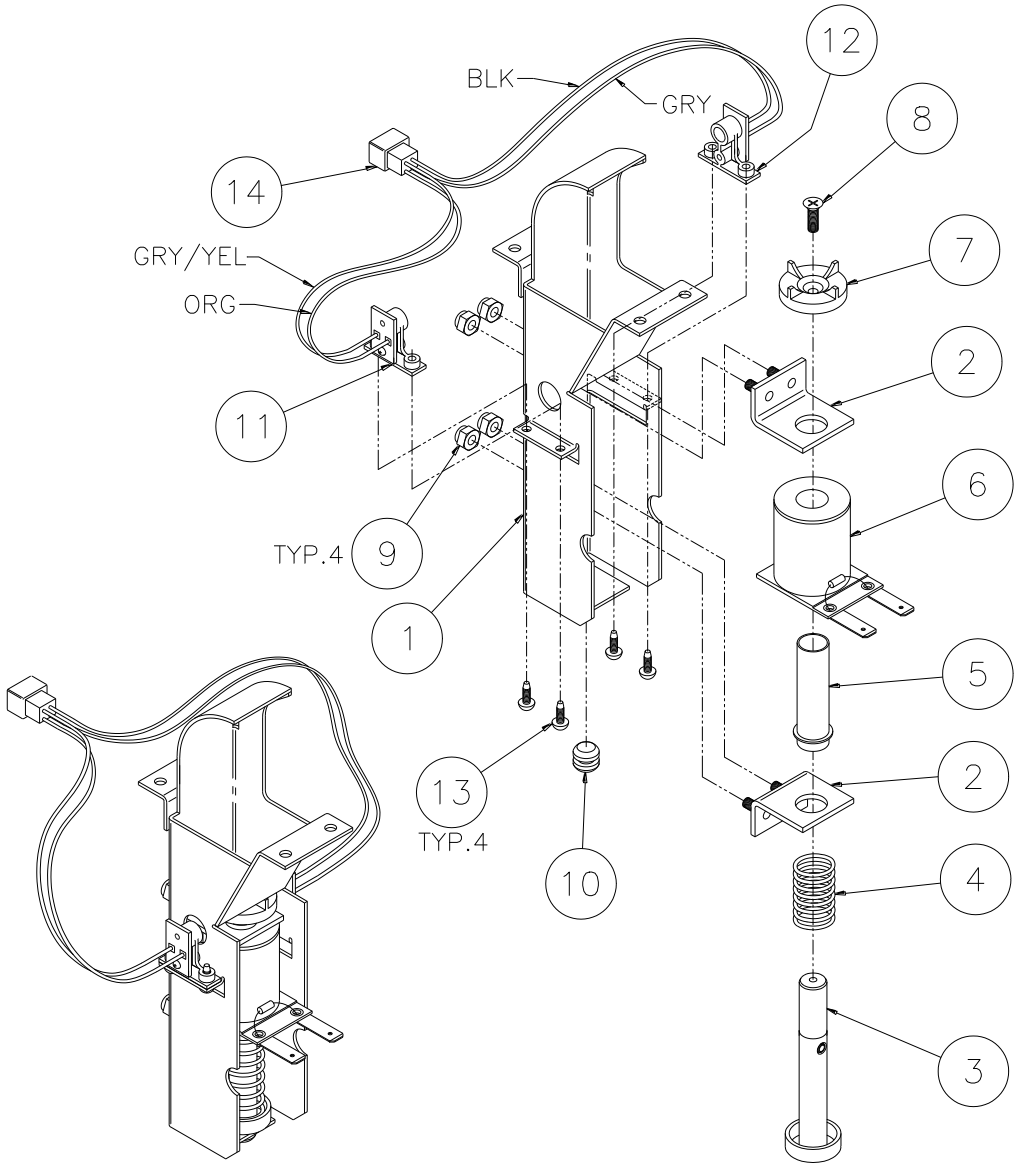


Item	Part Number	Description
1	**01-15142**	Main Bracket
2	**04-12615**	Ramp Bracket
3	**04-12616.2**	Skip Ramp Armature Assembly
4	**23-6420**	Grommet
5	**FL1-11753**	Flipper Coil – Yellow
6	**03-7067-6**	Coil Tubing
7	**04-10911-1**	Coil Centering Bracket
8	**5647-12693-07**	Sub-Mini Micro Switch
9	**01-8600**	Insulator
10	**01-8240**	Plate Nut 2-56
11	**4002-01105-08**	MS 2-56 x ½ P-PH-S
12	**4408-01119-01**	Nut 8-32 ESNA/NTM
13	**4008-01227-06**	MS 8-32 x 3/8 Pin Hd SEMS
14	**H-23019-2**	2 – Pin Cable
15	**5070-09054-00**	Diode – 1N4004 1.0A

Associated Parts: (Not Shown)

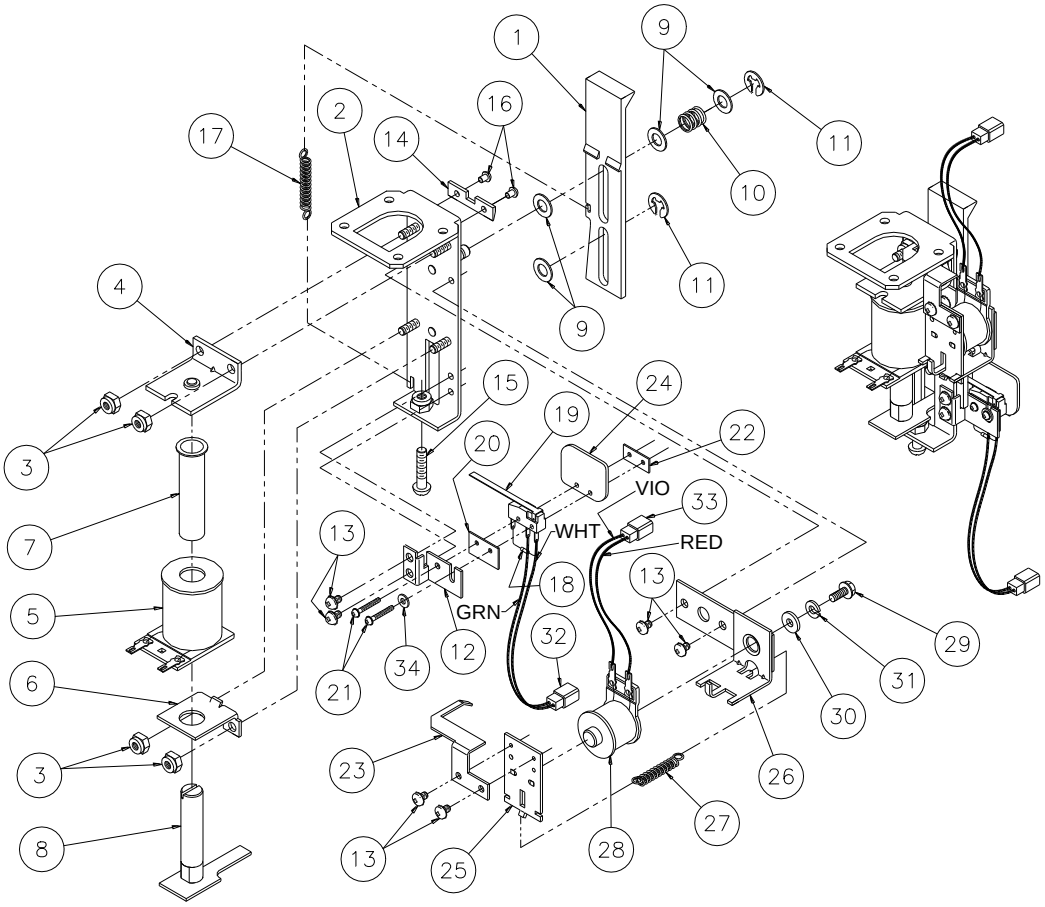
Part Number	Description
A-22988	Skip Ramp Flap Assembly
01-15140	Flap Assembly Mounting Bracket

A-23156 Popper Assembly



Item	Part Number	Description
1	**04-12741**	Popper Bracket
2	**04-10322-2**	Coil Bracket
3	**04-10291**	Armature & Extension Assembly
4	**10-135**	Plunger Spring
5	**03-7067**	Coil Tubing
6	**AE1-25-1000**	Coil Assembly
7	**03-8561**	Ball Popper Cup
8	**4106-01152-08**	SMS 6-32 x ½ TFP – FLH
9	**4408-01119-01**	Nut 8-32 ESNA – NTM
10	**23-6420**	Grommet
11	**A-16909**	Opto P/T Assembly
12	**A-16908**	Opto LED Assembly
13	**4106-01013-06**	SMS 6-32 x 3/8 P- PH T-25
14	**H-23022-5**	4-Pin Cable

A-15211-1
1 Bank Drop Target Assembly

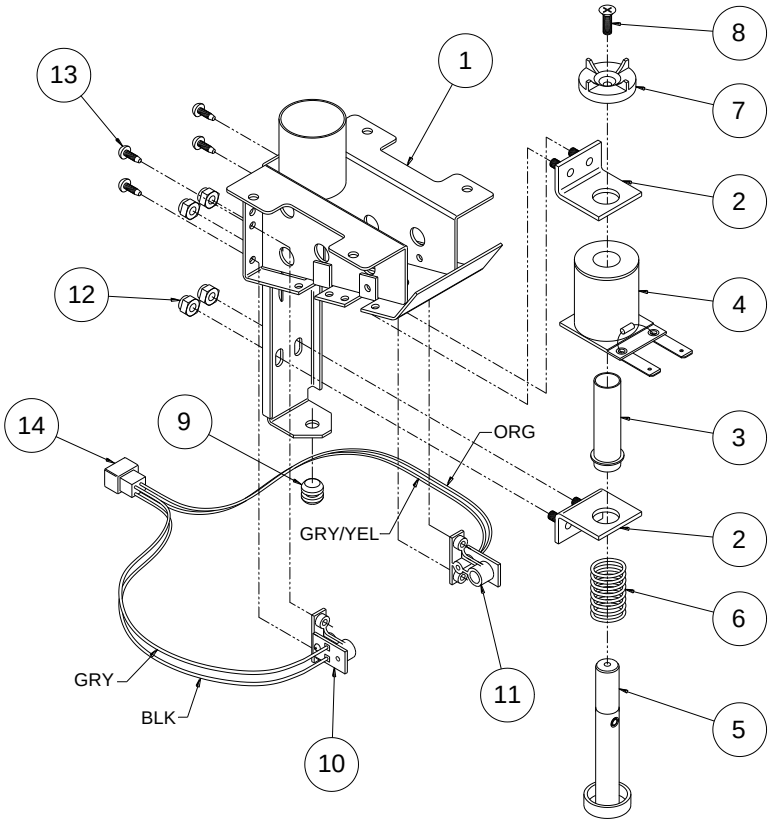


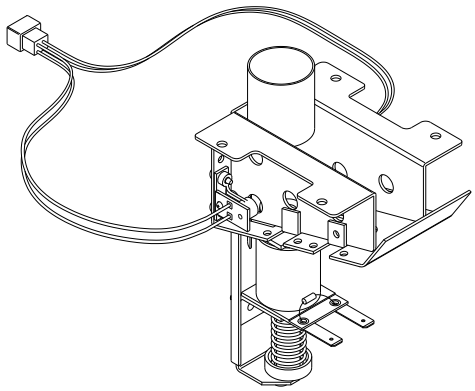
Item	Part Number	Description
1	**03-8750**	Flush Target
2	**A-14617**	Bracket & Post Assembly
3	**4408-01119-01**	Nut 8-32 ESNA-NTM
4	**A-11397**	Drop Target Stop Bracket
5	**AE1-26-1200**	Coil Assembly
6	**01-8413**	Coil Mounting Bracket
7	**03-7066-4**	Coil Tubing
8	**A-11388-2**	Plunger & Reset Plate Assembly
9	**4700-00072-00**	FW .256 x .500 x .032
10	**10-392**	Compression Spring
11	**20-8712-25**	E Ring – ¼" Shaft
12	**01-10183**	Switch Bracket
13	**4006-01003-04**	MS 6-32 x 1/4" P-PH-S
14	**03-8034**	Single Target Stop
15	**4010-01025-14**	MS 10-32 x 7/8" P-RH
16	**07-6688-17N**	Rivet 5/32" x 1/8" Nickel
17	**10-433**	Extension Spring

Item	Part Number	Description
18	**5070-09054-00**	Diode 1N4004 1.0A
19	**5647-12693-31**	Sub Mini Micro Switch
20	**01-8600**	Insulator
21	**4002-01105-10**	MS 2-56 x 5/8" P-PH-S
22	**01-8240**	Plate Nut 2-56
23	**01-8647-R**	Actuator
24	**03-8630**	Actuator Guide
25	**A-15821**	Armature Assembly
26	**A-15209**	Frame & Eyelet Assembly
27	**10-363**	Extension Spring
28	**SM1-26-600**	Coil Assembly
29	**4008-01021-06**	MS 8-32 x 3/8" P-RH Brass
30	**4700-00089-00**	FW .172 x .437 x .062 Brass
31	**4701-00003-00**	LW #8 Split
32	**H-23019-2**	Switch 2 Pin Cable
33	**H-19523**	Solenoid 2 Pin Cable
34	**4700-00003-00**	FW .125 x .281 x .032

Note: *Apply Loctite 242 to screws, items 13 (6 places) and 29 (1 place) prior to assembly.*

A-20680-1
3 Ball Popper Assembly



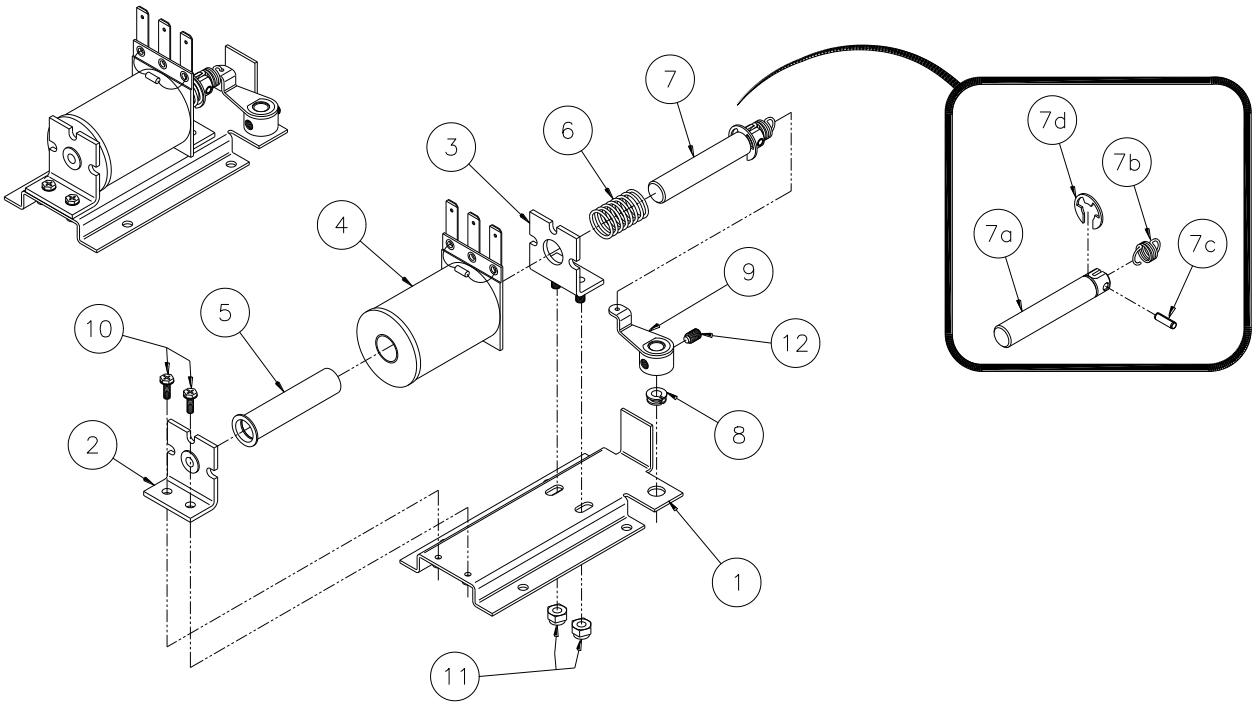


Item	Part Number	Description
1	**04-10323**	3 Ball Popper Frame
2	**04-10322-2**	Coil Bracket
3	**03-7067**	Coil Tubing
4	**AE1-23-800**	Coil Assembly
5	**04-10291**	Armature & Extension Assembly
6	**10-135**	Plunger Spring
7	**03-8561**	Ball Popper Cup
8	**4106-01152-08**	SMS 6-32 x ½" TFP-FLH
9	**23-6420**	Grommet
10	**A-16908**	OPTO LED Assembly
11	**A-16909**	OPTO P/T Assembly
12	**4408-01119-01**	Nut 8-32 ESNA/NTM
13	**4106-01013-06**	SMS 6-32 x 3/8 P-PH T-25
14	**H-23022-5**	4-Pin Cable

Associated Parts: (Not Shown)

Part Number	Description
04-12598	Popper Hood

A-22993
Diverter Mechanism Assembly

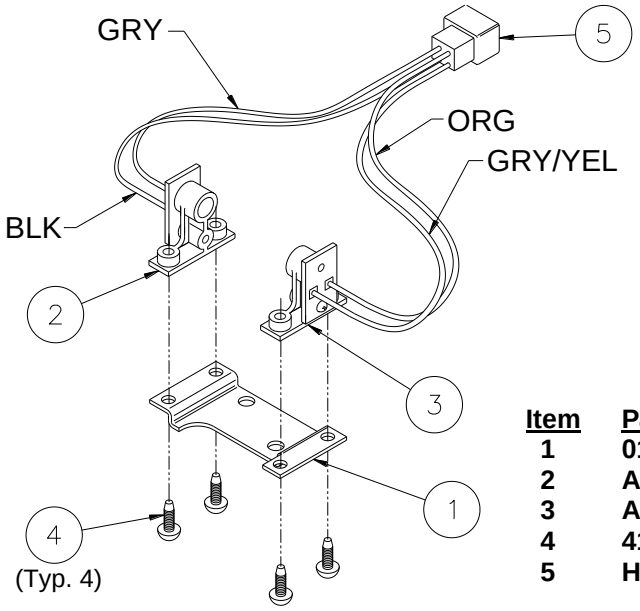


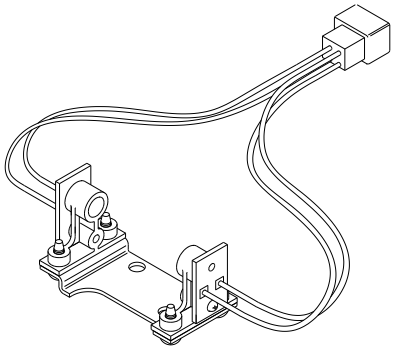
Item	Part Number	Description
1	**04-12619**	Diverter Mech Bracket
2	**04-10911-6**	¾ Coil Centering Bracket
3	**04-10911-1**	¾ Coil Centering Bracket
4	**FL1-22241**	Flipper Coil
5	**03-7066-5**	Coil Tubing
6	**10-303**	Master Spring
7	**A-16636**	Diverter Plunger Assembly
7a	**02-4793**	Diverter Plunger
7b	**10-389**	Gate Spring
7c	**20-8716-5**	Roll Pin, ∅1/8” x 7/16”
7d	**20-8712-43**	E-ring, ∅7/16” Shaft
8	**20-8790**	Nyliner Bearing
9	**04-10408**	Drive Arm Assembly
10	**4008-01227-06**	MS 8-32 x 3/8 PIN HD - S
11	**4408-01119-01**	Nut 8-32 ESNA – NTM
12	**4010-01196-04**	SS 10-32 x ¼ SH – CP – N

Associated Parts: (Not Shown)

Part Number	Description
A-22970	Diverter Paddle Assembly

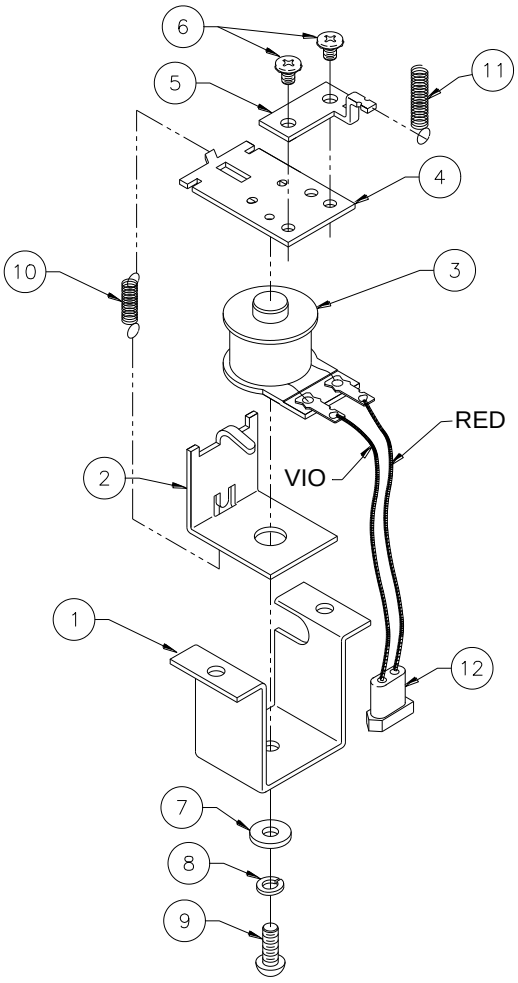
A-23012 Opto Bracket Assembly

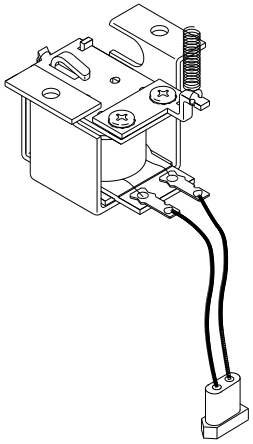




Item	Part Number	Description
1	**01-15162.1**	Opto Mounting Bracket
2	**A-16908**	Opto LED Assembly
3	**A-16909**	Opto P/T Assembly
4	**4106-01013-06**	SMS #6 x 3/8” P-PH T-25
5	**H-23022-5**	4 Pin Square Opto Cable

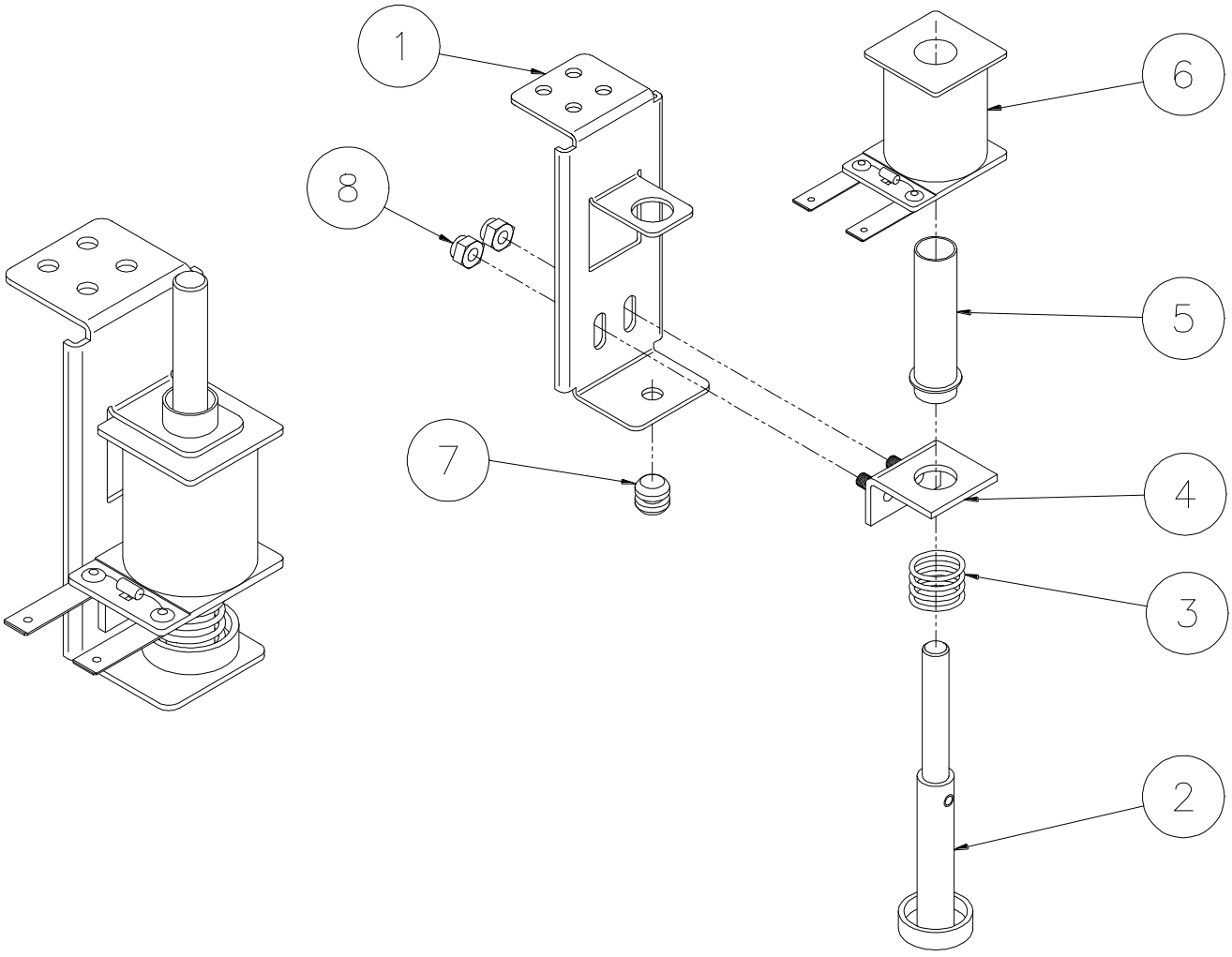
A-17796 Ball Gate Actuator Assembly





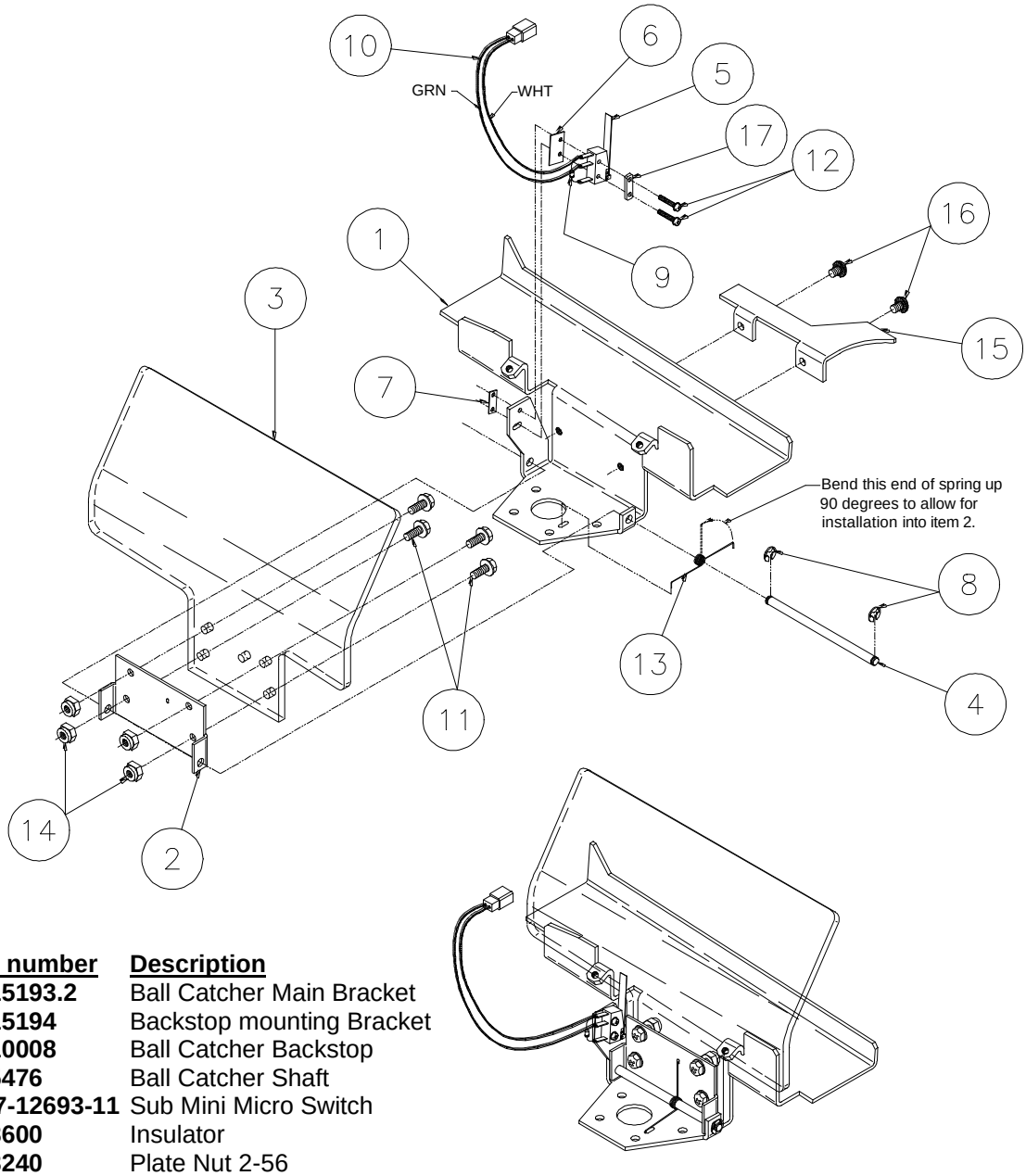
Item	Part Number	Description
1	**01-12348**	Ball Gate Coil Bracket
2	**A-6892**	Frame & Eyelet
3	**A-14406**	Coil Assembly
4	**A-11146**	Ball Gate Armature Assembly
5	**01-8391**	Armature Tip
6	**4006-01003-03**	MS 6-32 x 3/16” P-PH-S
7	**4700-00089-00**	**F**W .172 x 7/16 x .062 Brass
8	**4701-00003-00**	LW #8 Split
9	**4008-01021-07**	MS 8-32 x 7/16” P-RH-Brass
10	**10-120**	Spring
11	**10-194**	Extension Spring
12	**H-19523**	2 Pin Cable

A-22977
Up/Down Post Assembly



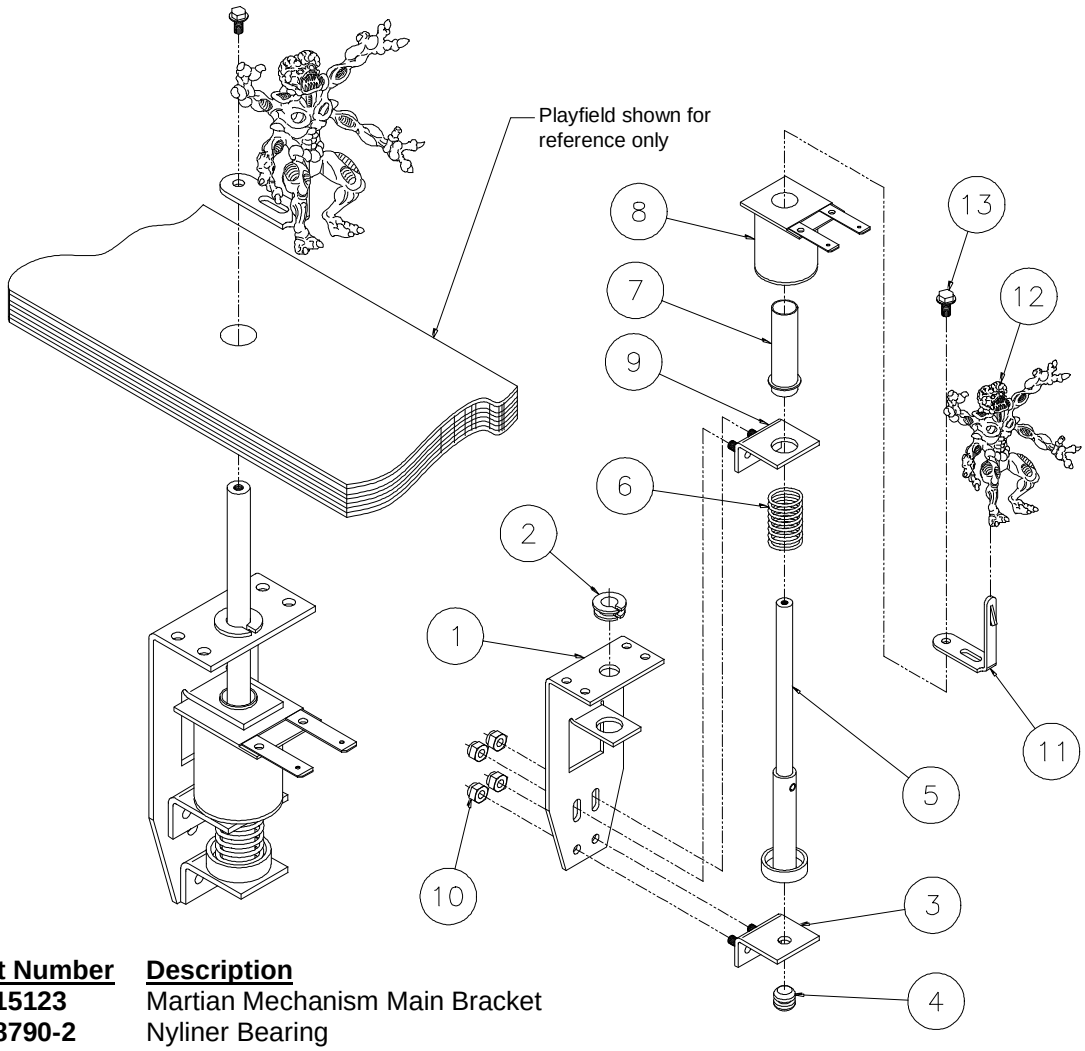
Item	Part Number	Description
1	**01-15122**	Up/Down Post Assembly Bracket
2	**A-17986**	Bell Armature Assembly
3	**10-135**	Plunger Spring
4	**04-10910-1**	Coil Centering Bracket
5	**03-7067-5**	Coil Tubing
6	**AE1-26-1500**	Coil
7	**23-6420**	Grommet
8	**4408-01119-01**	Nut 8-32 ESNA/NTM

A-22992
Ball Catcher Assembly



Item	Part number	Description
1	**01-15193.2**	Ball Catcher Main Bracket
2	**01-15194**	Backstop mounting Bracket
3	**03-10008**	Ball Catcher Backstop
4	**02-5476**	Ball Catcher Shaft
5	**5647-12693-11**	Sub Mini Micro Switch
6	**01-8600**	Insulator
7	**01-8240**	Plate Nut 2-56
8	**20-8712-18**	E-Ring 3/16" Shaft
9	**5070-09054-00**	Diode 1N4004 1.0A
10	**H-23019-8**	2 Pin Cable
11	**4008-01220-08**	MS 8-32 x ½" Pin Hd
12	**4002-01105-08**	MS 2-56 x ½" P-PH-S
13	**10-562.1**	Ball Catcher Spring
14	**4408-01119-01**	Nut 8-32 ESNA/NTM
15	**01-15204.2**	Ball Catcher Anti Ball Trap Bracket
16	**4008-01227-04**	MS 8-32 x ¼" Pin Hd-SEMS
17	**01-15218**	Switch Protector Plate

A-22960
Martian Mechanism Assembly

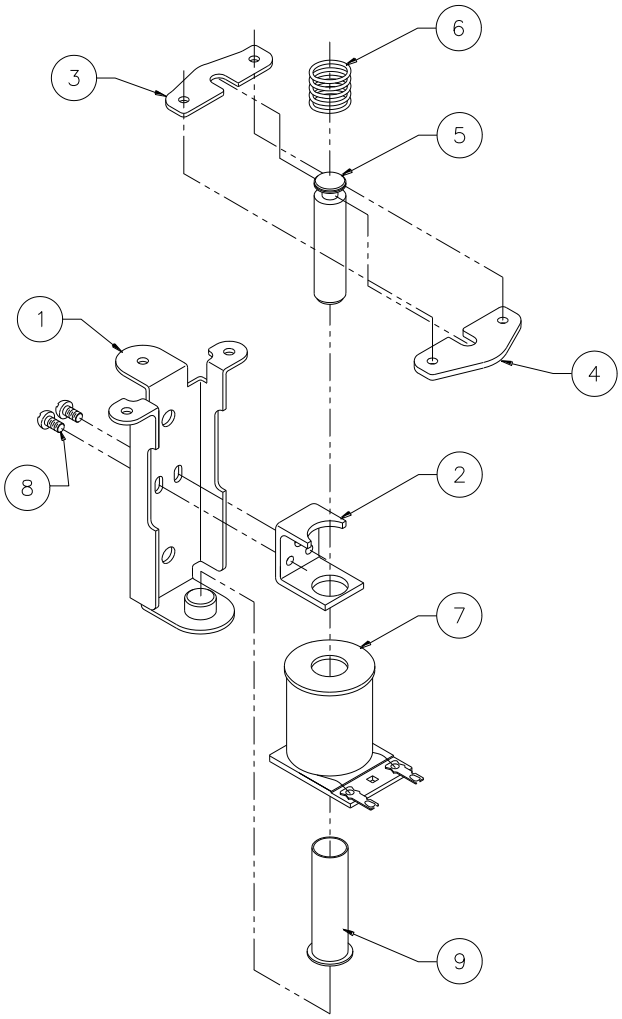


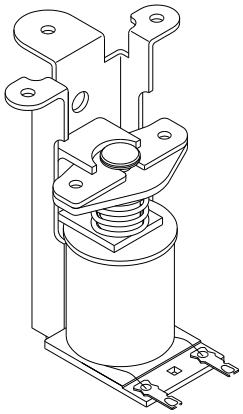
Item	Part Number	Description
1	**01-15123**	Martian Mechanism Main Bracket
2	**20-8790-2**	Nyliner Bearing
3	**04-10910-7**	Coil Centering Bracket
4	**23-6420**	Rubber Grommet
5	**04-12595.2**	Martian Plunger Assembly
6	**10-135**	Plunger Spring
7	**03-7067**	Coil Tubing
8	**AE1-26-1500**	Coil
9	**04-10910-1**	Coil Centering Bracket
10	**4408-01119-01**	Nut 8-32 ESNA-NTM

Associated Parts:

Item	Part Number	Description
11	**01-15124**	Martian Retaining Bracket
12	**23-6768**	Martian Figurine
13	**4008-01168-06**	Machine Screw, 8-32 x 3/8”

A-22205-4
Jet Coil & Bracket Assembly





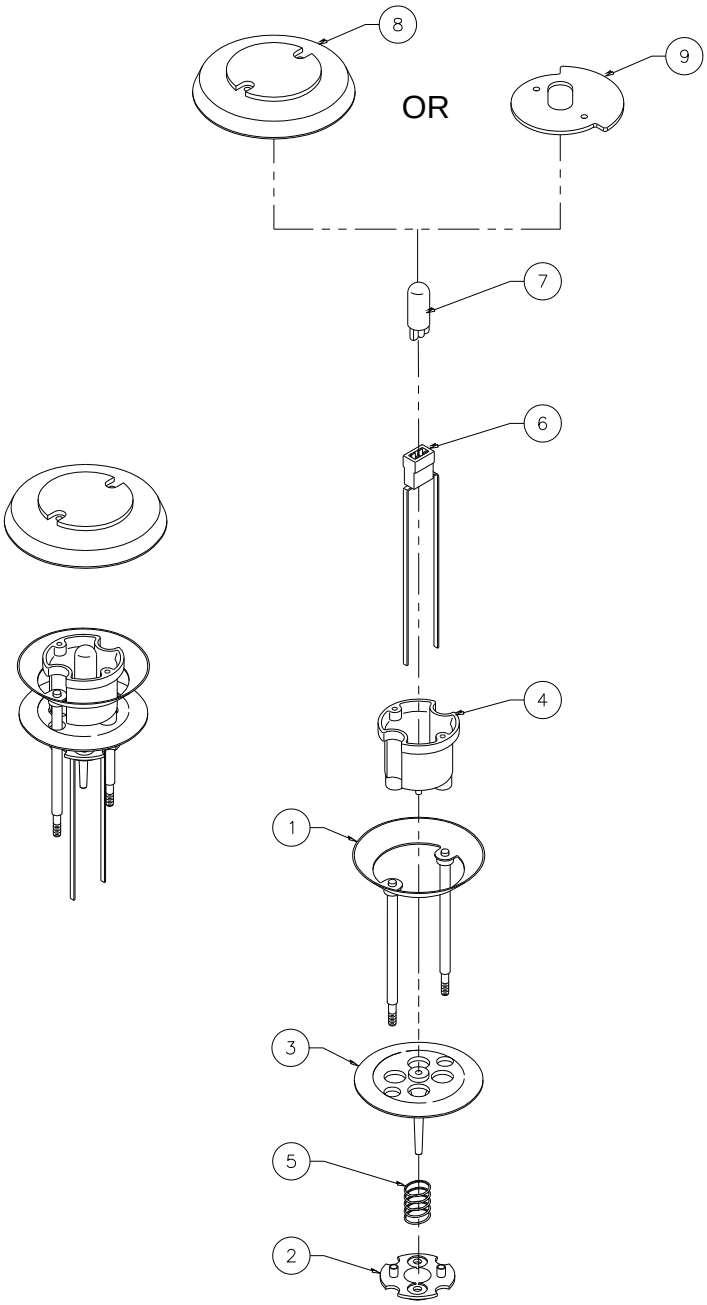
Item	Part Number	Description
1	**04-10888**	Bracket and Stop Assembly
2	**01-1747**	Coil Retainer Bracket
3	**01-5492**	Armature Link (Steel)
4	**01-5493**	Armature Link (Bakelite)
5	**02-3406-1**	Coil Plunger
6	**10-326**	Jet Bumper Spring
7	**AE1-26-1200**	Coil Assembly
8	**4006-01017-04**	MS 6-32 x ¼" P-RH-S
9	**03-7066**	Coil Tubing

Associated Parts: (Not Shown)

Item	Part Number	Description
10	**A-12030-3**	Jet Bumper Leaf Switch Assembly
10a)	**A-16443-1**	Jet Bumper Switch and Diode Assembly
10b)	**01-1168**	Bumper Switch Mounting Bracket
10c)	**01-3670**	Curved Switch Plate
10d)	**03-7395**	Switch Actuator
10e)	**4005-01003-12**	MS 5-40 x ¾" P-PHS
10f)	**4405-01117-00**	Nut 5-40 Hex

B-9414-6
Jet Bumper Assembly

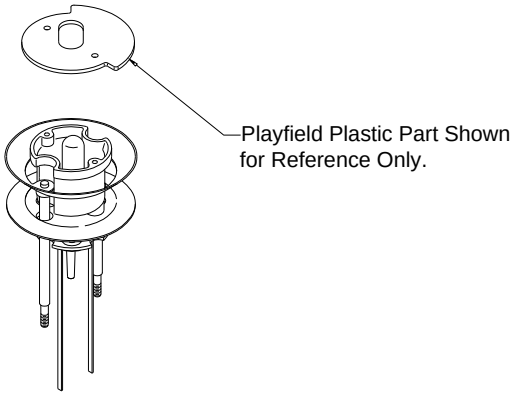
(Two B-9414-6 assemblies required. One to be used in conjunction with associated part number 03-8254-9, item 8, and the other to be used with item 9, part number 31-3437-24.)



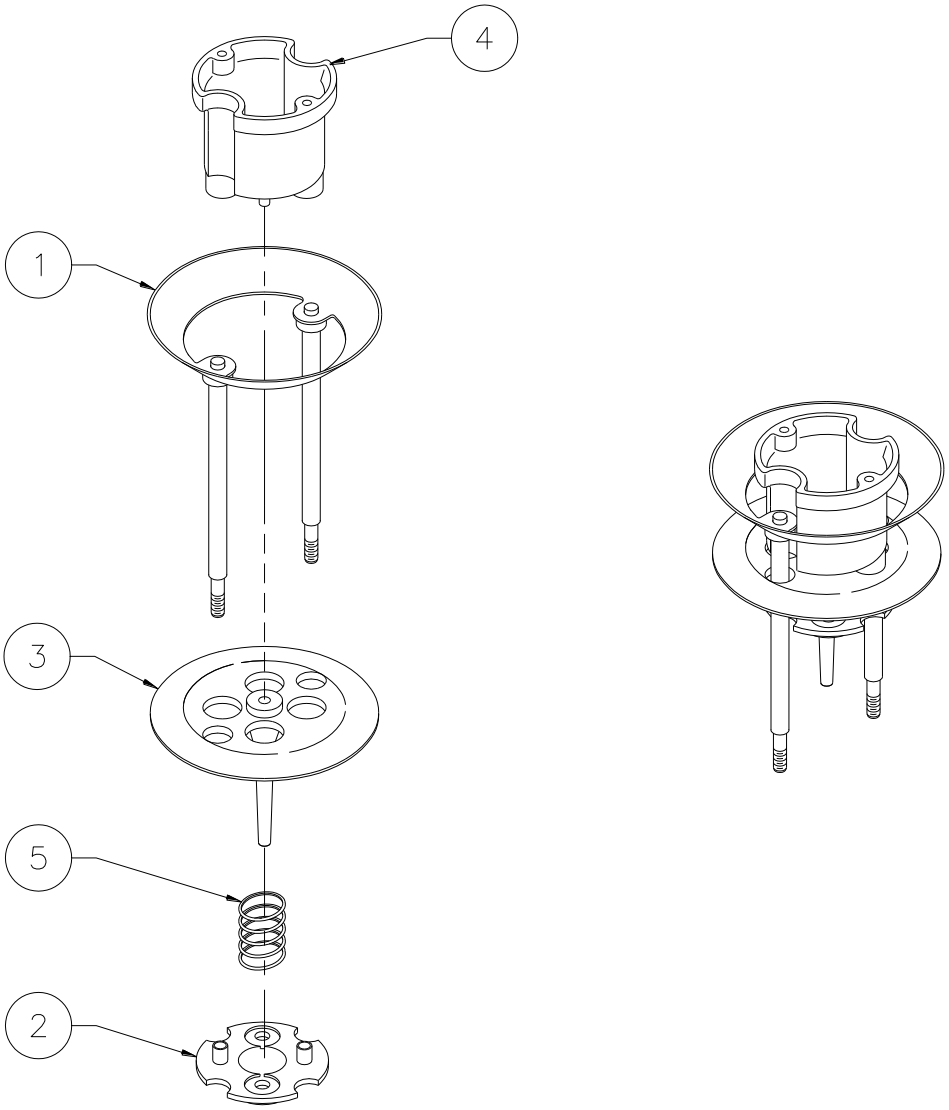
Item	Part Number	Description
1	**A-4754**	Bumper Ring Assembly
2	**03-6009-A5**	Bumper Base
3	**03-6035-7**	Bumper Wafer – Black
4	**03-7443-5**	Bumper Body
5	**10-7**	Spring
6	**24-8776**	Wedge Base Socket
7	**24-8768**	Bulb #555 (6.3V, 0.25A)

Associated Parts

Item	Part Number	Description
8	**03-8254-9**	Jet Bumper Cap, Red
9	**31-3437-24**	Playfield Plastic (Ref. Only)

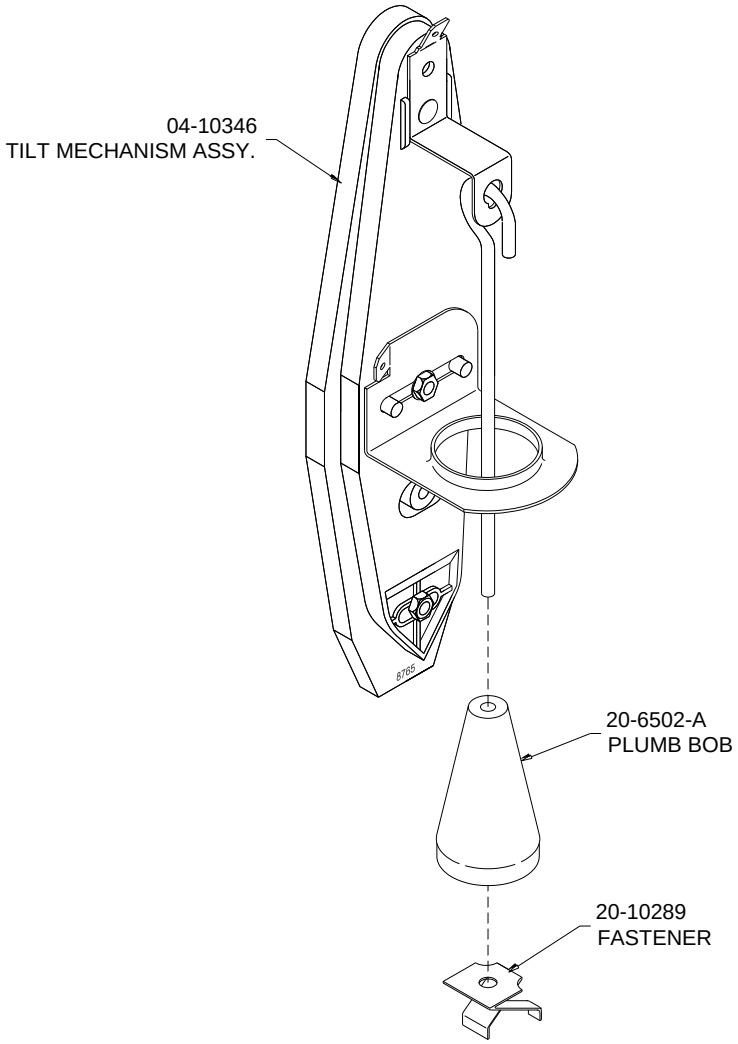


B-13123-2
Jet Bumper Assembly

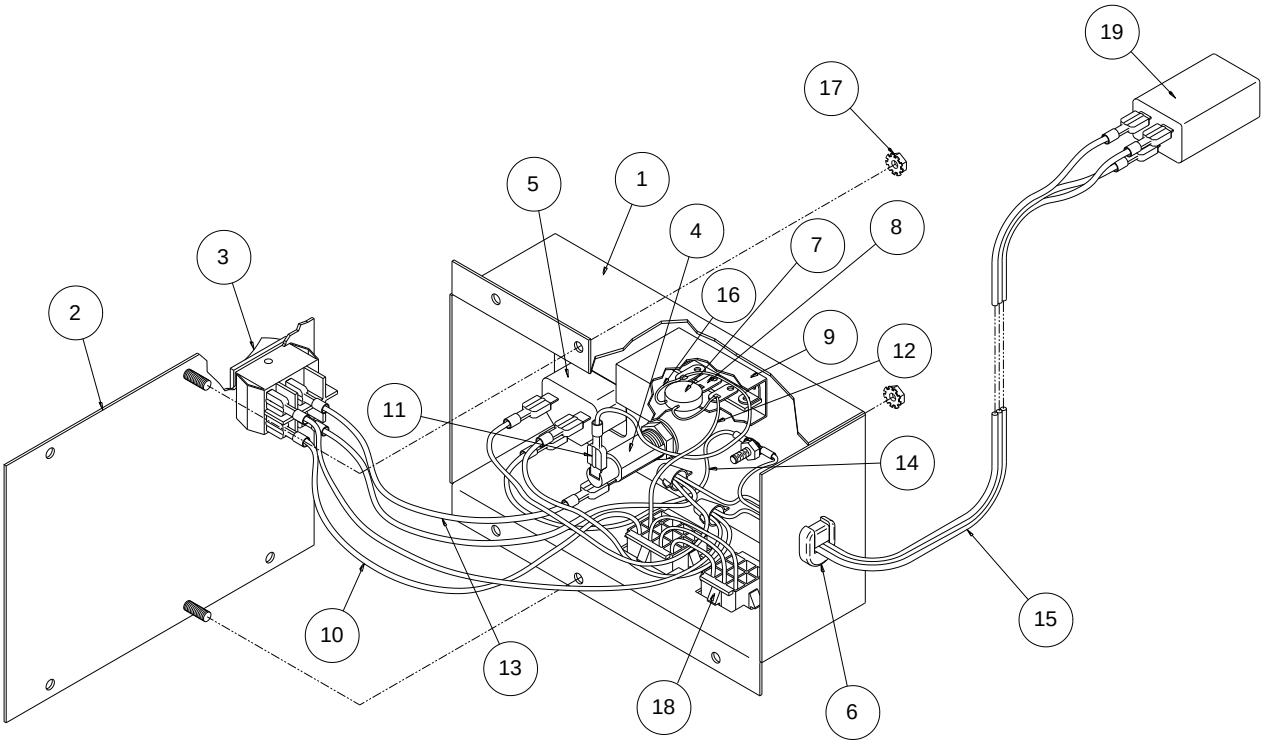


Item	Part Number	Description
1	A-4754	Bumper Ring Assembly
2	03-6009-A5	Bumper Base
3	03-6035-7	Bumper Wafer – Black
4	03-7443-5	Bumper Body
5	10-7	Spring

04-10346.1
Tilt Mechanism Assembly



A-22966
Power Control Box Assembly



Item	Part Number	Description
1	**01-15126**	Power Control Box
2	**04-12597**	Switch Mounting Plate
3	**5642-13935-00**	Rocker Switch – DPST
4	**5733-14734-00**	Fuse Holder, 5 x 20mm
5	**5851-13867-00**	Outlet, IEC Connector 237 Socket
6	**03-8712**	Strain Relief Bushing
7	**5016-12978-00**	Thermistor, 8A, 2.5R25
8	**5822-13865-00**	Terminal Strip, 3 CKT, 2 Mounting
9	**01-12299**	Terminal Strip Insulator
10	**H-23015**	Jumper, Neutral SW/1FC

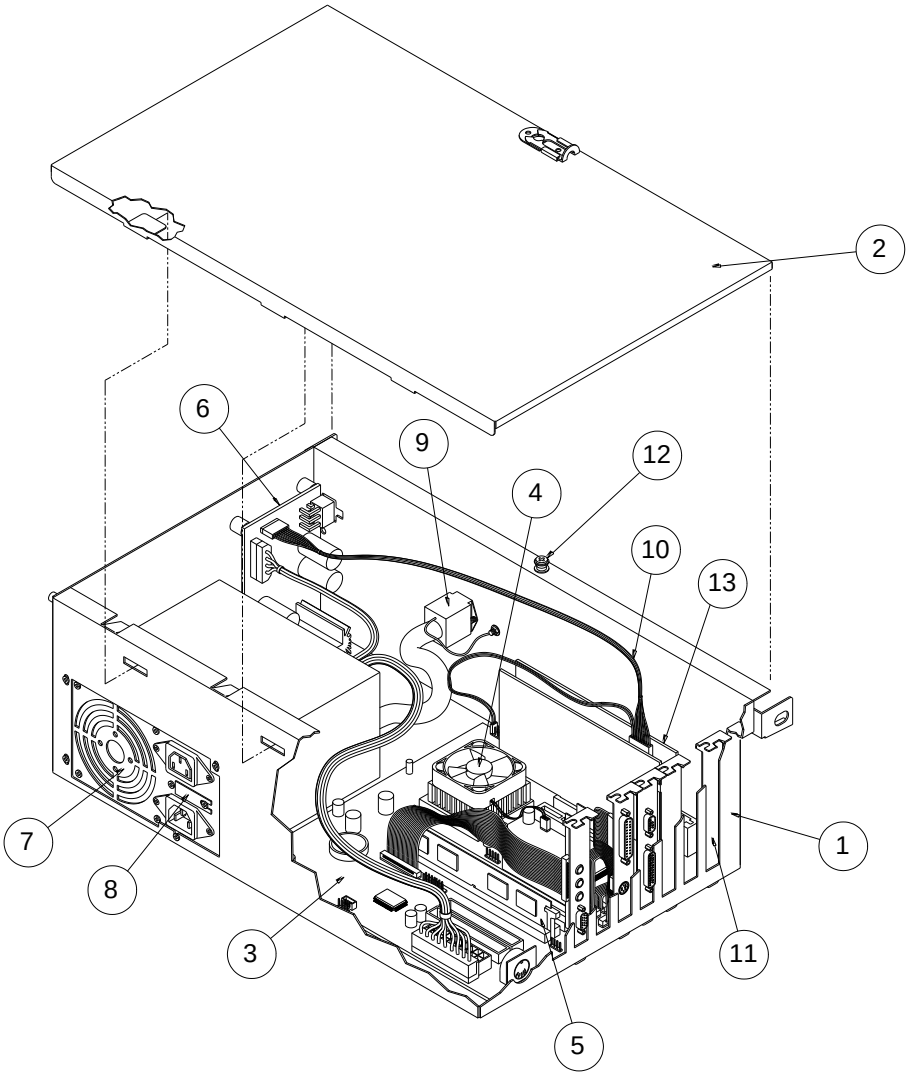
Item	Part Number	Description
11	**H-17543**	Jumper, Hot Cable
12	**H-23014**	Jumper, Hot Interface
13	**H-17545**	Jumper, SW/Fuse Cable
14	**H-17542**	Jumper, Ground Cable
15	**H-23005**	3 Conductor Power Cable Assembly
16	**RM-21-06**	#18 Vinyl Sleeving
17	**4408-01128-00**	Nut, 8-32 KEPS
18	**H-22991**	Transformer Programming Cable
19	**5102-14793-00**	Line Filter

Note: *This drawing is not to be used for the purpose of wiring the Power Control box Assembly. Refer to the appropriate schematic diagram when attempting the wiring of this assembly.*

Power Interface/Cordset Application Chart

COUNTRY	UNIVERSAL PWR. INTERFACE ASSEMBLY	VOLTAGE PROGRAMMING JUMP CABLE			4AMP FUSE	5AMP FUSE	POWER ADAPTER CORD	CORDSET								
	A-22966	H-22990-1	H-22990-2	H-22990-3	5731-14530-00	5731-14046-00	5850-14052-00	5850-13271-00	5850-13272-00	5850-13273-00	5850-13274-00	5850-13275-00	5850-13276-00	5850-13277-00	5850-13278-00	A-17175-2
UNITED STATES	X	X				X	X	X								
CANADA	X	X				X		X								
TAIWAN	X	X				X		X								
MEXICO	X	X				X		X								
CENTRAL AMERICA	X	X				X		X								
SOUTH KOREA	X	X				X		X								
PUERTO RICO	X	X				X		X								
AUSTRIA	X		X		X				X							
BELGIUM	X		X		X				X							
FINLAND	X		X		X				X							
FRANCE	X		X		X				X							
GREECE	X		X		X				X							
HOLLAND	X		X		X				X							
HUNGARY	X		X		X				X							
NETHERLANDS	X		X		X				X							
NETH. ANTILLES	X		X		X				X							
NORWAY	X		X		X				X							
POLAND	X		X		X				X							
PORTUGAL	X		X		X				X							
SPAIN	X		X		X				X							
SWEDEN	X		X		X				X							
TURKEY	X		X		X				X							
WEST GERMANY	X		X		X				X							
UNITED KINGDOM	X		X		X					X						
IRELAND	X		X		X					X						
HONG KONG	X		X		X					X						
DENMARK	X		X		X						X					
ITALY	X		X		X							X				
CHILE	X		X		X							X				
PEOPLE'S REP. OF CHINA	X		X		X							X				
SWITZERLAND	X		X		X								X			
AUSTRALIA	X		X		X									X		
NEW ZEALAND	X		X		X									X		
ARGENTINA	X		X		X									X		
JAPAN	X			X		X									X	X
CROATIA	X		X		X				X							

A-22724
Computer Case Assembly

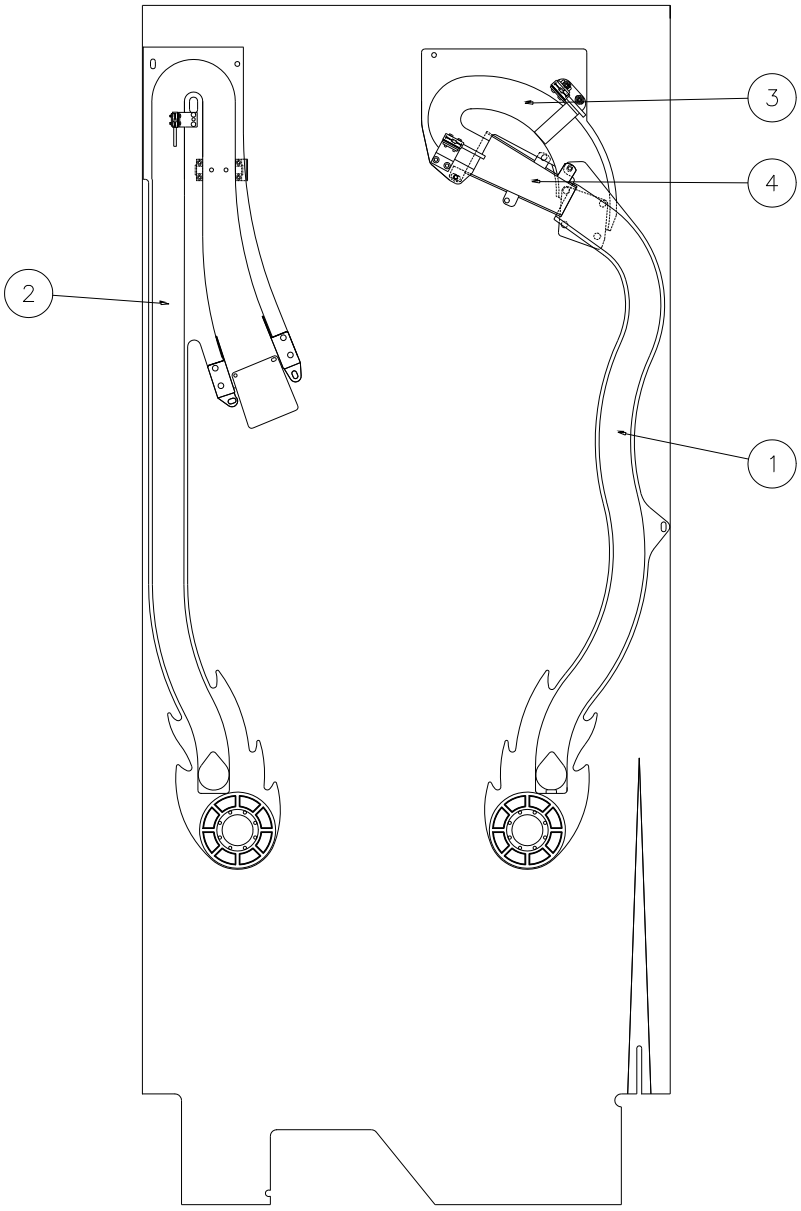


Item	Part Number	Description
1	04-12370.5	Computer Case
2	A-23301	Computer Case Top Assembly
3	04-12604	Mother Board
4	20-10698	Fan/Heat Sink – Socket 7
5	04-12618	SDRAM DIMM Memory Module – 168 Pin
6	04-12621	Audio Amp Assembly
7	20-10578	Power Supply
8	01-15195	Power Supply Switch Cover
9	03-7601-4	Button Switch - Red
10	H-22978	Audio/Reset Cable
11	01-15165	Computer Case Slot Cover
12	02-5477	Bushing

Associated Parts

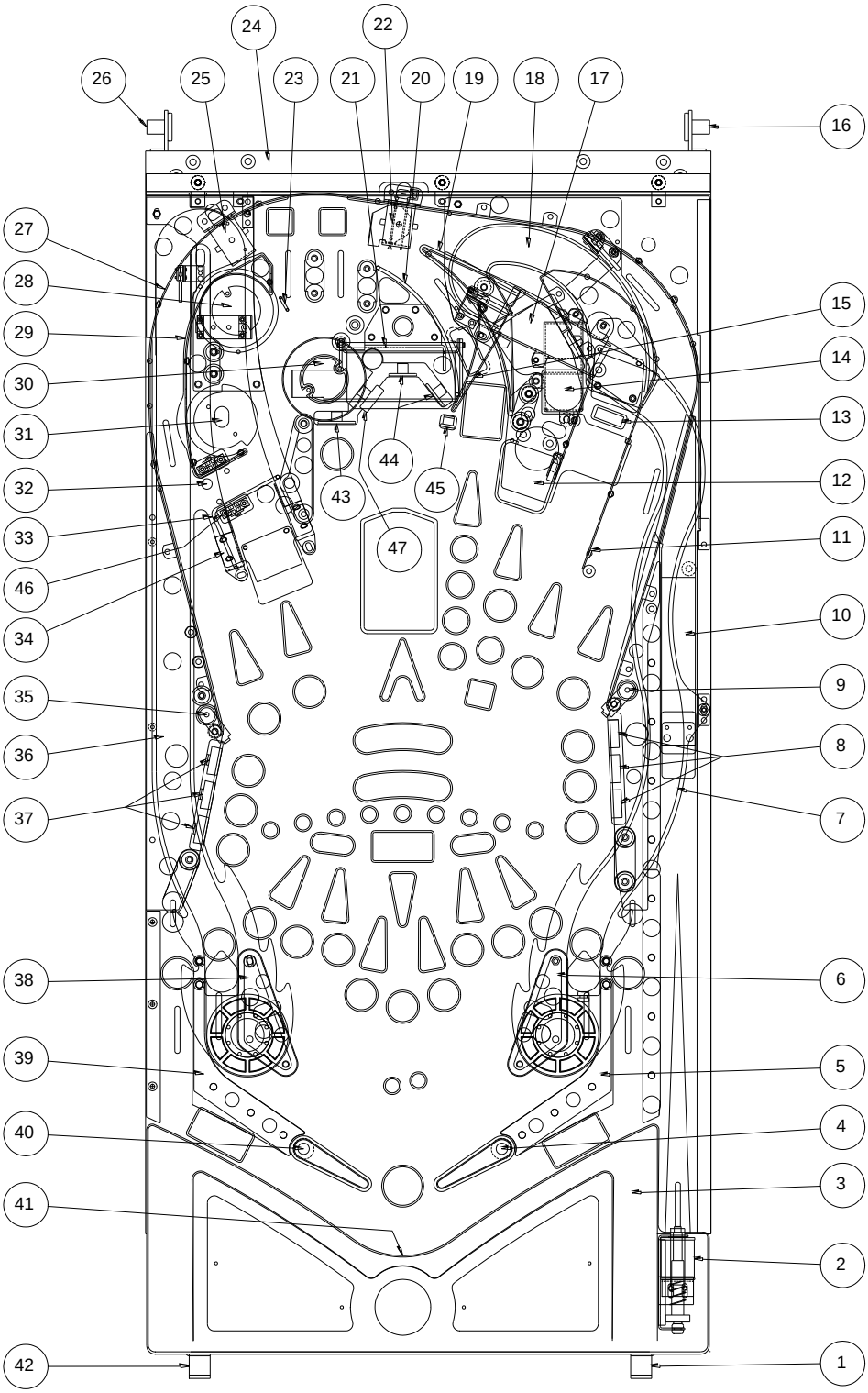
Item	Part Number	Description
13	A-22994-50070	Prism Card Sandwich Assembly
13a	A-23171-50070	Programmed Prism Card
13b	A-22995-50070	Daughter Card w/ROM

50070-PL
Upper Playfield Ramps



Item	Part Number	Description
1	**A-23017**	Lower Right Ramp Assembly
2	**A-23013**	Left ramp Assembly
3	**A-23016**	Upper Right Ramp Assembly
4	**01-15166**	Bridge

50070-PL
Upper Playfield Parts



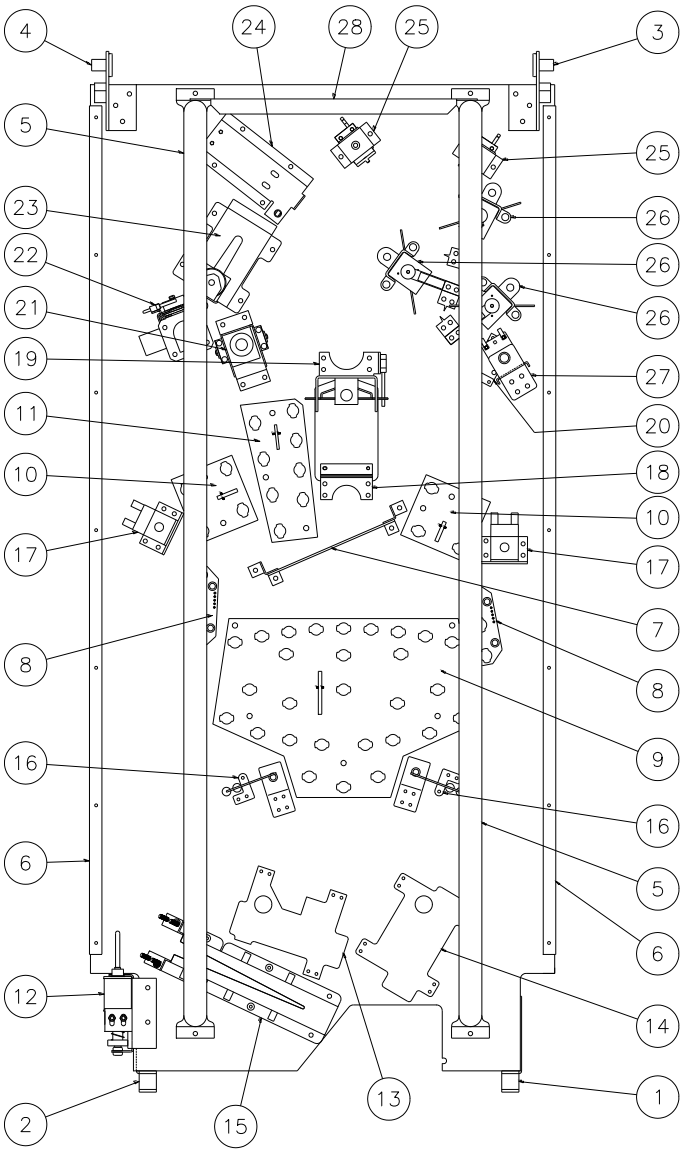
Upper Playfield Parts

Item #	Part Number	Description
1	01-14003-2	Right Playfield Hanger
2	A-22429-4	Auto-fire Kicker
	AE1-23-800	Coil
3	A-23006	Bottom Arch
	03-8544-1	Plastic Stud Grommet
	03-9883.1-2	Transparent Red Arch
4	20-10110-5	Flipper Bat & Shaft
	A-23004-R-2	Right Flipper Assembly
	FL1-11629	Coil
	SW-1A-194	Switch
5	A-22739-1	Right Flipper Ball Guide
6	A-22207-6	Slingshot Coil & Bracket
	AE1-26-1200	Coil
	A-17801	Slingshot Switch Assembly
	A-17811	Slingshot Kicker Assembly
7	A-23017	Right Ramp
	03-9481-1	Saucer Disk w/Grn Dome
8	A-18018-21	Green Standup Targets (3)
9	A-22960	Alien Mechanism
	AE1-26-1500	Coil
	23-6768	Alien Figurine (not shown)
	01-15124	Retaining Bracket
10	A-23136	Shooter Ramp
11	A-22969	Ball Guide #4
12	A-23156	Right Popper Assembly
	AE1-25-1000	Coil
	A-16908	LED Board
	A-16909	Photo Trans. Board
13	A-15211-1	Drop Target Assembly
	AE1-26-1200	Up Coil
	SM1-26-600	Down Coil
	5647-12693-31	Switch
14	A-20680-1	Right Lockup Popper
	AE1-23-800	Coil
	A-16908	LED Board
	A-16909	Photo Trans. Board
	04-12598	Popper Hood
15	A-22993	Diverter Mechanism
	FL1-22241	Flipper Coil-Brown
	A-22970	Diverter Paddle Assembly
16	04-12608.1-1	Right Playfield Foot
17	01-15166	Bridge
18	A-23016	Upper Right Ramp
	A-23140	Rollunder Gate
	5647-12693-36	Switch
	12-7436.1	Wireform
	5647-12693-21	Switch
19	04-12600	Ball Guide #3
20	12-7448	Wire Ball Guide #2
21	A-22992	Ball Catcher
	5647-12693-11	Switch
	03-10008	Ball Catcher Backstop
22	A-17796	Gate Actuator Assembly
	A-14406	Coil
	A-17797-2	Ball Gate Special Assy.
	12-6657.1-R	Wireform
23	12-7455	Pound-in Wireform
24	A-22852	Back Panel
25	A-17796	Gate Actuator Assembly
	A-14406	Coil
	A-19001	Ball Gate Special Assy.
	12-6657.1-L	Wireform
26	04-12608.1-2	Left Playfield Foot
27	04-12486.2	Ball Guide #1

Item #	Part Number	Description
28	A-22205-4	Jet Bumper Coil Assembly
	AE1-26-1200	Coil
	A-12030-3	Jet Bumper Switch Assy.
	B-9414-6	Jet Bumper Assembly
	03-8254-9	Red Cap
29	04-12599	Ball Guide #2
30	A-22205-4	Jet Bumper Coil Assembly
	AE1-26-1200	Coil
	A-12030-3	Jet Bumper Switch Assy.
	B-13123-2	Jet Bumper Assembly
31	A-22205-4	Jet Bumper Coil Assembly
	AE1-26-1200	Coil
	A-12030-3	Jet Bumper Switch Assy.
	B-9414-6	Jet Bumper Assembly
32	A-22977	Up/Down Post Assembly
	AE1-26-1500	Coil
	A-17986	Bell Armature
33	12-7447	Wire Ball Guide #1
34	A-23161	One-way Gate Assembly
	12-7454	Wireform
35	A-22960	Alien Mechanism
	AE1-26-1500	Coil
	23-6768	Alien Figurine (not shown)
	01-15124	Retaining Bracket
36	A-23013	Left Ramp
	5647-12693-21	Switch
	A-16908	LED Board
	A-16909	Photo Trans. Board
37	A-21576-21	Green Standup Targets (3)
38	A-22207-6	Slingshot Coil & Bracket
	AE1-26-1200	Coil
	A-17801	Slingshot Switch Assembly
	A-17811	Slingshot Kicker Assembly
39	A-22739-2	Left Flipper Ball Guide
40	20-10110-5	Flipper Bat & Shaft
	A-23001-L-2	Right Flipper Assembly
	FL1-11629	Coil
	SW-1A-194	Switch
41	04-12477.1	Bottom Arch Handle
42	01-14003-2	Left Playfield Hanger
43	A-18019-11	Green Standup Target
44	A-18060-4	Red Standup Targets (2)
45	A-18530-4	Red Standup Target
46	A-23012	Opto Bracket
	A-16908	LED Board
	A-16909	Photo Trans. Board
47	A-20846-4	Red Standup Target

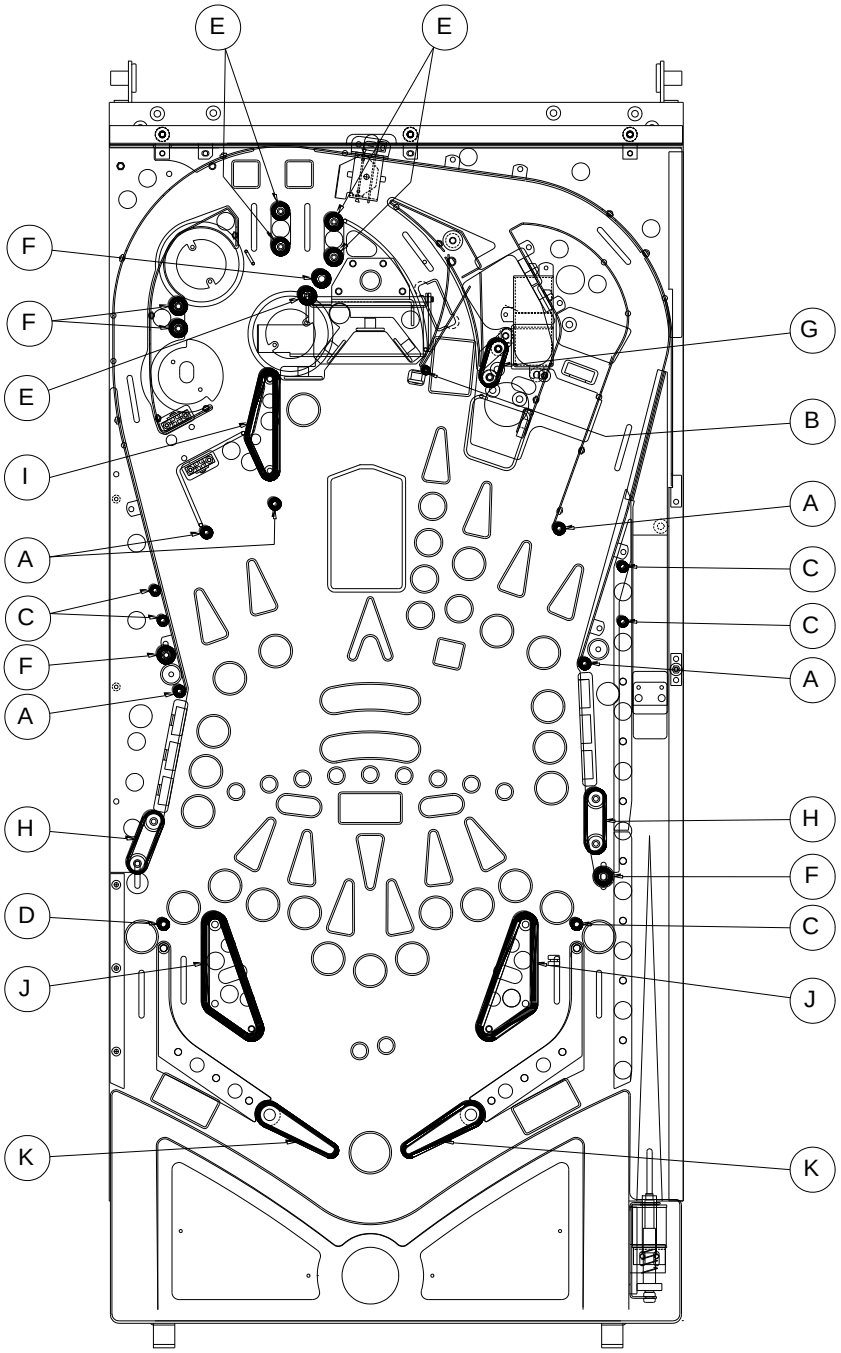
Not Shown:	
A-23167	Bottom Arch Support Bracket
01-15201.1	Playfield Side Rails (2)
03-8633	Level Mount
04-12739.1	Tinted Playfield Glass w/Decal
20-6500	1-1/16" Steel Ball (4)
20-9691	Level

50070-PL
Lower Playfield Parts



Item	Part Number	Description
1	**01-14003-1**	Playfield Hanging Bracket - Right
2	**01-14003-2**	Playfield Hanging Bracket - Left
3	**04-12608-1**	Playfield Slide Foot – Right
4	**04-12608-2**	Playfield slide Foot – Left
5	**01-15133.2**	Playfield Support Bracket
6	**01-15201.1**	Playfield Side Rails
7	**A-20246**	10 Opto PCB Assembly w/Brackets
8	**A21808-1**	3-Lamp PCB Assembly w/Spacer
9	**A-22987**	28-Lamp PCB Assembly w/Feet
10	**A-22983**	4-Lamp PCB Assembly w/Feet
11	**A-22986**	8-Lamp PCB Assembly w/Feet
12	**A-22429-4**	Auto-Fire Assembly
13	**A-23004-R-2**	Flipper Assembly – Right
14	**A-23001-L-2**	Flipper Assembly – Left
15	**A-19963-4**	Ball Trough Assembly
16	**A-17811**	Kicker Assembly
	A-22206-6	Coil & Bracket Assembly
	10-128	Spring
17	**A-22960**	Martian Mechanism Assembly
18	**A-22988**	Skip Ramp Flap Assembly
	01-15140	Flap Assembly Mounting Bracket
19	**A-22989**	Skip Ramp Assembly
20	**A-23012**	Opto Bracket Assembly
21	**A-23156**	Popper Assembly
22	**A-15211-1**	1 Bank Drop Target Assembly
23	**A-20680-1**	3 Ball Popper Assembly
24	**A-22993**	Diverter Mechanism Assembly
25	**A-17796**	Ball Gate Actuator Assembly
26	**A-22205-4**	Jet Coil & Bracket Assembly
27	**A-22977**	Up/Down post Assembly
28	**01-15125.4**	Playfield Cable Bracket

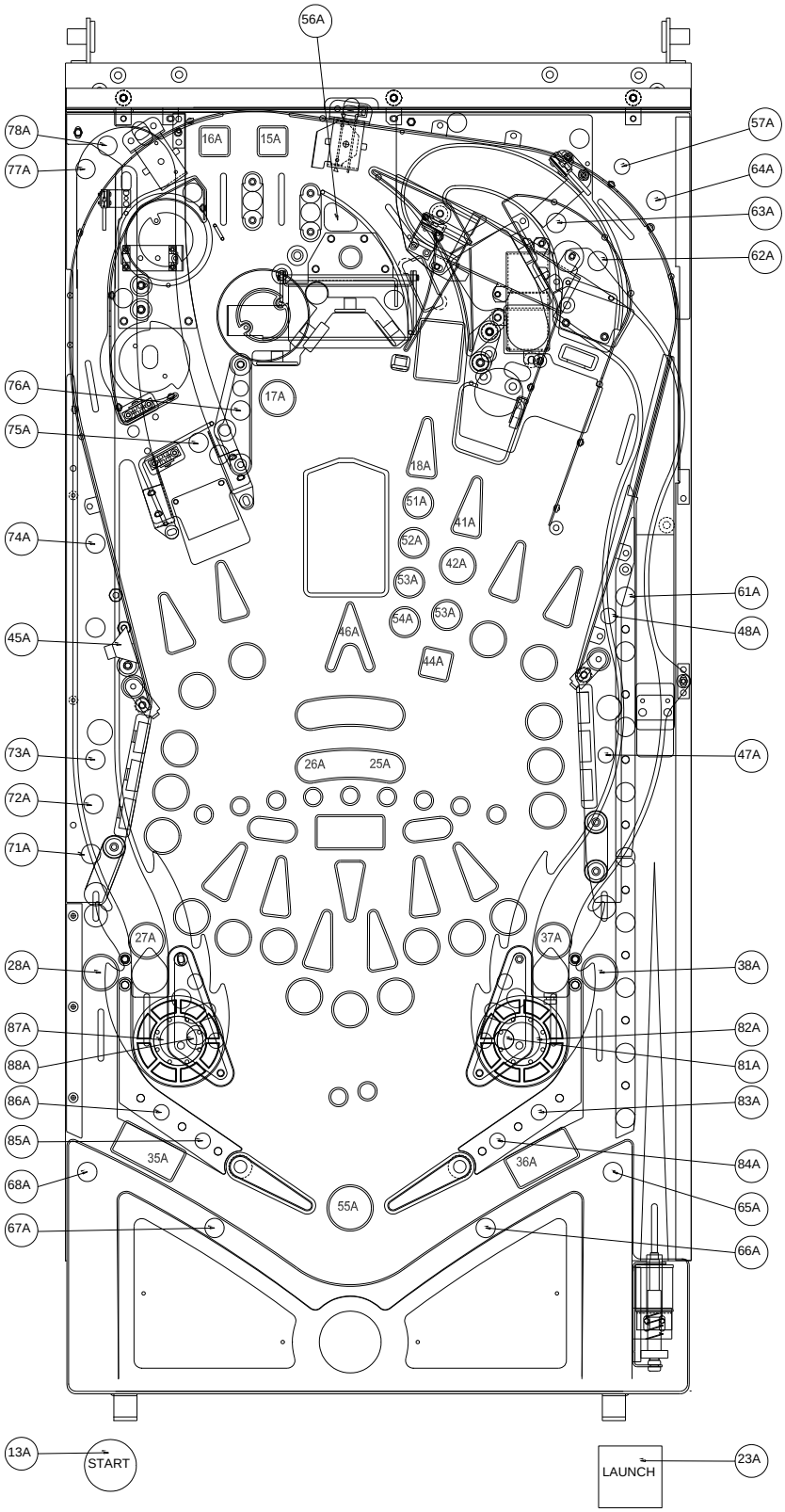
50070-PL Rubber Parts



Item	Part Number	Description	Qty.	Item	Part Number	Description	Qty.
A	**23-6552**	Yellow Rubber Bumper Sleeve	5	**G**	**23-6694-6**	1” Black Rubber Ring	1
B	**23-6556**	Black Rubber Bumper Sleeve	1	**H**	**23-6694-7**	1-1/4” Black Rubber Ring	2
C	**23-6641**	Black Rubber Bumper	5	**I**	**23-6694-9**	2” Black Rubber Ring	1
D	**23-6694-1**	3/32” Black Rubber Grommet	1	**J**	**23-6694-10**	2-1/2” Black Rubber Ring	2
E	**23-6694-3**	5/16” Black Rubber Ring	5	**K**	**23-6695**	Black Flipper Rubber Ring	2
F	**23-6694-5**	3/4” Black Rubber Ring	5				

50070-PL
Lamp Locations (Matrix A)

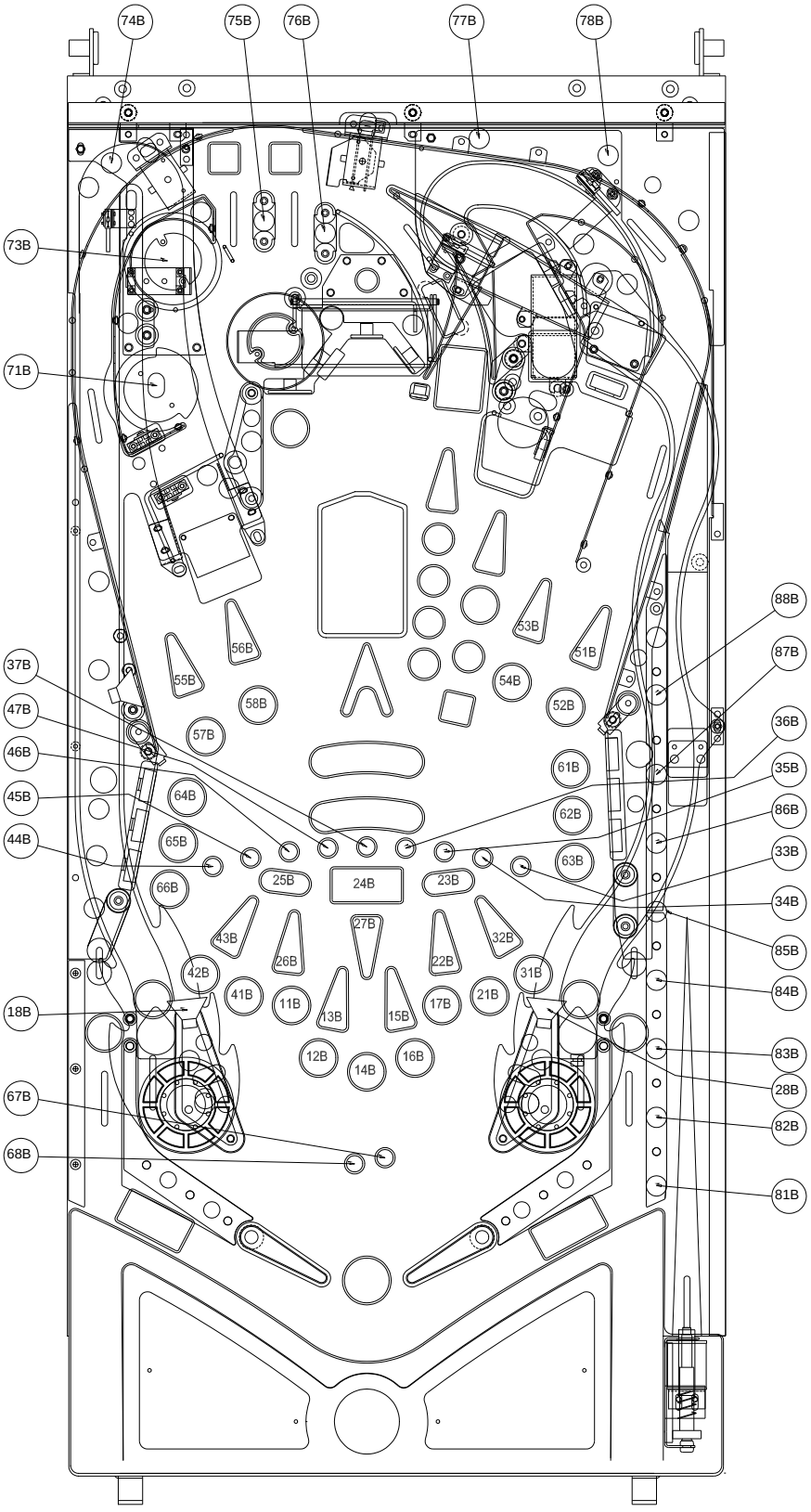
Note:
Lamp, item number 24A not shown.



Lamp Matrix A Locations

Item Number	Lamp Assembly Part Number	Bulb Type	Bulb Part Number	Socket Part Number	Description
11A	Not Used	Not Used	Not Used	Not Used	NOT USED
12A	Not Used	Not Used	Not Used	Not Used	NOT USED
13A	20-10696	Not Sold Separate	Not Sold Separate	Not Sold Separate	START BUTTON
14A	Not Used	Not Used	Not Used	Not Used	NOT USED
15A	A-17835	#44	24-6549	Not Sold Separate	RIGHT TOP LANE
16A	A-17807	#44	24-6549	Not Sold Separate	LEFT TOP LANE
17A	A-17807	#44	24-6459	Not Sold Separate	MARTIAN TARGET 4 (CENTER)
18A	A-17835	#44	24-6549	Not Sold Separate	CENTER LOOP ARROW
21A	Not Used	Not Used	Not Used	Not Used	TICKETS LOW
22A	Not Used	Not Used	Not Used	Not Used	NOT USED
23A	20-10692-1	Not Sold Separate	Not Sold Separate	Not Sold Separate	LAUNCH BUTTON
24A	Not Sold Separate	Not Sold Separate	Not Sold Separate	Not Sold Separate	COIN DOOR ILLUMINATION
25A	A-17807	#44	24-6549	Not Sold Separate	MOTHERSHIP MULTIBALL (RIGHT)
26A	A-17807	#44	24-6549	Not Sold Separate	MOTHERSHIP MULTIBALL (LEFT)
27A	A-17835	#44	24-6549	Not Sold Separate	LEFT RETURN LANE
28A	A-17807	#44	24-6549	Not Sold Separate	LEFT OUTLANE
31A	Not Used	Not Used	Not Used	Not Used	NOT USED
32A	Not Used	Not Used	Not Used	Not Used	NOT USED
33A	Not Used	Not Used	Not Used	Not Used	NOT USED
34A	Not Used	Not Used	Not Used	Not Used	NOT USED
35A	A-17807	#44	24-6549	Not Sold Separate	LEFT DRAIN TO TROUGH
36A	A-17807	#44	24-6549	Not Sold Separate	RIGHT DRAIN TO TROUGH
37A	A-17835	#44	24-6549	Not Sold Separate	RIGHT RETURN LANE
38A	A-17835	#44	24-6549	Not Sold Separate	RIGHT OUTLANE
41A	04-12613	#555	24-8768	24-8767	RIGHT POPPER ARROW
42A	04-12613	#555	24-8768	24-8767	EXTRA BALL
43A	04-12613	#555	24-8768	24-8767	MARTIAN ATTACK
44A	04-12613	#555	24-8768	24-8767	STROKE OF LUCK
45A	04-11037-8	#555	24-8768	Not Sold Separate	LEFT SIDE SPOTLIGHT
46A	A-17807	#44	24-6549	Not Sold Separate	CENTER ARROW
47A	A-23163	#44	24-6549	Not Sold Separate	RIGHT MARTIAN (HIGH)
48A	A-23163	#44	24-6549	Not Sold Separate	RIGHT MARTIAN (LOW)
51A	04-12613	#555	24-8768	24-8767	MULTIBALL
52A	04-12613	#555	24-8768	24-8767	CAPTURE 2
53A	04-12613	#555	24-8768	24-8767	CAPTURE 1
54A	04-12613	#555	24-8768	24-8767	CATURE ZONE ACTIVE
55A	A-17807	#44	24-6549	Not Sold Separate	SHOOT AGAIN
56A	A-23163	#44	24-6549	Not Sold Separate	BEHIND CENTER TARGETS
57A	A-23163	#44	24-6549	Not Sold Separate	UPPER RIGHT CORNER (MIDDLE)
58A	Not Used	Not Used	Not Used	Not Used	NOT USED
61A	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 9 (TOP)
62A	A-23163	#44	24-6549	Not Sold Separate	UNDER RIGHT RAMP (LOW)
63A	A-23163	#44	24-6549	Not Sold Separate	UNDER RIGHT RAMP (HIGH)
64A	A-23163	#44	24-6549	Not Sold Separate	UPPER RIGHT CORNER (LOW)
65A	A-23163	#44	24-6549	Not Sold Separate	RIGHT ARCH (RIGHT)
66A	A-23164	#44	24-6549	Not Sold Separate	RIGHT ARCH (LEFT)
67A	A-23163	#44	24-6549	Not Sold Separate	LEFT ARCH (RIGHT)
68A	A-23163	#44	24-6549	Not Sold Separate	LEFT ARCH (LEFT)
71A	A-23163	#44	24-6549	Not Sold Separate	LEFT SIDE 1 (BOTTOM)
72A	A-23163	#44	24-6549	Not Sold Separate	LEFT SIDE 2
73A	A-23163	#44	24-6549	Not Sold Separate	LEFT SIDE 3
74A	A-23163	#44	24-6549	Not Sold Separate	LEFT SIDE 4 (TOP)
75A	A-23163	#44	24-6549	Not Sold Separate	UNDER LEFT RAMP (BOTTOM)
76A	A-23163	#44	24-6549	Not Sold Separate	UNDER LEFT RAMP (TOP)
77A	A-23163	#44	24-6549	Not Sold Separate	BETWEEN LEFT & BOTTOM JETS
78A	A-23163	#44	24-6549	Not Sold Separate	UPPER LEFT CORNER
81A	A-23163	#44	24-6549	Not Sold Separate	RIGHT SLINGSHOT (BOTTOM)
82A	A-23163	#44	24-6549	Not Sold Separate	RIGHT SLINGSHOT (SAUCER)
83A	A-23163	#44	24-6549	Not Sold Separate	RIGHT RETURN LANE (RIGHT)
84A	A-23163	#44	24-6549	Not Sold Separate	RIGHT RETURN LANE (LEFT)
85A	A-23163	#44	24-6549	Not Sold Separate	LEFT RETURN LANE (RIGHT)
86A	A-23163	#44	24-6549	Not Sold Separate	LEFT RETURN LANE (LEFT)
87A	A-23163	#44	24-6549	Not Sold Separate	LEFT SLINGSHOT (SAUCER)
88A	A-23163	#44	24-6549	Not Sold Separate	LEFT SLINGSHOT (BOTTOM)

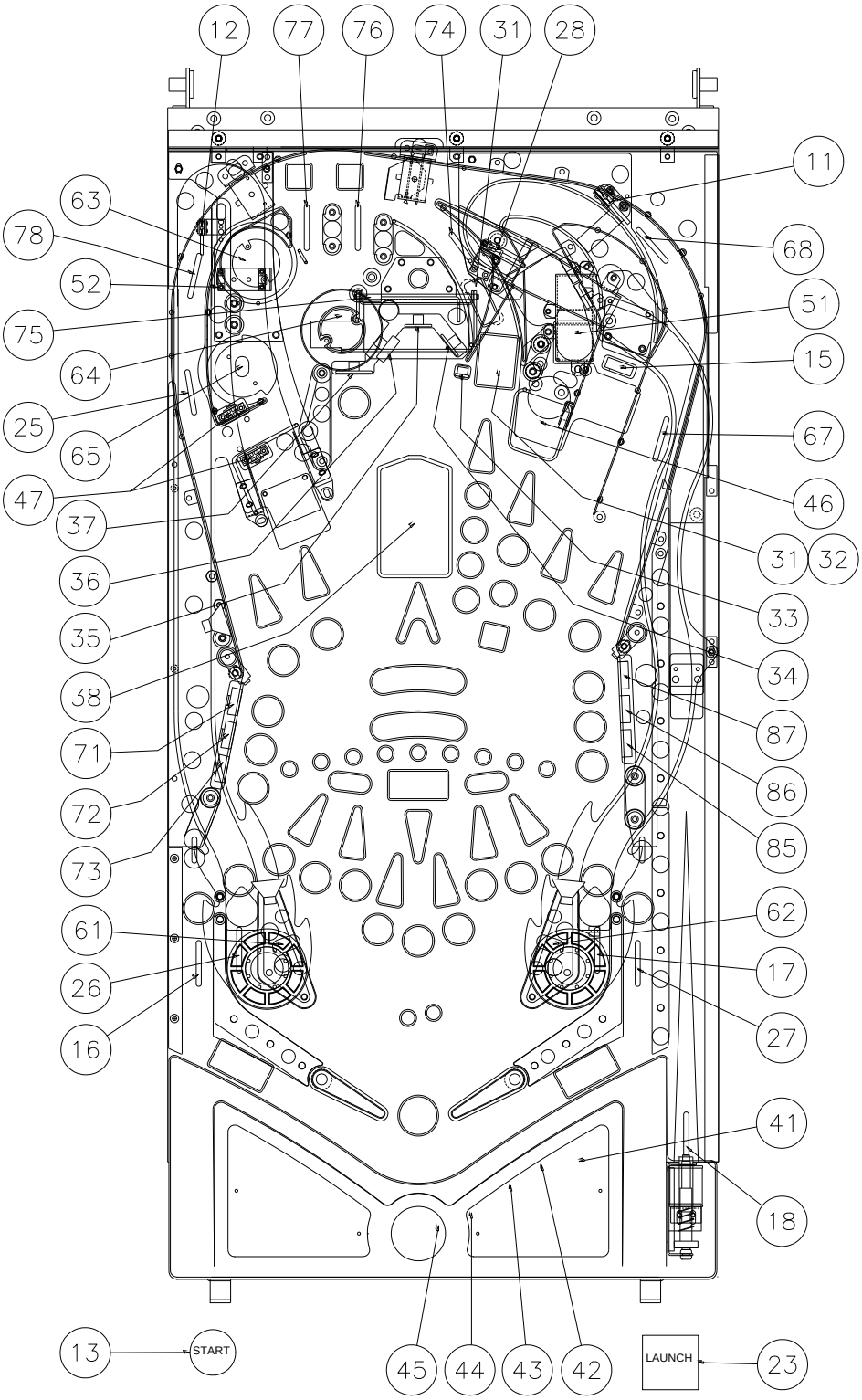
50070-PL
Lamp Locations (Matrix B)



Lamp Matrix B Locations

Item Number	Lamp Assembly Part Number	Bulb Type	Bulb Part Number	Socket Part Number	Description
11B	04-12614	#555	24-8768	24-8767	SECRET WEAPON
12B	04-12614	#555	24-8768	24-8767	TOWER STRUGGLE
13B	04-12614	#555	24-8768	24-8767	CENTER SAUCER BEAM (LEFT)
14B	04-12614	#555	24-8768	24-8767	QUESTION MARK
15B	04-12614	#555	24-8768	24-8767	CENTER SAUCER BEAM (RIGHT)
16B	04-12614	#555	24-8768	24-8767	DRIVE-IN DEMOLITION
17B	04-12614	#555	24-8768	24-8767	PARIS IN PERIL
18B	04-12860-2	#555	24-8768	Not Sold Separate	LEFT SLINGSHOT SPOTLIGHT
21B	04-12614	#555	24-8768	24-8767	BIG-O-BEAM
22B	04-12614	#555	24-8768	24-8767	RIGHT SAUCER BEAM (LEFT)
23B	04-12614	#555	24-8768	24-8767	WEAPONS
24B	04-12614	#555	24-8768	24-8767	SAUCER
25B	04-12614	#555	24-8768	24-8767	FUEL
26B	04-12614	#555	24-8768	24-8767	LEFT SAUCER BEAM (RIGHT)
27B	04-12614	#555	24-8768	24-8767	CENTER SAUCER BEAM (CENTER)
28B	04-12860-2	#555	24-8768	Not Sold Separate	RIGHT SLINGSHOT SPOTLIGHT
31B	04-12614	#555	24-8768	24-8767	MARS KNEADS WOMEN
32B	04-12614	#555	24-8768	24-8767	RIGHT SAUCER BEAM (RIGHT)
33B	04-12614	#555	24-8768	24-8767	SAUCER RIM 9 (RIGHT)
34B	04-12614	#555	24-8768	24-8767	SAUCER RIM 8
35B	04-12614	#555	24-8768	24-8767	SAUCER RIM 7
36B	04-12614	#555	24-8768	24-8767	SAUCER RIM 6
37B	04-12614	#555	24-8768	24-8767	SAUCER RIM 5
38B	Not Used	Not Used	Not Used	Not Used	NOT USED
41B	04-12614	#555	24-8768	24-8767	MARTIAN HAPPY HOUR
42B	04-12614	#555	24-8768	24-8767	ALIEN ABDUCTION
43B	04-12614	#555	24-8768	24-8767	LEFT SAUCER BEAM (LEFT)
44B	04-12614	#555	24-8768	24-8767	SAUCER RIM 1 (LEFT)
45B	04-12614	#555	24-8768	24-8767	SAUCER RIM 2
46B	04-12614	#555	24-8768	24-8767	SAUCER RIM 3
47B	04-12614	#555	24-8768	24-8767	SAUCER RIM 4
48B	Not Used	Not Used	Not Used	Not Used	NOT USED
51B	04-12611	#555	24-8768	24-8767	RIGHT LOOP ARROW
52B	04-12611	#555	24-8768	24-8767	RIGHT LOOP CIRCLE
53B	04-12611	#555	24-8768	24-8767	RIGHT RAMP ARROW
54B	04-12611	#555	24-8768	24-8767	RIGHT RAMP CIRCLE
55B	04-12611	#555	24-8768	24-8767	LEFT LOOP ARROW
56B	04-12611	#555	24-8768	24-8767	LEFT RAMP ARROW
57B	04-12611	#555	24-8768	24-8767	LEFT LOOP CIRCLE
58B	04-12611	#555	24-8768	24-8767	LEFT RAMP CIRCLE
61B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 5 (RIGHT TOP)
62B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 6 (RIGHT MID.)
63B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 7 (RIGHT BOT.)
64B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 3 (LEFT TOP)
65B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 2 (LEFT MID.)
66B	A-21808	#555	24-8768	24-8767	MARTIAN TARGET 1 (LEFT BOT.)
67B	A-17835	#44	24-6549	Not Sold Separate	RIGHT MARTIAN EYE
68B	A-17835	#44	24-6549	Not Sold Separate	LEFT MARTIAN EYE
71B	-----	#555	24-8768	24-8776	BOTTOM JET BUMPER
72B	Not Used	Not Used	Not Used	Not Used	NOT USED
73B	-----	#555	24-8768	24-8776	LEFT JET BUMPER
74B	A-23163	#44	24-6549	Not Sold Separate	LEFT OF LEFT TOP LANE
75B	A-23163	#44	24-6549	Not Sold Separate	BETWEEN LEFT/RIGHT TOP LANES
76B	A-23163	#44	24-6549	Not Sold Separate	RIGHT OF RIGHT TOP LANES
77B	A-23163	#44	24-6549	Not Sold Separate	TOP OF CENTER LOOP
78B	A-23163	#44	24-6549	Not Sold Separate	UPPER RIGHT CORNER (HIGH)
81B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 1 (BOTTOM)
82B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 2
83B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 3
84B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 4
85B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 5
86B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 6
87B	A-23163	#44	24-6549	Not Sold Separate	SHOOTER LANE 7
88B	A-17807	#44	24-6549	Not Sold Separate	SHOOTER LANE 8

50070-PL
Playfield Switch Locations

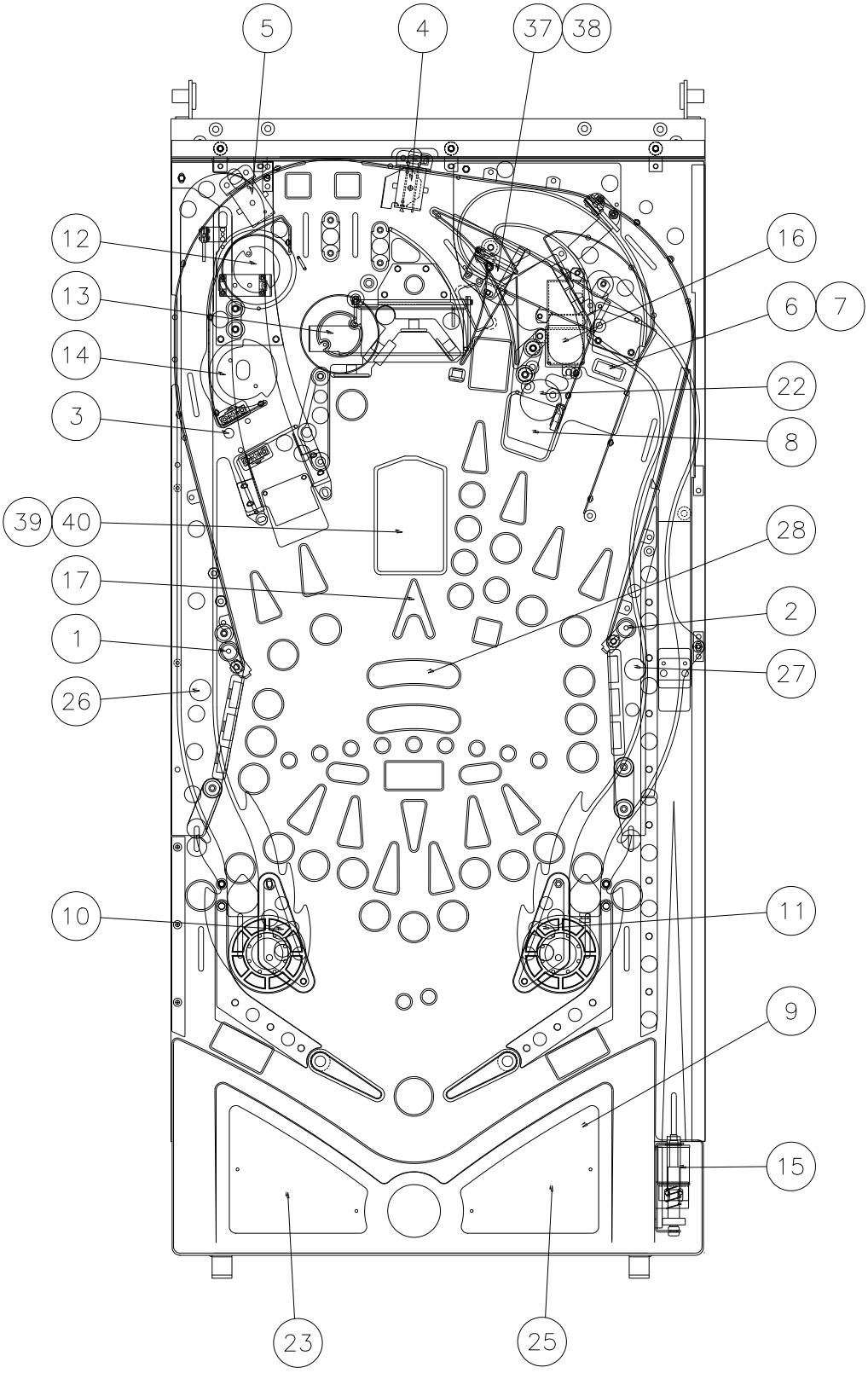


Switch Locations

Item Number	Switch Assembly Part Number ***OR*** Opto Assembly Part Number	Switch Part Number	Description
D13	-----	SW-1A-194	*LOWER RIGHT FLIPPER E.O.S.
D14	-----	SW-1A-194	*LOWER LEFT FLIPPER E.O.S.
D15	NOT USED	NOT USED	UPPER RIGHT FLIPPER E.O.S
D16	NOT USED	NOT USED	UPPER LEFT FLIPPER E.O.S.
D21	A-22984	-----	*LOWER RIGHT FLIPPER CABINET BUTTON
D22	A-22984	-----	*LOWER LEFT FLIPPER CABINET BUTTON
D23	A-22984	-----	RIGHT ACTION BUTTON
D24	A-22984	-----	LEFT ACTION BUTTON
11	A-23140	5647-12693-36	RIGHT RAMP ENTRANCE
12	-----	5647-12693-21	LEFT RAMP EXIT
13	20-10696	-----	START BUTTON
14	NOT USED	NOT USED	NOT USED
15	-----	5647-12693-31	DROP TARGET DOWN
16	A-17813-1	5647-12693-19	LEFT OUTLANE
17	A-17813	5647-12693-19	RIGHT RETURN LANE
18	A-18973	5647-12693-32	SHOOTER LANE
21	NOT USED	NOT USED	NOT USED
22	NOT USED	NOT USED	NOT USED
23	20-10692-1	-----	LAUNCH BUTTON
24	NOT USED	NOT USED	NOT USED
25	A-17813-1	5647-12693-19	LEFT LOOP (LOW)
26	A-17813	5647-12693-19	LEFT RETURN LANE
27	A-17813	5647-12693-19	RIGHT OUTLANE
28	-----	5647-12693-21	RIGHT RAMP EXIT
31	20-10293	-----	CENTER LOOP REED (BOTTOM)
32	20-10293	-----	CENTER LOOP REED (TOP)
33	A-18530-4	-----	CENTER TARGET 4
34	A-18060-4	-----	CENTER TARGET 3
35	A-18060-4	-----	CENTER TARGET 2
36	A-20846-4	-----	CENTER TARGET 1
37	A-18019-11	-----	MARTIAN TARGET 4 (CENTER)
38	-----	5647-12693-07	UP/DOWN RAMP UP
41	A-18617-1 (LED) A-18618-1 (PHOTO TRANS)	----- -----	TROUGH EJECT
42	A-18617-1 (LED) A-18618-1 (PHOTO TRANS)	----- -----	TROUGH BALL 1
43	A-18617-1 (LED) A-18618-1 (PHOTO TRANS)	----- -----	TROUGH BALL 2
44	A-18617-1 (LED) A-18618-1 (PHOTO TRANS)	----- -----	TROUGH BALL 3
45	A-18617-1 (LED) A-18618-1 (PHOTO TRANS)	----- -----	TROUGH BALL 4
46	A-16908 (LED) A-16909 (PHOTO TRANS)	----- -----	RIGHT POPPER
47	A-16908 (LED) A-16909 (PHOTO TRANS)	----- -----	JET EXIT
48	NOT USED	NOT USED	NOT USED
51	A-16908 (LED) A-16909 (PHOTO TRANS)	----- -----	RIGHT LOCKUP 1
52	A-16908 (LED) A-16909 (PHOTO TRANS)	----- -----	LEFT RAMP ENTRANCE
53	NOT USED	NOT USED	NOT USED
54	NOT USED	NOT USED	NOT USED
55	NOT USED	NOT USED	NOT USED
56	NOT USED	NOT USED	NOT USED
57	NOT USED	NOT USED	NOT USED
58	NOT USED	NOT USED	NOT USED
61	A-17801	A-17800 (KICK) A-17794 (SCORE)**	LEFT SLINGSHOT
62	A-17801	A-17800 (KICK) A-17794 (SCORE)**	RIGHT SLINGSHOT
63	B-12030-3	A-16443-1	LEFT JET BUMPER
64	B-12030-3	A-16443-1	RIGHT JET BUMPER
65	B-12030-3	A-16443-1	BOTTOM JET BUMPER
66	NOT USED	NOT USED	NOT USED
67	A-17813	5647-12693-19	RIGHT LOOP (LOW)
68	A-17813	5647-12693-19	RIGHT LOOP (HIGH)
71	A-21576-21	-----	MARTIAN TARGET 3 (LEFT TOP)
72	A-21576-21	-----	MARTIAN TARGET 2 (LEFT MIDDLE)
73	A-21576-21	-----	MARTIAN TARGET 1 (LEFT BOTTOM)
74	A-17813-1	5647-12693-19	CENTER LOOP ROLLOVER
75	-----	5647-12693-11	CENTER DEFLECTOR PANEL
76	A-17813	5647-12693-19	RIGHT TOP LANE
77	A-17813	5647-12693-19	LEFT TOP LANE
78	A-17813	5647-12693-19	LEFT LOOP (HIGH)
81	NOT USED	NOT USED	NOT USED
82	NOT USED	NOT USED	NOT USED
83	NOT USED	NOT USED	NOT USED
84	NOT USED	NOT USED	NOT USED
85	A-18018-21	-----	MARTIAN TARGET 7 (RIGHT BOTTOM)
86	A-18018-21	-----	MARTIAN TARGET 6 (RIGHT MIDDLE)
87	A-18018-21	-----	MARTIAN TARGET 5 (RIGHT TOP)
88	NOT USED	NOT USED	NOT USED

***NOT SHOWN. **SCORE SWITCHES HAVE DIODES ATTACHED.**

50070-PL
Solenoid/Flasher Locations



Solenoid/Flasher Locations

Item Number	Assembly Part Number	Coil or Flasher Part Number	Description
01	A-22960	AE1-26-1500	LEFT MARTIAN
02	A-22960	AE1-26-1500	RIGHT MARTIAN
03	A-22977	AE1-26-1500	JET EXIT POST
04	A-17796	A-14406	RIGHT GATE
05	A-17796	A-14406	LEFT GATE
06	A-15211-1	SM1-26-600	DROP TARGET DOWN
07	A-15211-1	AE1-26-1200	DROP TARGET UP
08	A-23156	AE1-25-1000	RIGHT POPPER
09	A-19963-4	AE1-26-1500	TROUGH EJECT
10	A-22207-6	AE1-26-1200	LEFT SLINGSHOT
11	A-22207-6	AE1-26-1200	RIGHT SLINGSHOT
12	A-22205-4	AE1-26-1200	LEFT JET BUMPER
13	A-22205-4	AE1-26-1200	RIGHT JET BUMPER
14	A-22205-4	AE1-26-1200	BOTTOM JET BUMPER
15	A-22429-4	AE1-23-800	AUTO PLUNGER
16	A-20680-1	AE1-23-800	RIGHT LOCKUP
17	A-17802	24-8802 (#906)	CENTER ARROW FLASHER
18	NOT USED	NOT USED	NOT USED
19	NOT USED	NOT USED	NOT USED
20	NOT USED	NOT USED	NOT USED
21	NOT USED	NOT USED	NOT USED
22	-----	24-8802 (#906)	RIGHT POPPER FLASHER
23	A-17983	24-8704 (#89)	LEFT ARCH FLASHER
24	NOT USED	NOT USED	NOT USED
25	A-17983	24-8704 (#89)	RIGHT ARCH FLASHER
26	A-17983	24-8704 (#89)	LEFT MARTIAN FLASHER
27	A-17983	24-8704 (#89)	RIGHT MARTIAN FLASHER
28	A-17802	24-8802 (#906)	ATTACK MARS FLASHER

Flipper Circuits

	Assembly Part Number	Coil Part Number	Description
33-34	A-23004-R-2	FL1-11629	LOWER RIGHT FLIPPER
35-36	A-23001-L-2	FL1-11629	LOWER LEFT FLIPPER
37-38	A-22993	FL1-22241	LOCK DIVERTER
39-40	A-22989	FL1-11753	UP/DOWN RAMP

LAMP MATRIX A

COLUMN / ROW	1A YEL-BRN J108-9 Q5	2A YEL-RED J108-10 Q9	3A YEL-ORG J108-11 Q13	4A YEL-BLK J108-12 Q17	5A YEL-GRN J108-13 Q21	6A YEL-BLU J108-14 Q25	7A YEL-VIO J108-15 Q29	8A YEL-GRY J108-16 Q33
1A BRN-BLK J108-1 Q3	NOT USED 11A	TICKETS LOW 21A	NOT USED 31A	RIGHT POPPER ARROW 41A	MULTIBALL 51A	SHOOTER LANE 9 (TOP) 61A	LEFT SIDE 1 (BOTTOM) 71A	RIGHT SLINGSHOT (BOTTOM) 81A
2A BRN-RED J108-2 Q7	NOT USED 12A	NOT USED 22A	NOT USED 32A	EXTRA BALL 42A	CAPTURE 2 52A	UNDER RIGHT RAMP (LOW) 62A	LEFT SIDE 2 72A	RIGHT SLINGSHOT (SAUCER) 82A
3A BRN-ORG J108-3 Q11	START BUTTON 13A	LAUNCH BUTTON 23A	NOT USED 33A	MARTIAN ATTACK 43A	CAPTURE 1 53A	UNDER RIGHT RAMP (HIGH) 63A	LEFT SIDE 3 73A	RIGHT RETURN LANE (RIGHT) 83A
4A BRN-YEL J108-4 Q15	NOT USED 14A	COIN DOOR ILLUMINA. 24A	NOT USED 34A	STROKE OF LUCK 44A	CAPTURE ZONE ACTIVE 54A	UPPER RT. CORNER (LOW) 64A	LEFT SIDE 4 (TOP) 74A	RIGHT RETURN LANE (LEFT) 84A
5A BRN-GRN J108-5 Q19	RIGHT TOP LANE 15A	MOTHERSHIP MULTIBALL (RIGHT) 25A	LEFT DRAIN TO TROUGH 35A	LEFT SIDE SPOTLIGHT 45A	SHOOT AGAIN 55A	RIGHT ARCH (RIGHT) 65A	UNDER LEFT RAMP (BOTTOM) 75A	LEFT RETURN LANE (RIGHT) 85A
6A BRN-BLU J108-6 Q23	LEFT TOP LANE 16A	MOTHERSHIP MULTIBALL (LEFT) 26A	RIGHT DRAIN TO TROUGH 36A	CENTER ARROW 46A	BEHIND CENTER TARGETS 56A	RIGHT ARCH (LEFT) 66A	UNDER LEFT RAMP (TOP) 76A	LEFT RETURN LANE (LEFT) 86A
7A BRN-VIO J108-7 Q27	MARTIAN TARGET 4 (CENTER) 17A	LEFT RETURN LANE 27A	RIGHT RETURN LANE 37A	RIGHT MARTIAN (HIGH) 47A	UPPER RT. CORNER (MIDDLE) 57A	LEFT ARCH (RIGHT) 67A	BETWEEN L/B JETS 77A	LEFT SLINGSHOT (SAUCER) 87A
8A BRN-GRY J108-8 Q31	CENTER LOOP ARROW 18A	LEFT OUTLANE 28A	RIGHT OUTLANE 38A	RIGHT MARTIAN (LOW) 48A	NOT USED 58A	LEFT ARCH (LEFT) 68A	UPPER LEFT CORNER 78A	LEFT SLINGSHOT (BOTTOM) 88A

LAMP MATRIX B

COLUMN / ROW	1B YEL-BRN J107-10 Q6	2B YEL-RED J107-11 Q10	3B YEL-ORG J107-12 Q14	4B YEL-BLK J107-13 Q18	5B YEL-GRN J107-14 Q22	6B YEL-BLU J107-15 Q26	7B YEL-VIO J107-16 Q30	8B YEL-GRY J107-17 Q34
1B RED-BRN J107-1 Q4	SECRET WEAPON 11B	BIG-O-BEAM 21B	MARS KNEADS WOMEN 31B	MARTIAN HARRY HOUR 41B	RIGHT LOOP ARROW 51B	MARTIAN TARGET 5 (RT. TOP) 61B	BOTTOM JET BUMPER 71B	SHOOTER LANE 1 (BOTTOM) 81B
2B RED-BLK J107-2 Q8	TOWER STRUGGLE 12B	RIGHT SAUCER BEAM (LEFT) 22B	RIGHT SAUCER BEAM (RT) 32B	ALIEN ABDUCTION 42B	RIGHT LOOP CIRCLE 52B	MARTIAN TARGET 6 (RT. MID.) 62B	NOT USED 72B	SHOOTER LANE 2 82B
3B RED-ORG J107-3 Q12	CENTER SAUCER BEAM (LEFT) 13B	WEAPONS 23B	SAUCER RIM 9 (RIGHT) 33B	LEFT SAUCER BEAM (LEFT) 43B	RIGHT RAMP ARROW 53B	MARTIAN TARGET 7 (RT. BOT.) 63B	LEFT JET BUMPER 73B	SHOOTER LANE 3 83B
4B RED-YEL J107-4 Q16	QUESTION MARK 14B	SAUCER 24B	SAUCER RIM 8 34B	SAUCER RIM 1 (LEFT) 44B	RIGHT RAMP CIRCLE 54B	MARTIAN TARGET 3 (LEFT TOP) 64B	LEFT OF LEFT TOP LANE 74B	SHOOTER LANE 4 84B
5B RED-GRN J107-5 Q20	CENTER SAUCER BEAM (RT.) 15B	FUEL 25B	SAUCER RIM 7 35B	SAUCER RIM 2 45B	LEFT LOOP ARROW 55B	MARTIAN TARGET 2 (LEFT MID.) 65B	BETWEEN L/R TOP LANES 75B	SHOOTER LANE 5 85B
6B RED-BLU J107-6 Q24	DRIVE-IN DEMOLITION 16B	LEFT SAUCER BEAM (RT.) 26B	SAUCER RIM 6 36B	SAUCER RIM 3 46B	LEFT RAMP ARROW 56B	MARTIAN TARGET 1 (LEFT BOT.) 66B	RIGHT OF RIGHT TOP LANE 76B	SHOOTER LANE 6 86B
7B RED-VIO J107-7 Q28	PARIS IN PERIL 17B	CENTER SAUCER BEAM (CNTR) 27B	SAUCER RIM 5 37B	SAUCER RIM 4 47B	LEFT LOOP CIRCLE 57B	RIGHT MARTIAN EYE 67B	TOP OF CENTER LOOP 77B	SHOOTER LANE 7 87B
8B RED-GRY J107-8 Q32	LEFT SLINGSHOT SPOTLIGHT 18B	RIGHT SLINGSHOT SPOTLIGHT 28B	NOT USED 38B	NOT USED 48B	LEFT RAMP CIRCLE 58B	LEFT MARTIAN EYE 68B	UPPER RT. CORNER (HIGH) 78B	SHOOTER LANE 8 88B

SWITCH MATRIX

COLUMN / ROW	1 **GRN-BRN** J116-1 U45-18	2 **GRN-RED** J116-2 U45-17	3 **GRN-ORG** J116-3 U45-16	4 **GRN-WHT** J116-4 U45-15	5 **GRN-BLK** J116-5 U45-14	6 **GRN-BLU** J116-6 U45-13	7 **GRN-VIO** J116-7 U45-12	8 **GRN-GRY** J116-8 U45-11
1 **WHT-BRN** J116-12 U51-7	RIGHT RAMP ENTRANCE 11	NOT USED 21	CENTER LOOP REED (BOTTOM) 31	TROUGH JAM 41	RIGHT LOCKUP 1 51	LEFT SLINGSHOT 61	MARTIAN TARGET 3 (LEFT TOP) 71	NOT USED 81
2 **WHT-RED** J116-13 U51-5	LEFT RAMP EXIT 12	NOT USED 22	CENTER LOOP REED (TOP) 32	TROUGH BALL 1 42	LEFT RAMP ENTRANCE 52	RIGHT SLINGSHOT 62	MARTIAN TARGET 2 (LEFT MID.) 72	NOT USED 82
3 **WHT-ORG** J116-14 U51-9	START BUTTON 13	LAUNCH BUTTON 23	CENTER TARGET 4 33	TROUGH BALL 2 43	NOT USED 53	LEFT JET BUMPER 63	MARTIAN TARGET 1 (LEFT BOT.) 73	NOT USED 83
4 **WHT-YEL** J116-15 U51-11	NOT USED 14	NOT USED 24	CENTER TARGET 3 34	TROUGH BALL 3 44	NOT USED 54	RIGHT JET BUMPER 64	CENTER LOOP ROLLOVER 74	NOT USED 84
5 **WHT-GRN** J116-16 U57-7	DROP TARGET DOWN 15	LEFT LOOP (LOW) 25	CENTER TARGET 2 35	TROUGH BALL 4 45	NOT USED 55	BOTTOM JET BUMPER 65	CENTER DEFLECTOR PANEL 75	MARTIAN TARGET 7 (RT. BOT.) 85
6 **WHT-BLU** J116-17 U57-5	LEFT OUTLANE 16	LEFT RETURN LANE 26	CENTER TARGET 1 36	RIGHT POPPER 46	NOT USED 56	NOT USED 66	RIGHT TOP LANE 76	MARTIAN TARGET 6 (RT. MID.) 86
7 **WHT-VIO** J116-18 U57-9	RIGHT RETURN LANE 17	RIGHT OUTLANE 27	MARTIAN TARGET 4 (CENTER) 37	JET EXIT 47	NOT USED 57	RIGHT LOOP (LOW) 67	LEFT TOP LANE 77	MARTIAN TARGET 5 (RT. TOP) 87
8 **WHT-GRY** J116-19 U57-11	SHOOTER LANE 18	RIGHT RAMP EXIT 28	UP/DOWN RAMP UP 38	NOT USED 48	NOT USED 58	RIGHT LOOP (HIGH) 68	LEFT LOOP (HIGH) 78	NOT USED 88

= OPTO, TYPICALLY CLOSED

COIN SLOT SWITCHES

DIRECT GROUND	**ORN-BRN** J114-1 U43-7	**ORG-RED** J114-2 U43-5	**ORG-BLK** J114-3 U43-9	**ORG-YEL** J114-4 U43-11	**ORG-GRN** J114-5 U48-7	**ORG-BLU** J114-6 U48-5	**ORG-VIO** J114-8 U48-9	**ORG-GRY** J114-9 U48-11
BLACK J114-14	LEFT COIN SLOT D1	CENTER COIN SLOT D2	RIGHT COIN SLOT D3	4TH COIN OPTION D4	NOT USED D5	NOT USED D6	NOT USED D7	NOT USED D8

DIAGNOSTIC SWITCHES (E.O.S.)

DIRECT GROUND	**GRY-BLK** J114-10 U50-7	**GRY-ORG** J114-11 U50-5	**GRY-RED** J114-12 U50-9	**GRY-BRN** J114-13 U50-11
BLACK J114-14	'ESCAPE' BUTTON D9	'DOWN' BUTTON D10	'UP' BUTTON D11	'ENTER' BUTTON D12

PLAYFIELD FLIPPER SWITCHES

DIRECT GROUND	**BLK-GRN** J115-9 U56-7	**BLK-BLU** J115-10 U56-5	**BLK-VIO** J115-20 U56-9	**BLK-GRY** J115-21 U56-11
BLACK J115-22	LOWER RIGHT FLIPPER E.O.S. D13	LOWER LEFT FLIPPER E.O.S. D14	NOT USED D15	NOT USED D16

CABINET FLIPPER SWITCHES

DIRECT GROUND	**BLK-BRN** J113-1 U49-7	**BLK-RED** J113-2 U49-5	**BLK-ORG** J113-3 U49-9	**BLK-YEL** J113-4 U49-11	**BLK-GRN** J113-6 U55-7	**BLK-BLU** J113-7 U55-5	**BLK-VIO** J113-8 U55-9	**BLK-GRY** J113-9 U55-11
BLACK J113-10	SLAM TILT D17	COIN DOOR CLOSED D18	PLUMB BOB TILT D19	NOT USED D20	RIGHT FLIPPER BUTTON D21	LEFT FLIPPER BUTTON D22	RIGHT ACTION BUTTON D23	LEFT ACTION BUTTON D24

SOLENOID TABLE

Sol No.	Solenoid Function	Fuse	Power to Playfield	Power Wire Color	Drive Transistor	Drive to Playfield	Drive Wire Color	Solenoid Part Number or Flasher Type
01	LEFT MARTIAN	F101	J102-1	RED-BRN	Q59	J110-13	VIO-BRN	AE1-26-1500
02	RIGHT MARTIAN	F101	J102-1	RED-BRN	Q60	J110-14	VIO-RED	AE1-26-1500
03	JET EXIT POST	F101	J102-1	RED-BRN	Q61	J110-15	VIO-ORG	AE1-26-1500
04	RIGHT GATE	F101	J102-1	RED-BRN	Q62	J110-16	VIO-YEL	A-14406
05	LEFT GATE	F102	J102-2	RED-BLK	Q63	J110-17	VIO-GRN	A-14406
06	DROP TARGET DOWN	F102	J102-2	RED-BLK	Q64	J110-18	VIO-BLU	SM1-26-600
07	DROP TARGET UP	F102	J102-2	RED-BLK	Q65	J110-19	VIO-BLK	AE1-26-1200
08	RIGHT POPPER	F102	J102-2	RED-BLK	Q66	J110-20	VIO-GRY	AE1-25-1000
09	TROUGH EJECT	F103	J102-3	RED-ORG	Q51	J112-11	BRN-BLK	AE1-26-1500
10	LEFT SLINGSHOT	F103	J102-3	RED-ORG	Q52	J112-12	BRN-RED	AE1-26-1200
11	RIGHT SLINGSHOT	F103	J102-3	RED-ORG	Q53	J112-13	BRN-ORG	AE1-26-1200
12	LEFT JET BUMPER	F103	J102-3	RED-ORG	Q54	J112-14	BRN-YEL	AE1-26-1200
13	RIGHT JET BUMPER	F100	J102-7	RED-YEL	Q55	J112-15	BRN-GRN	AE1-26-1200
14	BOTTOM JET BUMPER	F100	J102-7	RED-YEL	Q56	J112-16	BRN-BLU	AE1-26-1200
15	AUTO PLUNGER	F100	J102-7	RED-YEL	Q57	J112-17	BRN-VIO	AE1-23-800
16	RIGHT LOCKUP	F100	J102-7	RED-YEL	Q58	J112-18	BRN-GRY	AE1-23-800
17	CENTER ARROW FLASHER	F109	J102-8	RED-WHT	Q43	J110-1	BLU-BRN	#906
18	NOT USED	F109			Q44	J110-2	BLU-RED	
19	NOT USED	F109			Q45	J110-3	BLU-ORG	
20	NOT USED	F109			Q46	J110-4	BLU-YEL	
21	NOT USED	F109			Q47	J110-5	BLU-GRN	
22	RIGHT POPPER FLASHER	F109	J102-8	RED-WHT	Q48	J110-6	BLU-BLK	#906
23	LEFT ARCH FLASHER	F109	J102-8	RED-WHT	Q49	J110-7	BLU-VIO	#89
24	NOT USED	F109			Q50	J110-8	BLU-GRY	
25	RIGHT ARCH FLASHER	F109	J102-8	RED-WHT	Q67	J112-9	BLK-BRN	#89
26	LEFT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q68	J112-10	BLK-RED	#89
27	RIGHT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q69	J112-19	BLK-ORG	#89
28	ATTACK MARS FLASHER	F109	J102-8	RED-WHT	Q70	J112-20	BLK-YEL	#906
	Flipper Circuits							
33	LOW RIGHT FLIPPER PWR				Q35	J112-1	YEL-GRN	
34	LOW RIGHT FLIPPER HOLD	F104	J103-1	RED-GRN	Q36	J112-2	ORG-GRN	FL1-11629
35	LOW LEFT FLIPPER PWR				Q37	J112-3	YEL-BLU	
36	LOW LEFT FLIPPER HOLD	F105	J103-2	RED-BLU	Q38	J112-4	ORG-BLU	FL1-11629
37	LOCK DIVERTER POWER				Q39	J112-5	YEL-VIO	
38	LOCK DIVERTER HOLD	F106	J103-3	RED-VIO	Q40	J112-6	ORG-VIO	FL1-22241
39	UP/DOWN RAMP POWER				Q41	J112-7	YEL-GRY	
40	UP/DOWN RAMP HOLD	F107	J103-4	RED-GRY	Q42	J112-8	ORG-GRY	FL1-11753

SECTION THREE

GAME WIRING

Schematics for standard PINBALL 2000 PC boards that are located in the cabinet and backbox are found in the PINBALL 2000 Schematic Manual. Schematics for PC boards attached to the playfield are found in this section.

LAMP MATRIX A

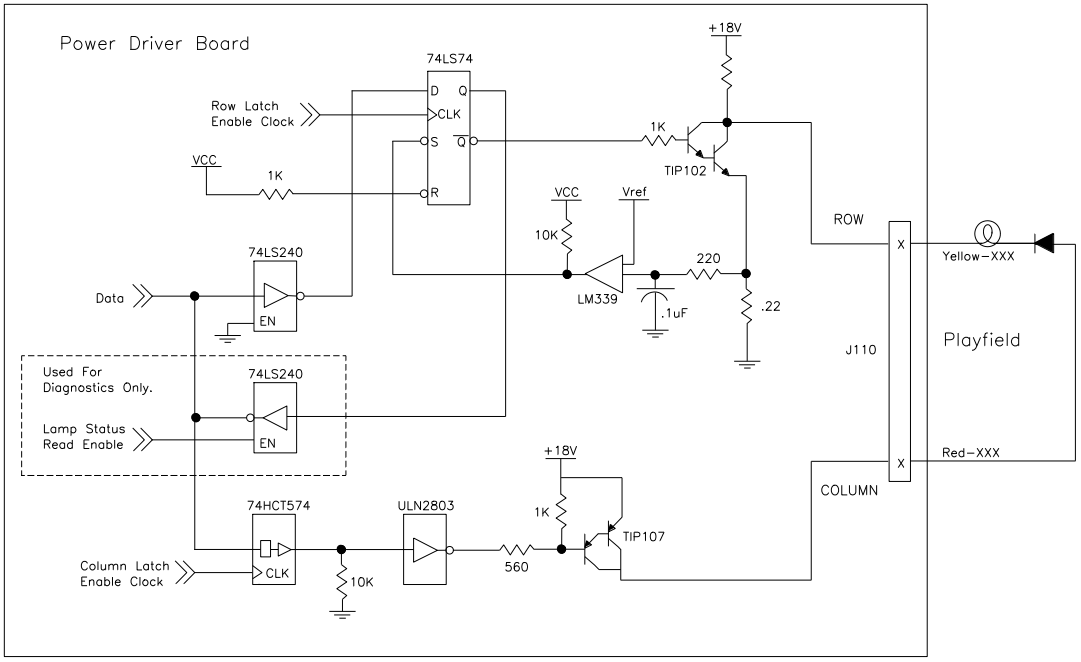
COLUMN / ROW	1A YEL-BRN J108-9 Q5	2A YEL-RED J108-10 Q9	3A YEL-ORG J108-11 Q13	4A YEL-BLK J108-12 Q17	5A YEL-GRN J108-13 Q21	6A YEL-BLU J108-14 Q25	7A YEL-VIO J108-15 Q29	8A YEL-GRY J108-16 Q33
1A BRN-BLK J108-1 Q3	NOT USED 11A	TICKETS LOW 21A	NOT USED 31A	RIGHT POPPER ARROW 41A	MULTIBALL 51A	SHOOTER LANE 9 (TOP) 61A	LEFT SIDE 1 (BOTTOM) 71A	RIGHT SLINGSHOT (BOTTOM) 81A
2A BRN-RED J108-2 Q7	NOT USED 12A	NOT USED 22A	NOT USED 32A	EXTRA BALL 42A	CAPTURE 2 52A	UNDER RIGHT RAMP (LOW) 62A	LEFT SIDE 2 72A	RIGHT SLINGSHOT (SAUCER) 82A
3A BRN-ORG J108-3 Q11	START BUTTON 13A	LAUNCH BUTTON 23A	NOT USED 33A	MARTIAN ATTACK 43A	CAPTURE 1 53A	UNDER RIGHT RAMP (HIGH) 63A	LEFT SIDE 3 73A	RIGHT RETURN LANE (RIGHT) 83A
4A BRN-YEL J108-4 Q15	NOT USED 14A	COIN DOOR ILLUMINA. 24A	NOT USED 34A	STROKE OF LUCK 44A	CAPTURE ZONE ACTIVE 54A	UPPER RT. CORNER (LOW) 64A	LEFT SIDE 4 (TOP) 74A	RIGHT RETURN LANE (LEFT) 84A
5A BRN-GRN J108-5 Q19	RIGHT TOP LANE 15A	MOTHERSHIP MULTIBALL (RIGHT) 25A	LEFT DRAIN TO TROUGH 35A	LEFT SIDE SPOTLIGHT 45A	SHOOT AGAIN 55A	RIGHT ARCH (RIGHT) 65A	UNDER LEFT RAMP (BOTTOM) 75A	LEFT RETURN LANE (RIGHT) 85A
6A BRN-BLU J108-6 Q23	LEFT TOP LANE 16A	MOTHERSHIP MULTIBALL (LEFT) 26A	RIGHT DRAIN TO TROUGH 36A	CENTER ARROW 46A	BEHIND CENTER TARGETS 56A	RIGHT ARCH (LEFT) 66A	UNDER LEFT RAMP (TOP) 76A	LEFT RETURN LANE (LEFT) 86A
7A BRN-VIO J108-7 Q27	MARTIAN TARGET 4 (CENTER) 17A	LEFT RETURN LANE 27A	RIGHT RETURN LANE 37A	RIGHT MARTIAN (HIGH) 47A	UPPER RT. CORNER (MIDDLE) 57A	LEFT ARCH (RIGHT) 67A	BETWEEN L/B JETS 77A	LEFT SLINGSHOT (SAUCER) 87A
8A BRN-GRY J108-8 Q31	CENTER LOOP ARROW 18A	LEFT OUTLANE 28A	RIGHT OUTLANE 38A	RIGHT MARTIAN (LOW) 48A	NOT USED 58A	LEFT ARCH (LEFT) 68A	UPPER LEFT CORNER 78A	LEFT SLINGSHOT (BOTTOM) 88A

J1XX = Power Driver Board

LAMP MATRIX B

COLUMN / ROW	1B YEL-BRN J107-10 Q6	2B YEL-RED J107-11 Q10	3B YEL-ORG J107-12 Q14	4B YEL-BLK J107-13 Q18	5B YEL-GRN J107-14 Q22	6B YEL-BLU J107-15 Q26	7B YEL-VIO J107-16 Q30	8B YEL-GRY J107-17 Q34
1B RED-BRN J107-1 Q4	SECRET WEAPON 11B	BIG-O-BEAM 21B	MARS KNEADS WOMEN 31B	MARTIAN HARRY HOUR 41B	RIGHT LOOP ARROW 51B	MARTIAN TARGET 5 (RT. TOP) 61B	BOTTOM JET BUMPER 71B	SHOOTER LANE 1 (BOTTOM) 81B
2B RED-BLK J107-2 Q8	TOWER STRUGGLE 12B	RIGHT SAUCER BEAM (LEFT) 22B	RIGHT SAUCER BEAM (RT) 32B	ALIEN ABDUCTION 42B	RIGHT LOOP CIRCLE 52B	MARTIAN TARGET 6 (RT. MID.) 62B	NOT USED 72B	SHOOTER LANE 2 82B
3B RED-ORG J107-3 Q12	CENTER SAUCER BEAM (LEFT) 13B	WEAPONS 23B	SAUCER RIM 9 (RIGHT) 33B	LEFT SAUCER BEAM (LEFT) 43B	RIGHT RAMP ARROW 53B	MARTIAN TARGET 7 (RT. BOT.) 63B	LEFT JET BUMPER 73B	SHOOTER LANE 3 83B
4B RED-YEL J107-4 Q16	QUESTION MARK 14B	SAUCER 24B	SAUCER RIM 8 34B	SAUCER RIM 1 (LEFT) 44B	RIGHT RAMP CIRCLE 54B	MARTIAN TARGET 3 (LEFT TOP) 64B	LEFT OF LEFT TOP LANE 74B	SHOOTER LANE 4 84B
5B RED-GRN J107-5 Q20	CENTER SAUCER BEAM (RT.) 15B	FUEL 25B	SAUCER RIM 7 35B	SAUCER RIM 2 45B	LEFT LOOP ARROW 55B	MARTIAN TARGET 2 (LEFT MID.) 65B	BETWEEN L/R TOP LANES 75B	SHOOTER LANE 5 85B
6B RED-BLU J107-6 Q24	DRIVE-IN DEMOLITION 16B	LEFT SAUCER BEAM (RT.) 26B	SAUCER RIM 6 36B	SAUCER RIM 3 46B	LEFT RAMP ARROW 56B	MARTIAN TARGET 1 (LEFT BOT.) 66B	RIGHT OF RIGHT TOP LANE 76B	SHOOTER LANE 6 86B
7B RED-VIO J107-7 Q28	PARIS IN PERIL 17B	CENTER SAUCER BEAM (CNTR) 27B	SAUCER RIM 5 37B	SAUCER RIM 4 47B	LEFT LOOP CIRCLE 57B	RIGHT MARTIAN EYE 67B	TOP OF CENTER LOOP 77B	SHOOTER LANE 7 87B
8B RED-GRY J107-8 Q32	LEFT SLINGSHOT SPOTLIGHT 18B	RIGHT SLINGSHOT SPOTLIGHT 28B	NOT USED 38B	NOT USED 48B	LEFT RAMP CIRCLE 58B	LEFT MARTIAN EYE 68B	UPPER RT. CORNER (HIGH) 78B	SHOOTER LANE 8 88B

LAMP MATRIX CIRCUIT



SWITCH MATRIX

COLUMN / ROW	1 GRN-BRN J116-1 U45-18	2 GRN-RED J116-2 U45-17	3 GRN-ORG J116-3 U45-16	4 GRN-WHT J116-4 U45-15	5 GRN-BLK J116-5 U45-14	6 GRN-BLU J116-6 U45-13	7 GRN-VIO J116-7 U45-12	8 GRN-GRY J116-8 U45-11
1 WHT-BRN J116-12 U51-7	RIGHT RAMP ENTRANCE 11	NOT USED 21	CENTER LOOP REED (BOTTOM) 31	TROUGH JAM 41	RIGHT LOCKUP 1 51	LEFT SLINGSHOT 61	MARTIAN TARGET 3 (LEFT TOP) 71	NOT USED 81
2 WHT-RED J116-13 U51-5	LEFT RAMP EXIT 12	NOT USED 22	CENTER LOOP REED (TOP) 32	TROUGH BALL 1 42	LEFT RAMP ENTRANCE 52	RIGHT SLINGSHOT 62	MARTIAN TARGET 2 (LEFT MID.) 72	NOT USED 82
3 WHT-ORG J116-14 U51-9	START BUTTON 13	LAUNCH BUTTON 23	CENTER TARGET 4 33	TROUGH BALL 2 43	NOT USED 53	LEFT JET BUMPER 63	MARTIAN TARGET 1 (LEFT BOT.) 73	NOT USED 83
4 WHT-YEL J116-15 U51-11	NOT USED 14	NOT USED 24	CENTER TARGET 3 34	TROUGH BALL 3 44	NOT USED 54	RIGHT JET BUMPER 64	CENTER LOOP ROLLOVER 74	NOT USED 84
5 WHT-GRN J116-16 U57-7	DROP TARGET DOWN 15	LEFT LOOP (LOW) 25	CENTER TARGET 2 35	TROUGH BALL 4 45	NOT USED 55	BOTTOM JET BUMPER 65	CENTER DEFLECTOR PANEL 75	MARTIAN TARGET 7 (RT. BOT.) 85
6 WHT-BLU J116-17 U57-5	LEFT OUTLANE 16	LEFT RETURN LANE 26	CENTER TARGET 1 36	RIGHT POPPER 46	NOT USED 56	NOT USED 66	RIGHT TOP LANE 76	MARTIAN TARGET 6 (RT. MID.) 86
7 WHT-VIO J116-18 U57-9	RIGHT RETURN LANE 17	RIGHT OUTLANE 27	MARTIAN TARGET 4 (CENTER) 37	JET EXIT 47	NOT USED 57	RIGHT LOOP (LOW) 67	LEFT TOP LANE 77	MARTIAN TARGET 5 (RT. TOP) 87
8 WHT-GRY J116-19 U57-11	SHOOTER LANE 18	RIGHT RAMP EXIT 28	UP/DOWN RAMP UP 38	NOT USED 48	NOT USED 58	RIGHT LOOP (HIGH) 68	LEFT LOOP (HIGH) 78	NOT USED 88

= OPTO, TYPICALLY CLOSED

COIN SLOT SWITCHES

DIRECT GROUND	ORN-BRN J114-1 U43-7	ORG-RED J114-2 U43-5	ORG-BLK J114-3 U43-9	ORG-YEL J114-4 U43-11	ORG-GRN J114-5 U48-7	ORG-BLU J114-6 U48-5	ORG-VIO J114-8 U48-9	ORG-GRY J114-9 U48-11
BLACK J114-14	LEFT COIN SLOT D1	CENTER COIN SLOT D2	RIGHT COIN SLOT D3	4TH COIN OPTION D4	NOT USED D5	NOT USED D6	NOT USED D7	NOT USED D8

DIAGNOSTIC SWITCHES

DIRECT GROUND	GRY-BLK J114-10 U50-7	GRY-ORG J114-11 U50-5	GRY-RED J114-12 U50-9	GRY-BRN J114-13 U50-11
BLACK J114-14	'ESCAPE' BUTTON D9	'DOWN' BUTTON D10	'UP' BUTTON D11	'ENTER' BUTTON D12

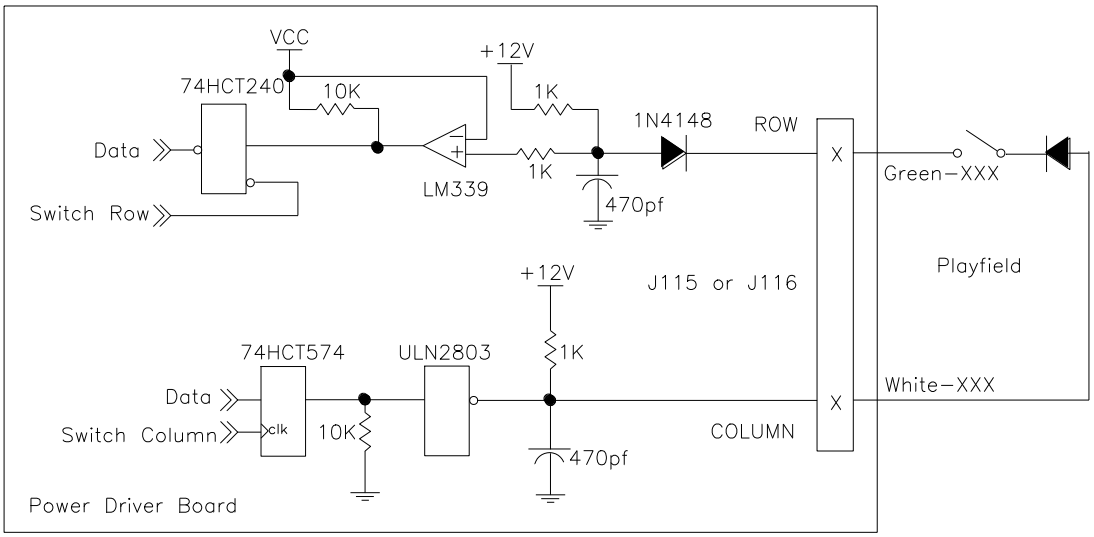
PLAYFIELD FLIPPER SWITCHES (E.O.S.)

DIRECT GROUND	BLK-GRN J115-9 U56-7	BLK-BLU J115-10 U56-5	BLK-VIO J115-20 U56-9	BLK-GRY J115-21 U56-11
BLACK J115-22	LOWER RIGHT FLIPPER E.O.S. D13	LOWER LEFT FLIPPER E.O.S. D14	NOT USED D15	NOT USED D16

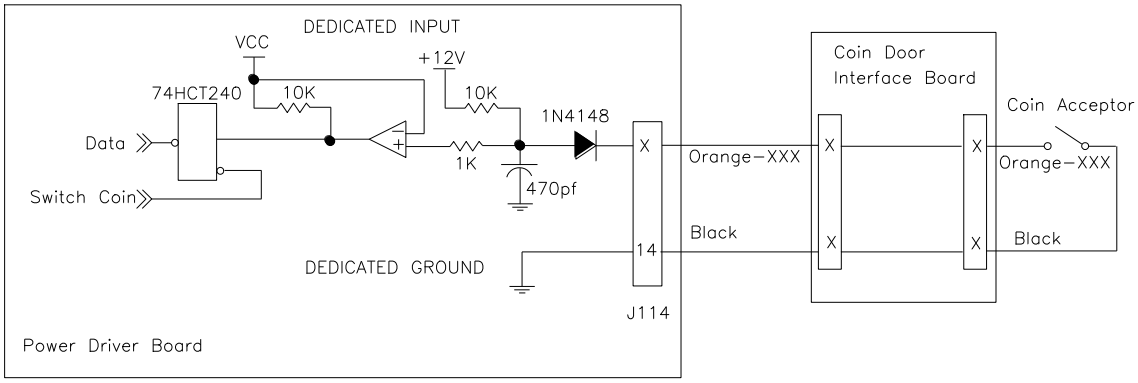
CABINET FLIPPER SWITCHES

DIRECT GROUND	BLK-BRN J113-1 U49-7	BLK-RED J113-2 U49-5	BLK-ORG J113-3 U49-9	BLK-YEL J113-4 U49-11	BLK-GRN J113-6 U55-7	BLK-BLU J113-7 U55-5	BLK-VIO J113-8 U55-9	BLK-GRY J113-9 U55-11
BLACK J113-10	SLAM TILT D17	COIN DOOR CLOSED D18	PLUMB BOB TILT D19	NOT USED D20	RIGHT FLIPPER BUTTON D21	LEFT FLIPPER BUTTON D22	RIGHT ACTION BUTTON D23	LEFT ACTION BUTTON D24

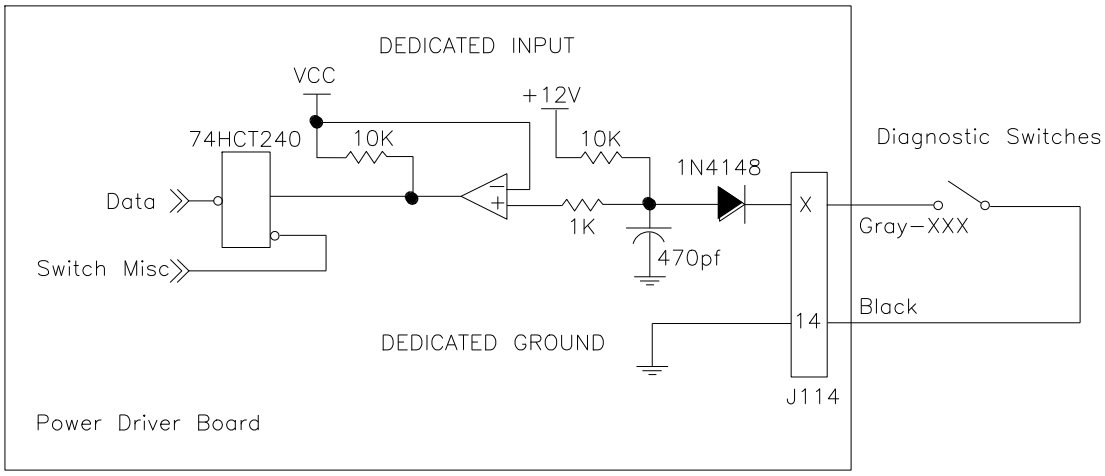
SWITCH MATRIX CIRCUIT



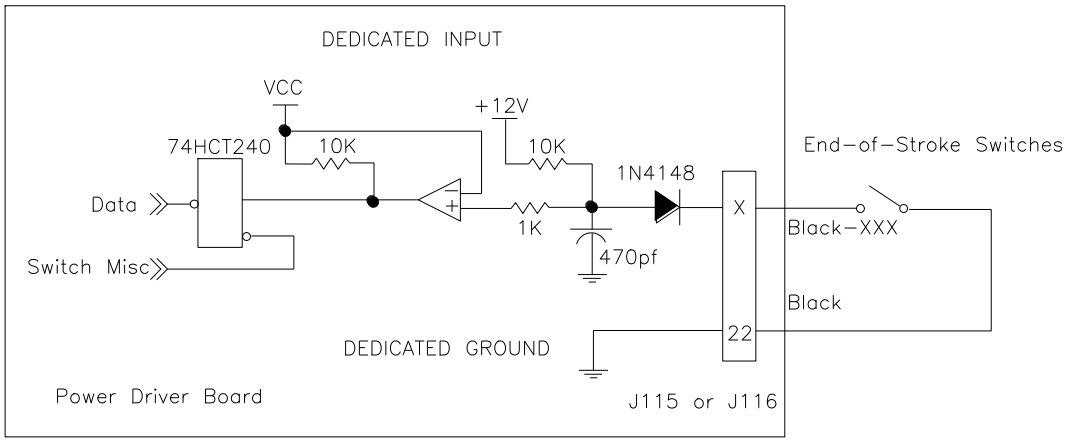
COIN SLOT SWITCH CIRCUIT



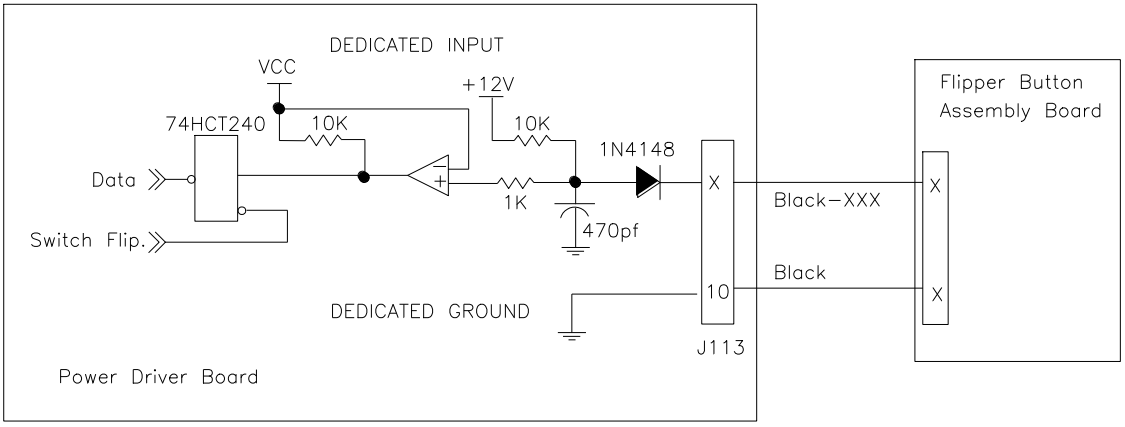
DIAGNOSTIC SWITCH CIRCUIT



PLAYFIELD FLIPPER SWITCH (E.O.S.) CIRCUIT



CABINET FLIPPER SWITCH CIRCUIT

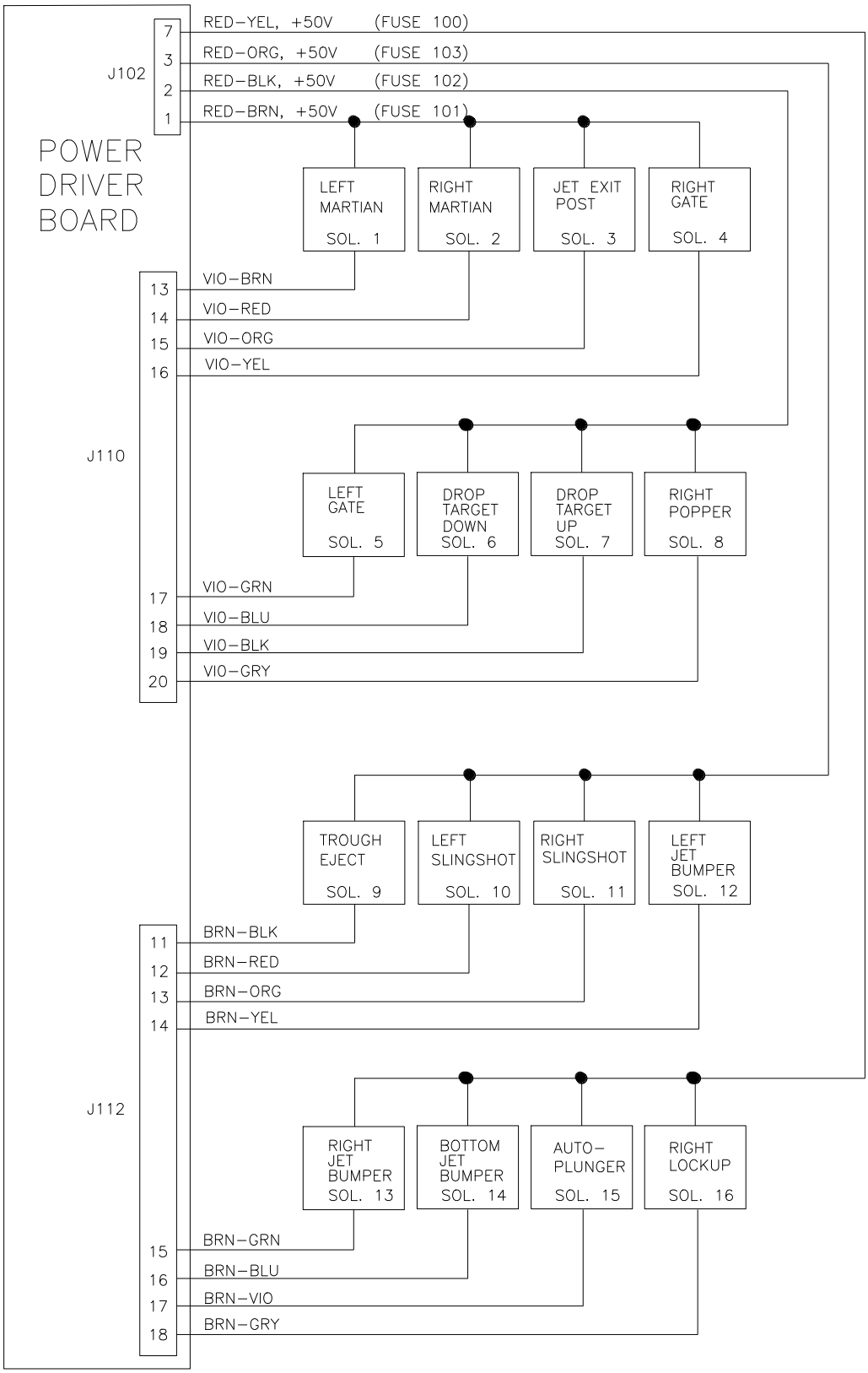


SOLENOID TABLE

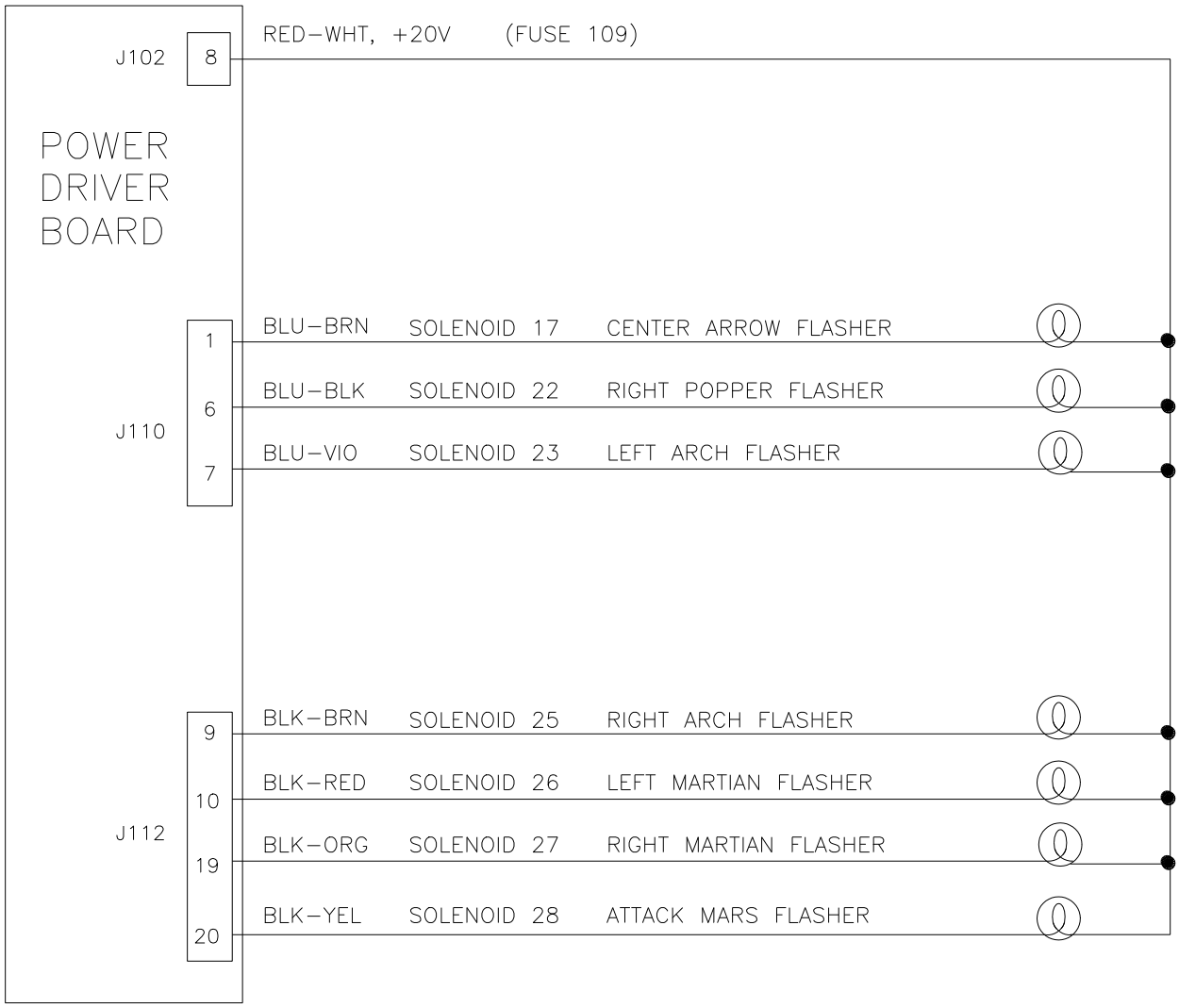
Sol No.	Solenoid Function	Fuse	Power to Playfield	Power Wire Color	Drive Transistor	Drive to Playfield	Drive Wire Color	Solenoid Part Number or Flasher Type
01	LEFT MARTIAN	F101	J102-1	RED-BRN	Q59	J110-13	VIO-BRN	AE1-26-1500
02	RIGHT MARTIAN	F101	J102-1	RED-BRN	Q60	J110-14	VIO-RED	AE1-26-1500
03	JET EXIT POST	F101	J102-1	RED-BRN	Q61	J110-15	VIO-ORG	AE1-26-1500
04	RIGHT GATE	F101	J102-1	RED-BRN	Q62	J110-16	VIO-YEL	A-14406
05	LEFT GATE	F102	J102-2	RED-BLK	Q63	J110-17	VIO-GRN	A-14406
06	DROP TARGET DOWN	F102	J102-2	RED-BLK	Q64	J110-18	VIO-BLU	SM1-26-600
07	DROP TARGET UP	F102	J102-2	RED-BLK	Q65	J110-19	VIO-BLK	AE1-26-1200
08	RIGHT POPPER	F102	J102-2	RED-BLK	Q66	J110-20	VIO-GRY	AE1-25-1000
09	TROUGH EJECT	F103	J102-3	RED-ORG	Q51	J112-11	BRN-BLK	AE1-26-1500
10	LEFT SLINGSHOT	F103	J102-3	RED-ORG	Q52	J112-12	BRN-RED	AE1-26-1200
11	RIGHT SLINGSHOT	F103	J102-3	RED-ORG	Q53	J112-13	BRN-ORG	AE1-26-1200
12	LEFT JET BUMPER	F103	J102-3	RED-ORG	Q54	J112-14	BRN-YEL	AE1-26-1200
13	RIGHT JET BUMPER	F100	J102-7	RED-YEL	Q55	J112-15	BRN-GRN	AE1-26-1200
14	BOTTOM JET BUMPER	F100	J102-7	RED-YEL	Q56	J112-16	BRN-BLU	AE1-26-1200
15	AUTO PLUNGER	F100	J102-7	RED-YEL	Q57	J112-17	BRN-VIO	AE1-23-800
16	RIGHT LOCKUP	F100	J102-7	RED-YEL	Q58	J112-18	BRN-GRY	AE1-23-800
17	CENTER ARROW FLASHER	F109	J102-8	RED-WHT	Q43	J110-1	BLU-BRN	#906
18	NOT USED	F109			Q44	J110-2	BLU-RED	
19	NOT USED	F109			Q45	J110-3	BLU-ORG	
20	NOT USED	F109			Q46	J110-4	BLU-YEL	
21	NOT USED	F109			Q47	J110-5	BLU-GRN	
22	RIGHT POPPER FLASHER	F109	J102-8	RED-WHT	Q48	J110-6	BLU-BLK	#906
23	LEFT ARCH FLASHER	F109	J102-8	RED-WHT	Q49	J110-7	BLU-VIO	#89
24	NOT USED	F109			Q50	J110-8	BLU-GRY	
25	RIGHT ARCH FLASHER	F109	J102-8	RED-WHT	Q67	J112-9	BLK-BRN	#89
26	LEFT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q68	J112-10	BLK-RED	#89
27	RIGHT MARTIAN FLASHER	F109	J102-8	RED-WHT	Q69	J112-19	BLK-ORG	#89
28	ATTACK MARS FLASHER	F109	J102-8	RED-WHT	Q70	J112-20	BLK-YEL	#906
	Flipper Circuits							
33	LOW RIGHT FLIPPER PWR				Q35	J112-1	YEL-GRN	
34	LOW RIGHT FLIPPER HOLD	F104	J103-1	RED-GRN	Q36	J112-2	ORG-GRN	FL1-11629
35	LOW LEFT FLIPPER PWR				Q37	J112-3	YEL-BLU	
36	LOW LEFT FLIPPER HOLD	F105	J103-2	RED-BLU	Q38	J112-4	ORG-BLU	FL1-11629
37	LOCK DIVERTER POWER				Q39	J112-5	YEL-VIO	
38	LOCK DIVERTER HOLD	F106	J103-3	RED-VIO	Q40	J112-6	ORG-VIO	FL1-22241
39	UP/DOWN RAMP POWER				Q41	J112-7	YEL-GRY	
40	UP/DOWN RAMP HOLD	F107	J103-4	RED-GRY	Q42	J112-8	ORG-GRY	FL1-11753

SOLENOID WIRING

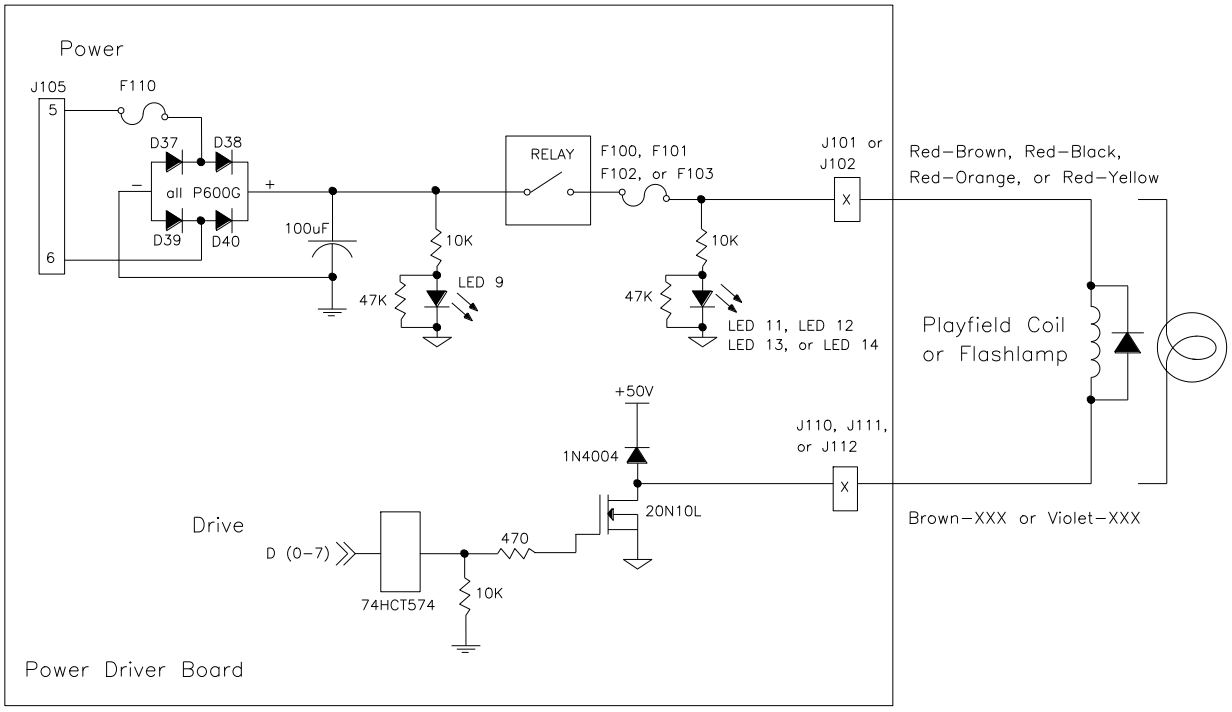
COILS



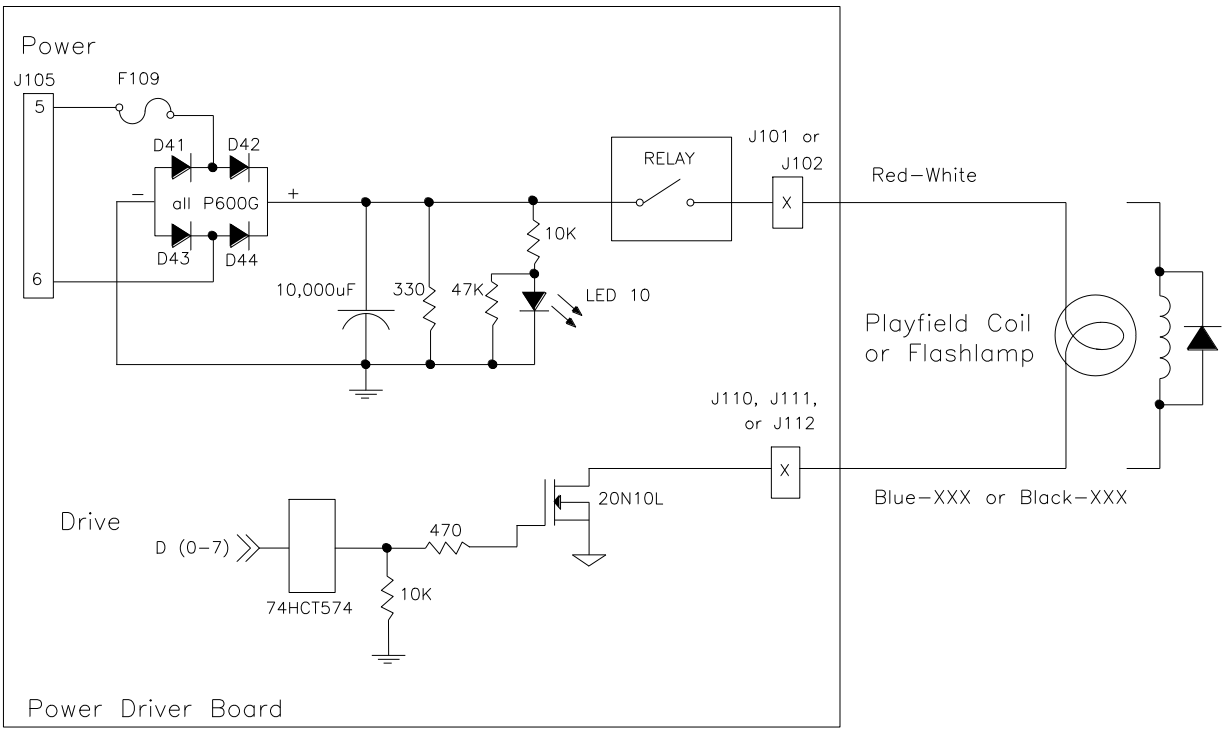
FLASHLAMPS


J102
8
RED–WHT, +20V (FUSE 109)
POWER
DRIVER
BOARD
J110
1
6
7
BLU–BRN SOLENOID 17 CENTER ARROW FLASHER
BLU–BLK SOLENOID 22 RIGHT POPPER FLASHER
BLU–VIO SOLENOID 23 LEFT ARCH FLASHER
J112
9
10
19
20
BLK–BRN SOLENOID 25 RIGHT ARCH FLASHER
BLK–RED SOLENOID 26 LEFT MARTIAN FLASHER
BLK–ORG SOLENOID 27 RIGHT MARTIAN FLASHER
BLK–YEL SOLENOID 28 ATTACK MARS FLASHER

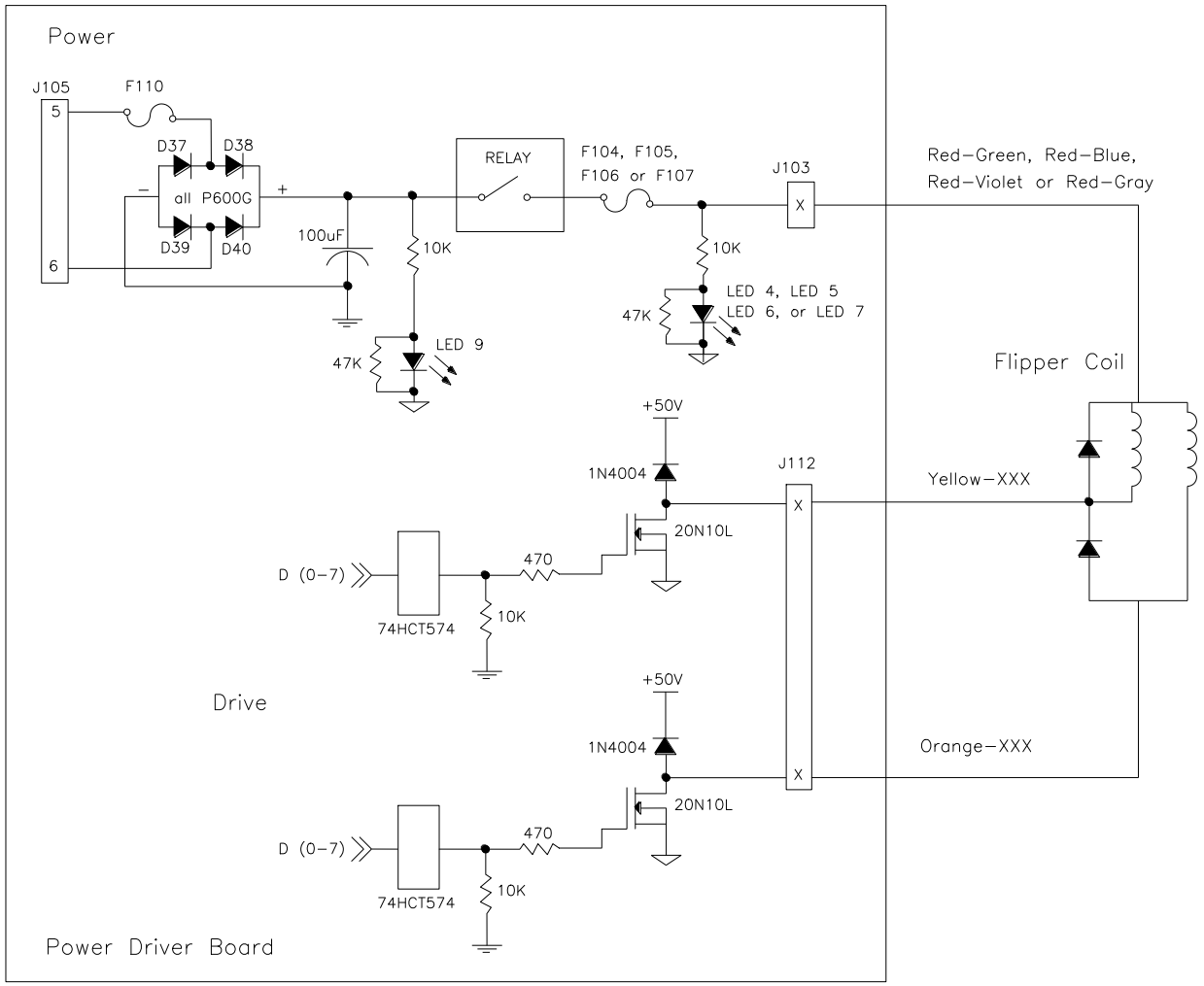
CIRCUIT for SOLENOIDS 1 to 16



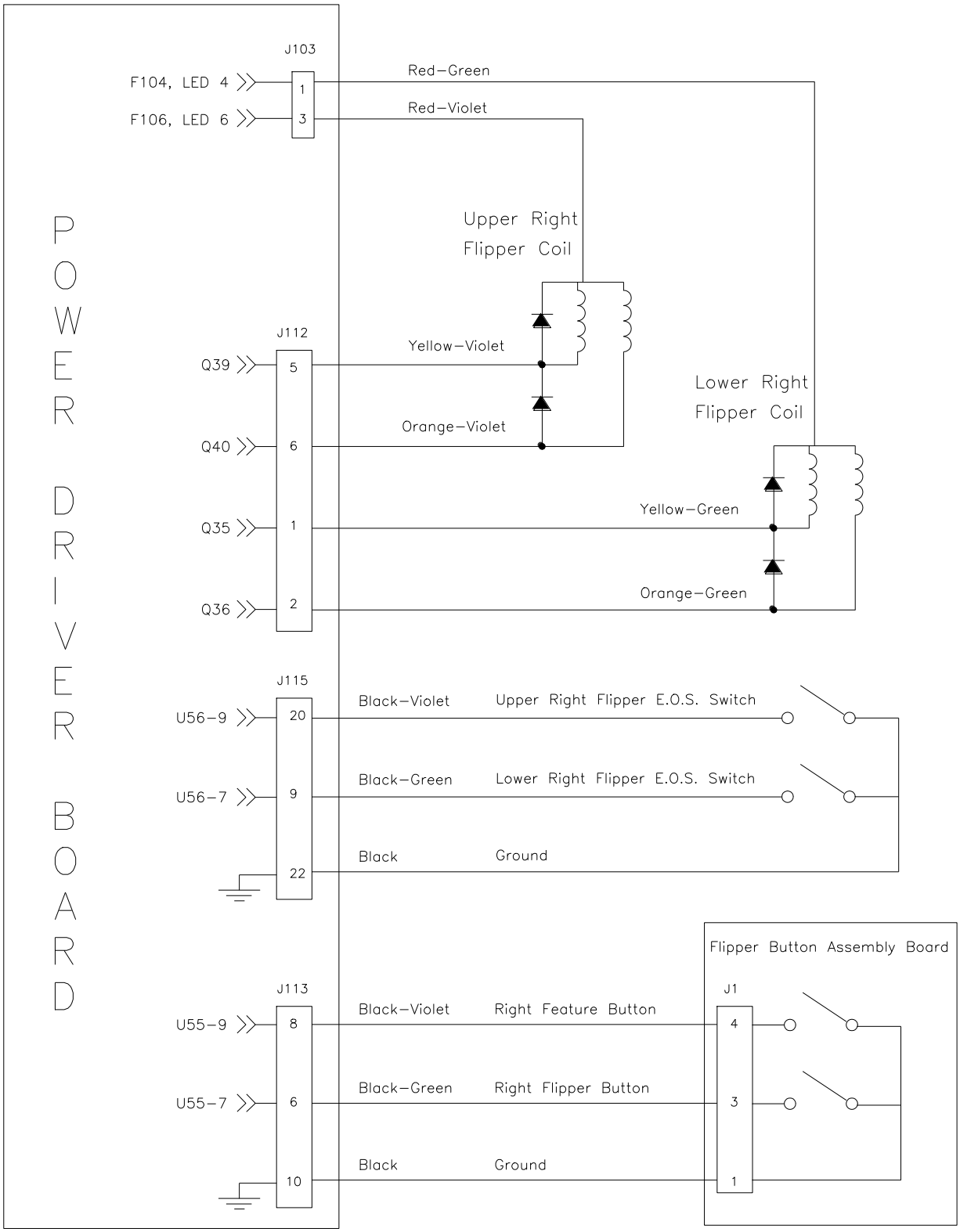
CIRCUIT for SOLENOIDS 17 to 28



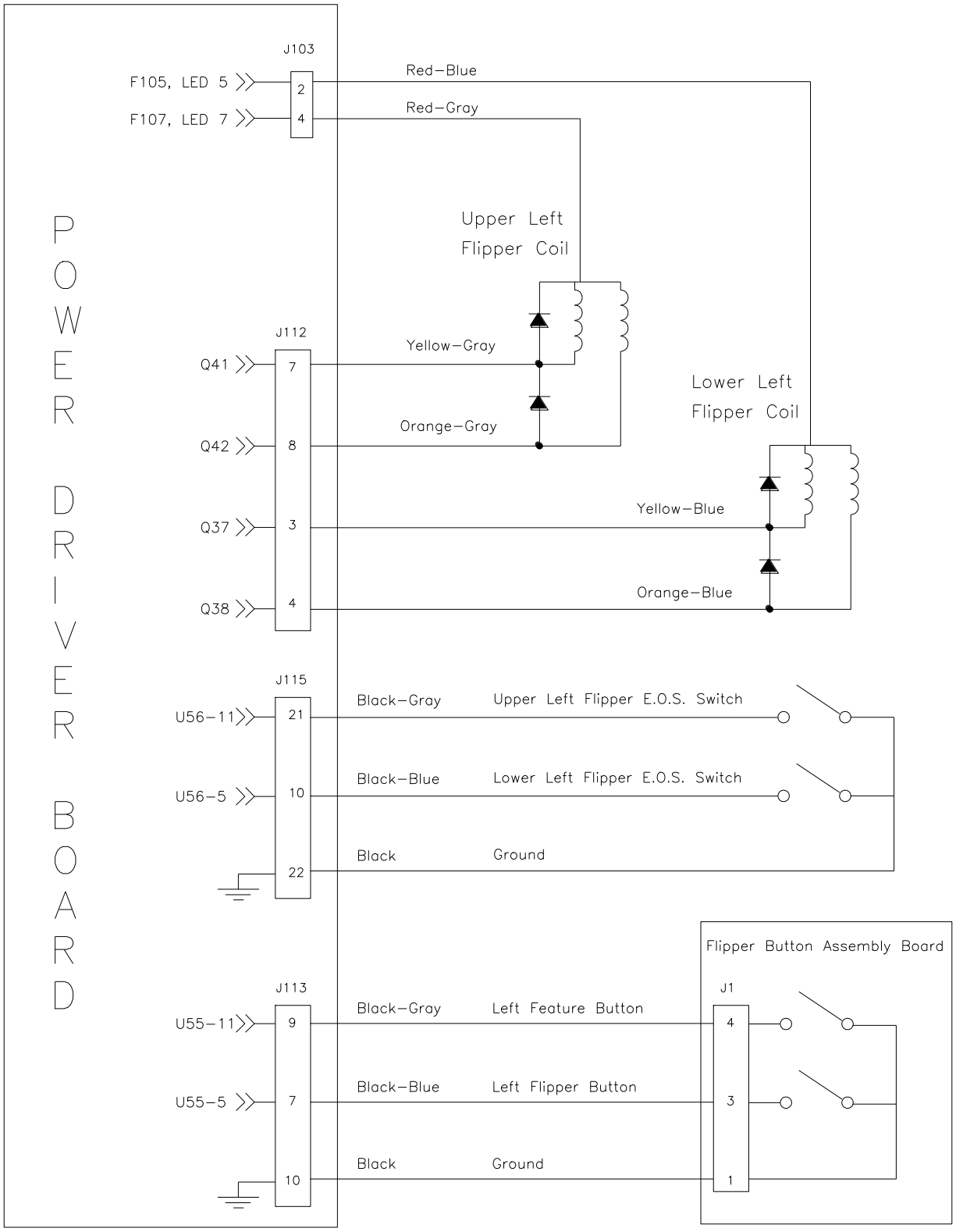
FLIPPER COIL CIRCUIT



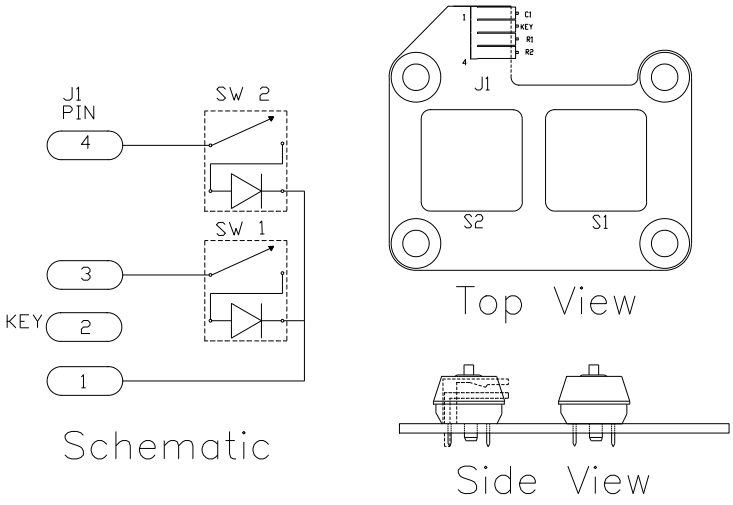
UPPER AND LOWER RIGHT FLIPPER CIRCUITS



UPPER AND LOWER LEFT FLIPPER CIRCUITS



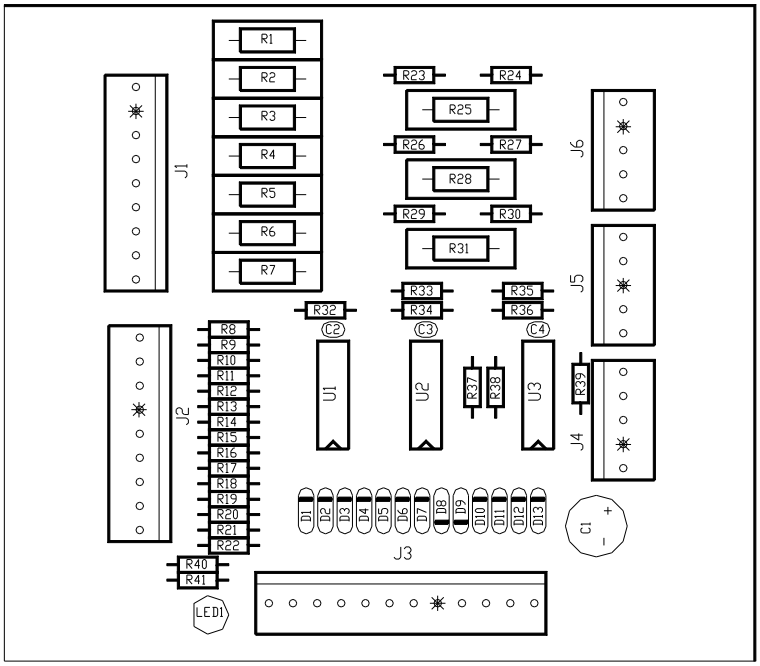
2 SWITCH FLIPPER BUTTON ASSEMBLY
04-12612



Left Side Flipper
J1-1 Black, Ground from Power Driver Board at J113-10
J1-2 Key
J1-3 Black-Blue, Flipper Button from Power Driver Board at J113-7
J1-4 Black-Gray, Feature Button from Power Driver Board at J113-9

Right Side Flipper
J1-1 Black, Ground from Power Driver Board at J113-10
J1-2 Key
J1-3 Black-Green, Flipper Button from Power Driver Board at J113-6
J1-4 Black-Violet, Feature Button from Power Driver Board at J113-8

10-Opto Switch Board Assembly
A-18159.1

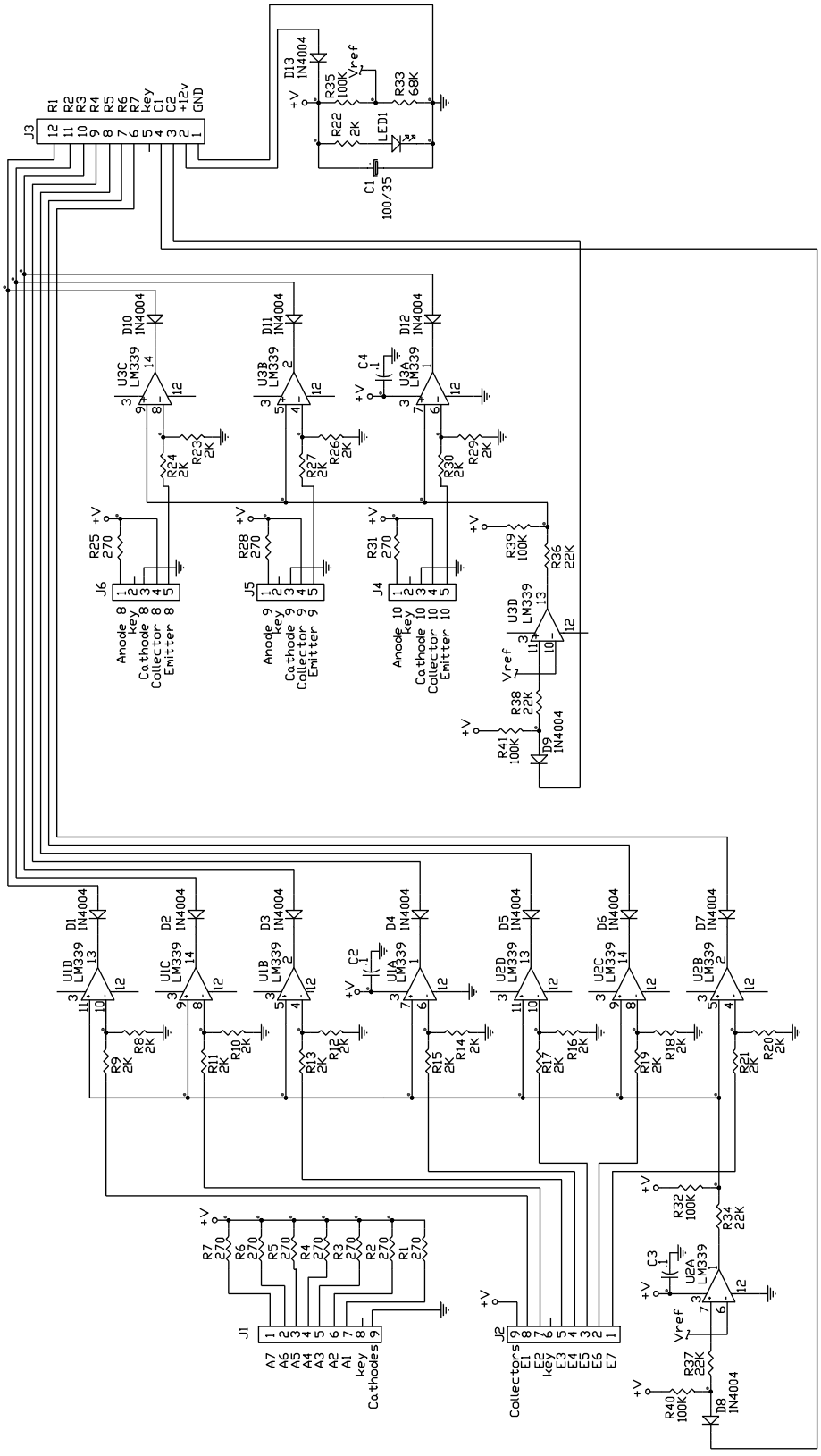


J1-1	GRY-VIO	To switch #47 LED board
J1-2	GRY-BLU	To switch #46 LED board
J1-3	GRY-GRN	To switch #45 LED board
J1-4	GRY-BLK	To switch #44 LED board
J1-5	GRY-ORG	To switch #43 LED board
J1-6	GRY-RED	To switch #42 LED board
J1-7	GRY-BRN	To switch #41 LED board
J1-8	KEY	
J1-9	BLK	Ground to LED boards
J2-1	ORG-VIO	To switch #47 Photo Trans. board
J2-2	ORG-BLU	To switch #46 Photo Trans. board
J2-3	ORG-GRN	To switch #45 Photo Trans. board
J2-4	ORG-YEL	To switch #44 Photo Trans. board
J2-5	ORG-BLK	To switch #43 Photo Trans. board
J2-6	KEY	
J2-7	ORG-RED	To switch #42 Photo Trans. board
J2-8	ORG-BRN	To switch #41 Photo Trans. board
J2-9	GRY-YEL	+12V to Photo Trans. boards

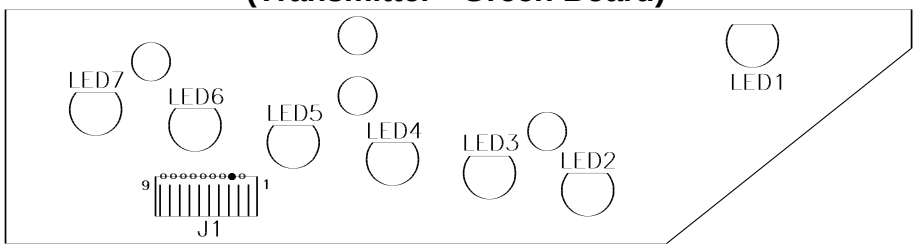
J3-1	BLK	Ground from Power Driver board J116-22
J3-2	GRY-YEL	+12V from Power Driver board J116-11
J3-3	GRN-BLK	Switch Row 5 from Power Driver board J116-5
J3-4	GRN-WHT	Switch Row 4 from Power Driver board J116-4
J3-5	KEY	
J3-6	WHT-VIO	Switch Column #7 from Power Driver board J116-18
J3-7	WHT-BLU	Switch Column #6 from Power Driver board J116-17
J3-8	WHT-GRN	Switch Column #5 from Power Driver board J116-16
J3-9	WHT-YEL	Switch Column #4 from Power Driver board J116-15
J3-10	WHT-ORG	Switch Column #3 from Power Driver board J116-14
J3-11	WHT-RED	Switch Column #2 from Power Driver board J116-13
J3-12	WHT-BRN	Switch Column #1 from Power Driver board J116-12

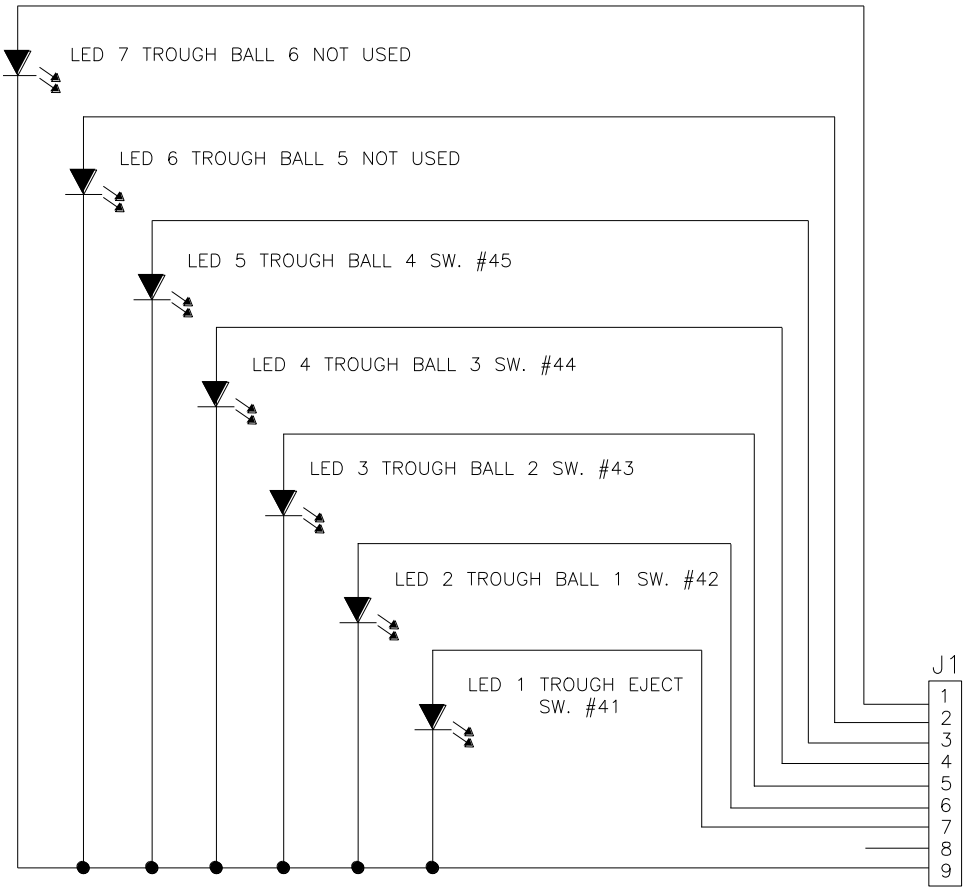
J4		NOT USED
J5-1	GRN-RED	To switch #52 LED board
J5-2	BLK	Ground to switch #52 LED board
J5-3	KEY	
J5-4	GRY-YEL	+12V to sw. #52 Photo Trans board
J5-5	WHT-RED	To switch #52 Photo Trans. board
J6-1	GRN-BRN	To switch #51 LED board
J6-2	KEY	
J6-3	BLK	Ground to switch #51 LED board
J6-4	GRY-YEL	+12V to sw. #51 Photo Trans board
J6-5	WHT-BRN	To switch #51 Photo Trans. board

10-Opto Switch Board Schematic
A-18159.1



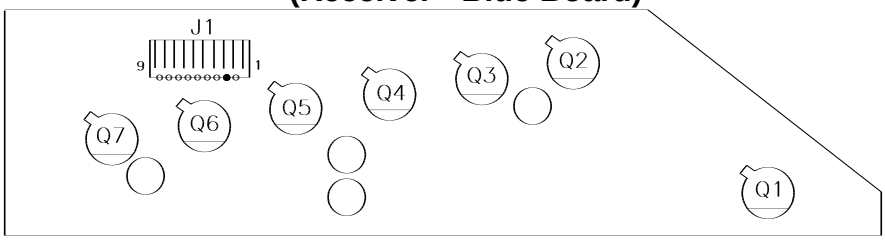
Ball Trough Infrared LED Board Assembly
A-18617-1
(Transmitter - Green Board)

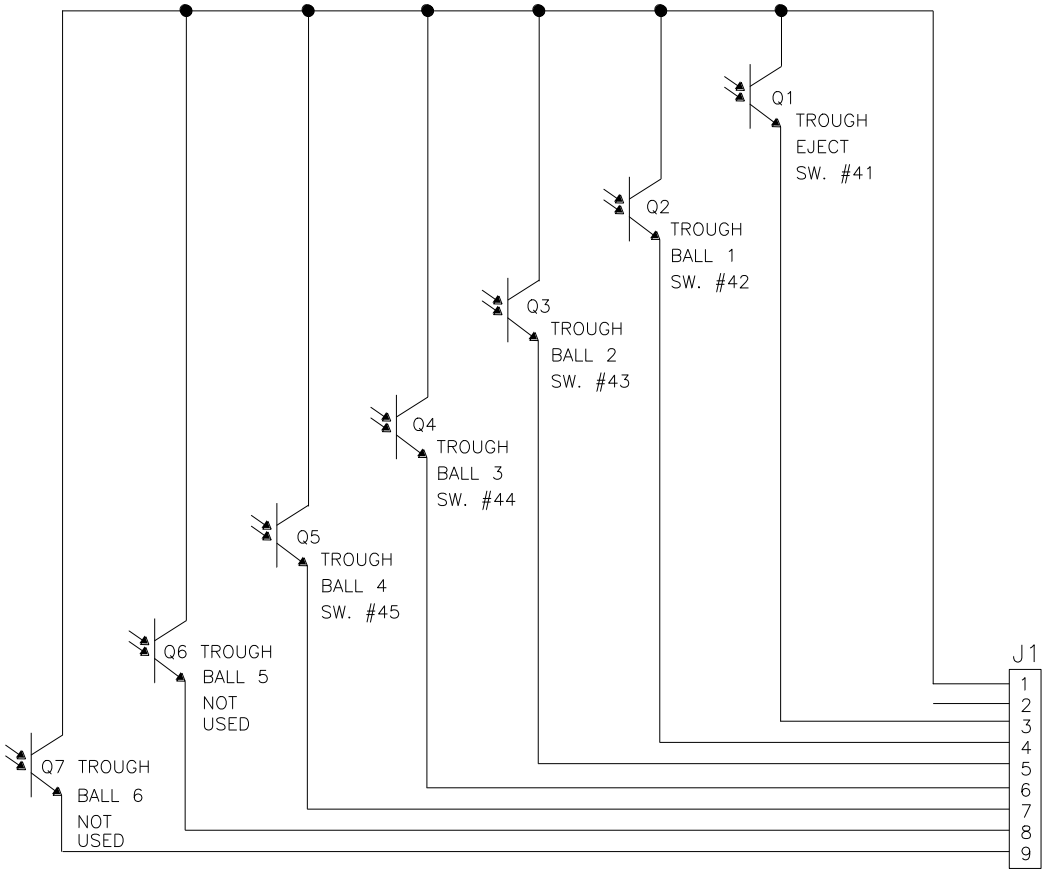




J1-1 N/C
J1-2 N/C
J1-3 GRY-GRN, For TROUGH BALL 4 switch #45 from 10-Opto Switch Board J1-3
J1-4 GRY-BLK, For TROUGH BALL 3 switch #44 from 10-Opto Switch Board J1-4
J1-5 GRY-ORG, For TROUGH BALL 2 switch #43 from 10-Opto Switch Board J1-5
J1-6 GRY-RED, For TROUGH BALL 1 switch #42 from 10-Opto Switch Board J1-6
J1-7 GRY-BRN, For TROUGH EJECT switch #41 from 10-Opto Switch Board J1-7
J1-8 KEY
J1-9 BLK, Ground from 10-Opto Switch Board J1-9

Ball Trough Infrared Photo Transistor Board Assembly
A-18618-1
(Receiver - Blue Board)





J1-1 GRY-YEL, +12V from 10-Opto Switch Board J2-9
J1-2 KEY
J1-3 ORG-BRN, For TROUGH EJECT switch #41 from 10-Opto Switch Board J2-8
J1-4 ORG-RED, For TROUGH BALL 1 switch #42 from 10-Opto Switch Board J2-7
J1-5 ORG-BLK, For TROUGH BALL 2 switch #43 from 10-Opto Switch Board J2-5
J1-6 ORG-YEL, For TROUGH BALL 3 switch #44 from 10-Opto Switch Board J2-4
J1-7 ORG-GRN, For TROUGH BALL 4 switch #45 from 10-Opto Switch Board J2-3
J1-8 N/C
J1-9 N/C

Wiring Diagram for the Ball Trough Assembly Opto Switches

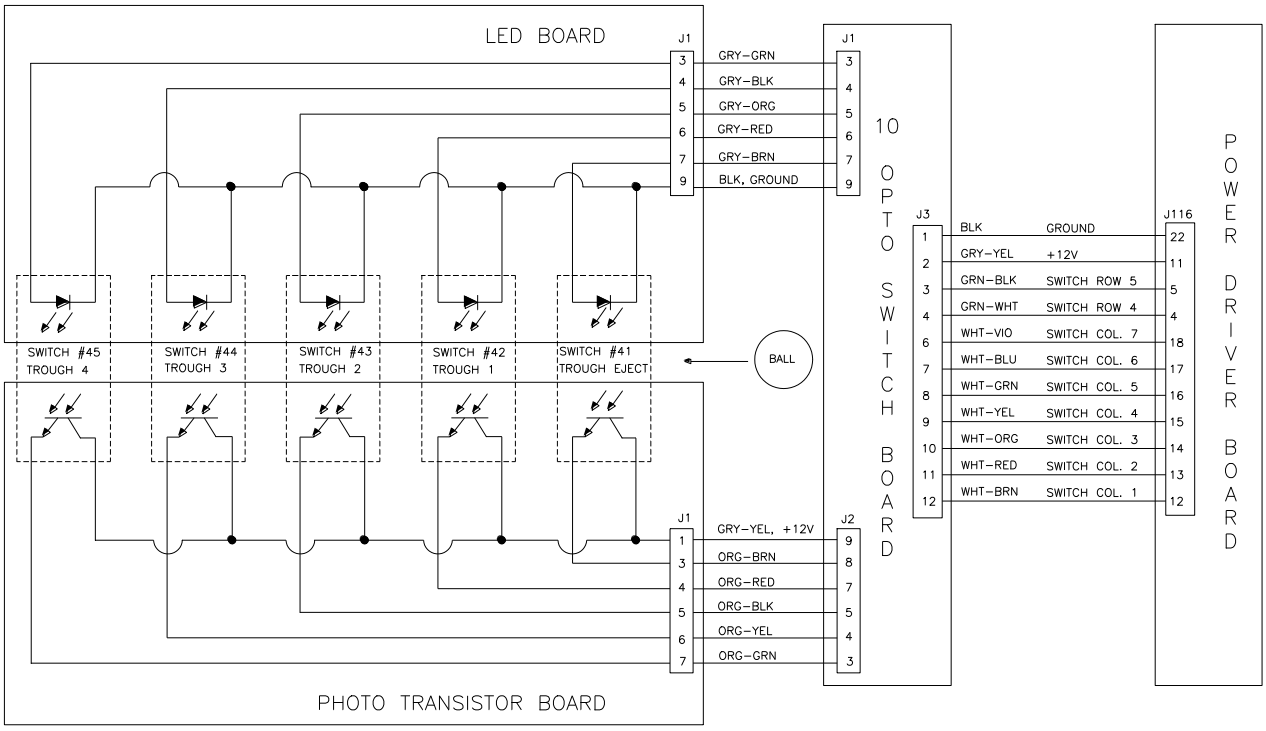
This diagram shows the opto switches in the Ball Trough Assembly only. See page 3-21 for the individual playfield opto switches.

The Ball Trough Assembly opto switches are:

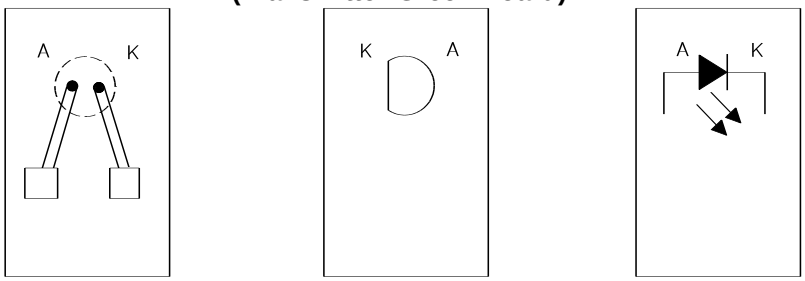
Switch #41	Trough Eject
Switch #42	Trough Ball 1
Switch #43	Trough Ball 2
Switch #44	Trough Ball 3
Switch #45	Trough Ball 4

The Ball Trough Assembly opto switches use a green Ball Trough LED board (P/N A-18617-1), and a blue Ball Trough Photo Transistor board (P/N A-18618-1).

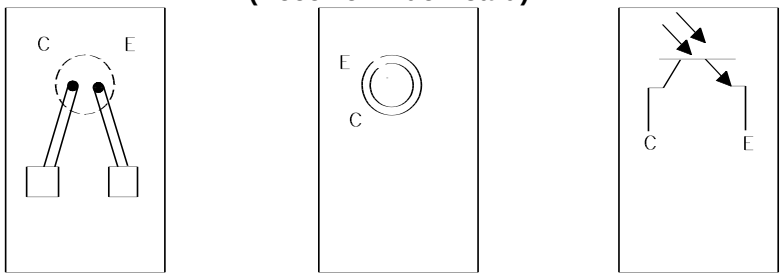
OPTO SWITCH OPERATION:
The ball rolls between the Ball Trough LED board and the Ball Trough Photo Transistor board and breaks the beam. The broken beam 'makes' the switch.



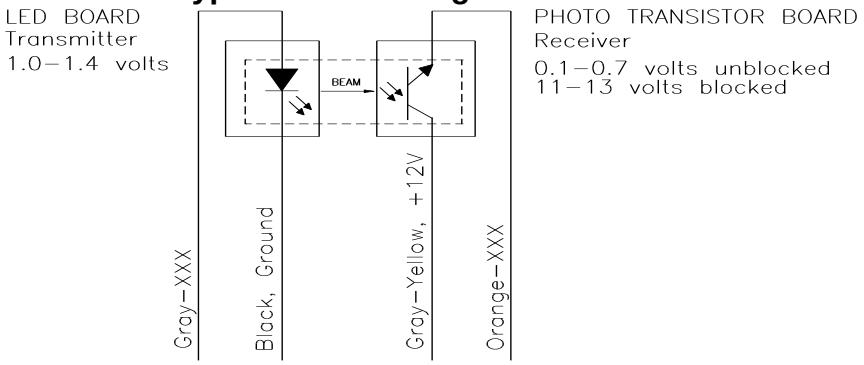
Individual LED Board Assembly
A-16908
(Transmitter-Green Board)

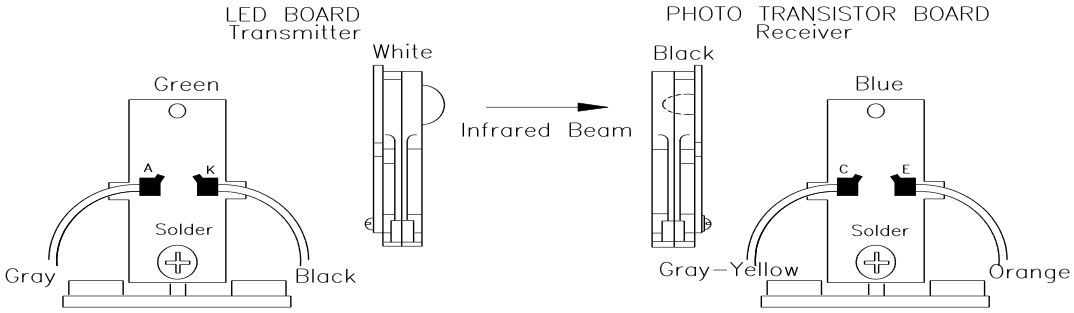


Individual Photo Transistor Board Assembly
A-16909
(Receiver-Blue Board)



Typical Circuit Diagram





Wiring Diagram for Individual Playfield Opto Switches

This diagram shows the wiring for individual playfield opto switches only. See page 3-19 for the wiring diagram for the Ball Trough Assembly opto switches.

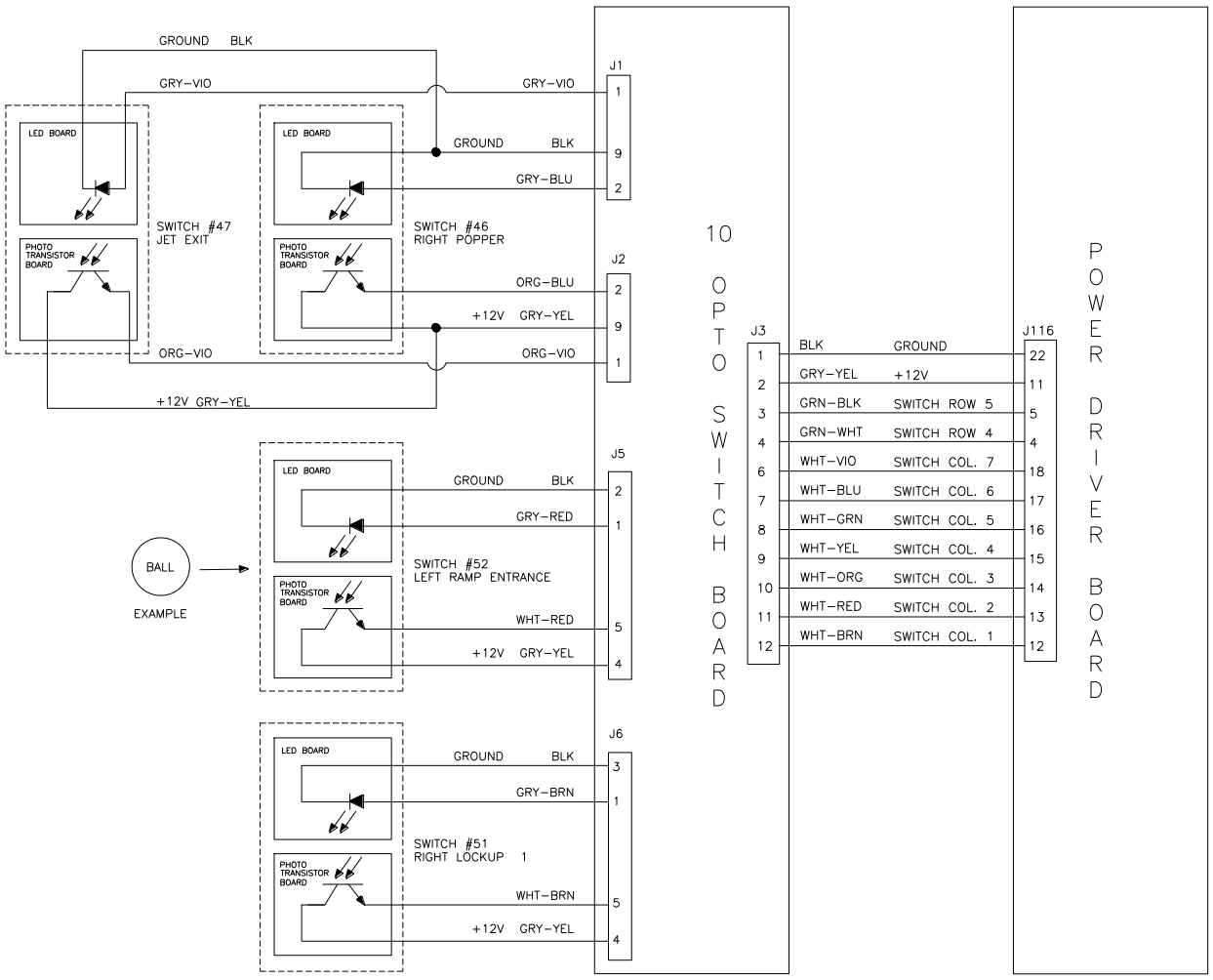
The individual playfield opto switches are:

Switch #46 Right Popper
Switch #47 Jet Eject
Switch #51 Right Lockup 1
Switch #52 Left Ramp Entrance

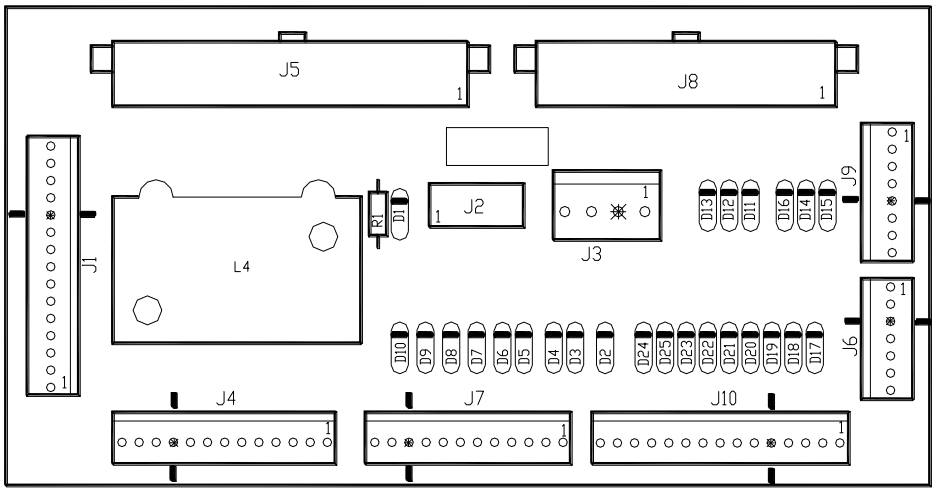
Each of these switches uses a green Individual LED board (P/N A-16908), and a blue Individual Photo Transistor board (P/N A-16909).

OPTO SWITCH OPERATION:

The ball rolls between the Individual LED board and the Individual Photo Transistor board and breaks the beam. The broken beam 'makes' the switch.



Coin Door Interface Board
04-12492



J1-1	BLK-YEL	Slam Tilt
J1-2	BLK-RED	Interlock Switch
J1-3	BLK-ORG	Plumb Bob Tilt
J1-4	N/C	
J1-5	BLK-GRN	Right Flipper 1
J1-6	BLK-BLU	Left Flipper 1
J1-7	BLK-VIO	Right Flipper 2
J1-8	BLK-GRY	Left Flipper 2
J1-9	N/C	
J1-10	N/C	
J1-11	Key	
J1-12	N/C	
J1-13	BLK	Flipper Switch Ground
J1-14	GRY-YEL	+12V
J1-15	N/C	Ground

J2	Ribbon Cable	

J3-1	BLU	For DBV
J3-2	Key	
J3-3	N/C	
J3-4	BRN	For DBV

J4-1	WHT-YEL	Lamp Row 4
J4-2	YEL-WHT	Lamp Column 2
J4-3	BLK	Switch Ground
J4-4	BLK-BRN	Coin 1 Left
J4-5	N/C	Coin 2 Center
J4-6	BLK-RED	Coin 3 Right
J4-7	WHT-GRY	Escape Button
J4-8	ORG-RED	Down Button
J4-9	ORG-GRN	Up Button
J4-10	Key	
J4-11	BLK-BLU	Enter Button
J4-12	GRN-BRN	Slam Column
J4-13	BLK-GRN	Slam Row

J5-1	BLK-BRN	Slam Switch
J5-2	BLK-RED	Interlock Switch
J5-3	BLK-ORG	Plumb Bob Tilt
J5-4	BLK-YEL	Misc.
J5-5	BLK-GRN	Right Flipper 1
J5-6	BLK-BLU	Left Flipper 1
J5-7	BLK-VIO	Right Flipper 2
J5-8	BLK-GRY	Left Flipper 2
J5-9	GRY-BRN	Enter Button
J5-10	GRY-RED	Up Button
J5-11	BLU	Counter Signal
J5-12	GRY-YEL	+12V
J5-13	ORG-BRN	Coin 1
J5-14	ORG-RED	Coin 2
J5-15	ORG-BLK	Coin 3
J5-16	ORG-YEL	Coin 4
J5-17	ORG-GRN	Coin 5
J5-18	ORG-BLU	Coin 6
J5-19	ORG-VIO	Coin 7
J5-20	ORG-GRY	Coin 8
J5-21	GRY-ORG	Down Button
J5-22	GRY-BLK	Escape Button
J5-23	VIO-GRY	Counter Power
J5-24	BLK	Ground

J6	Not Used	

J7-1	GRN-BRN	Switch Column 1
J7-2	GRN-RED	Switch Column 2
J7-3	N/C	
J7-4	N/C	
J7-5	N/C	
J7-6	N/C	
J7-7	N/C	
J7-8	WHT-ORG	Start Button
J7-9	WHT-ORG	Switch #23
J7-10	Key	
J7-11	N/C	
J7-12	N/C	

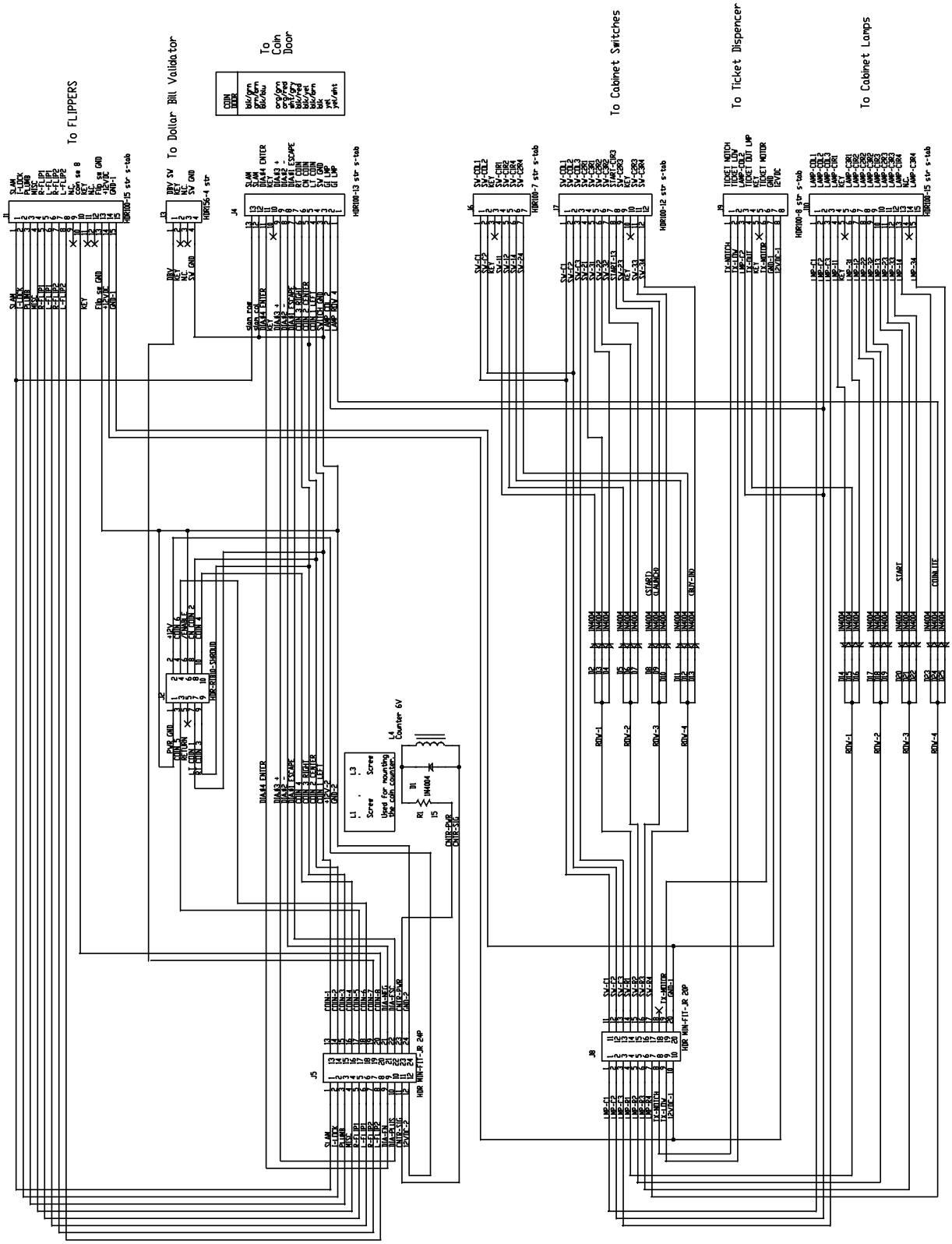
Coin Door Interface Board Continued…

J8-1	YEL-VIO	Lamp Column 1
J8-2	YEL-RED	Lamp Column 2
J8-3	YEL-ORG	Lamp Column 3
J8-4	BRN-BLK	Lamp Row 1
J8-5	BRN-RED	Lamp Row 2
J8-6	BRN-ORG	Lamp Row 3
J8-7	BRN-YEL	Lamp Row 4
J8-8	GRY-GRN	Ticket Notch
J8-9	GRY-BLU	Ticket Low
J8-10	GRY-YEL	+12V
J8-11	GRN-BRN	Switch Column 1
J8-12	GRN-RED	Switch Column 2
J8-13	GRN-ORG	Switch Column 3
J8-14	WHT-BRN	Switch Row 1
J8-15	WHT-RED	Switch Row 2
J8-16	WHT-ORG	Switch Row 3
J8-17	WHT-YEL	Switch Row 4
J8-18	Key	
J8-19	WHT	Ticket Motor
J8-20	BLK	Ground

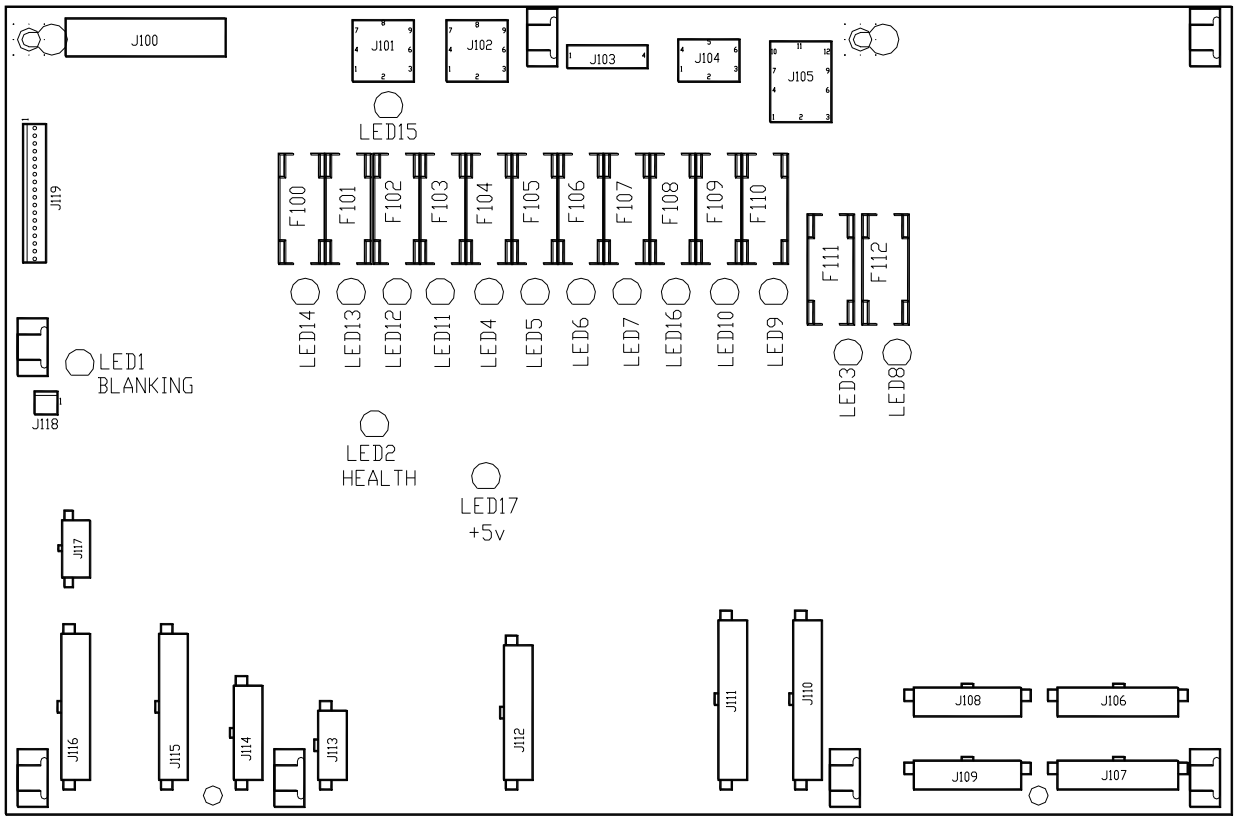
J9 Not Used

J10-1	N/C	
J10-2	YEL-RED	Lamp Column 2
J10-3	YEL-BRN	Lamp Column 3
J10-4	Key	
J10-5	N/C	
J10-6	N/C	
J10-7	N/C	
J10-8	N/C	
J10-9	N/C	
J10-10	BRN-ORG	Lamp 13
J10-11	ORG-BRN	Lamp 23
J10-12	N/C	
J10-13	N/C	
J10-14	N/C	
J10-15	N/C	

Coin Door Interface Board Schematic
04-12492



Power Driver Board Assembly
04-12329



J100	DB25 Parallel Port Extension Cable for Data to/from CPU motherboard	
J101-1/J102-1	RED-BRN	Power for solenoids 1 to 4
J101-2/J102-2	RED-BLK	Power for solenoids 5 to 8
J101-3/J102-3	RED-ORG	Power for solenoids 9 to 12
J101-4/J102-4	BLK	Ground
J101-5/J102-5	BLK	Ground
J101-6/J102-6	ORG	Signal Ground
J101-7/J102-7	RED-YEL	Power for solenoids 13 to 16
J101-8/J102-8	RED-WHT	+20V flashlamp power
J101-9/J102-9	GRY-YEL	+12V unregulated power
J103-1	RED-GRN	Power for lower right flipper
J103-2	RED-BLU	Power for lower left flipper
J103-3	RED-VIO	Power for upper right flipper
J103-4	RED-GRY	Power for upper left flipper
J104-1		50V AC
J104-2		50V AC
J104-3		No Connection
J104-4		20V AC
J104-5		20V AC
J104-6		No Connection

Power Driver Board Continued…

J105-1	BLU-WHT	18V AC Secondary for Lamp Matrix A
J105-2	BLU-WHT	18V AC Secondary for Lamp Matrix A
J105-3	WHT-BLU	18V AC Secondary for Lamp Matrix B
J105-4	WHT-BLU	18V AC Secondary for Lamp Matrix B
J105-5	BLK-YEL	51.4V AC Secondary
J105-6	BLK-YEL	51.4V AC Secondary
J105-7	WHT-RED	16V AC Secondary
J105-8	WHT-RED	16V AC Secondary
J105-9	WHT-GRN	9.8V AC Secondary
J105-10	WHT-GRN	9.8V AC Secondary
J105-11		No Connection
J105-12		No Connection

J106-1/J107-1	RED-BRN	Lamp Row 1B
J106-2/J107-2	RED-BLK	Lamp Row 2B
J106-3/J107-3	RED-ORG	Lamp Row 3B
J106-4/J107-4	RED-YEL	Lamp Row 4B
J106-5/J107-5	RED-GRN	Lamp Row 5B
J106-6/J107-6	RED-BLU	Lamp Row 6B
J106-7/J107-7	RED-VIO	Lamp Row 7B
J106-8/J107-8	RED-GRY	Lamp Row 8B
J106-9/J107-9		No Connection
J106-10/J107-10	YEL-BRN	Lamp Column 1B
J106-11/J107-11	YEL-RED	Lamp Column 2B
J106-12/J107-12	YEL-ORG	Lamp Column 3B
J106-13/J107-13	YEL-BLK	Lamp Column 4B
J106-14/J107-14	YEL-GRN	Lamp Column 5B
J106-15/J107-15	YEL-BLU	Lamp Column 6B
J106-16/J107-16	YEL-VIO	Lamp Column 7B
J106-17/J107-17	YEL-GRY	Lamp Column 8B
J106-18/J107-18		No Connection

J108-1/J109-1	BRN-BLK	Lamp Row 1A
J108-2/J109-2	BRN-RED	Lamp Row 2A
J108-3/J109-3	BRN-ORG	Lamp Row 3A
J108-4/J109-4	BRN-YEL	Lamp Row 4A
J108-5/J109-5	BRN-GRN	Lamp Row 5A
J108-6/J109-6	BRN-BLU	Lamp Row 6A
J108-7/J109-7	BRN-VIO	Lamp Row 7A
J108-8/J109-8	BRN-GRY	Lamp Row 8A
J108-9/J109-9	YEL-BRN	Lamp Column 1A
J108-10/J109-10	YEL-RED	Lamp Column 2A
J108-11/J109-11	YEL-ORG	Lamp Column 3A
J108-12/J109-12	YEL-BLK	Lamp Column 4A
J108-13/J109-13	YEL-GRN	Lamp Column 5A
J108-14/J109-14	YEL-BLU	Lamp Column 6A
J108-15/J109-15	YEL-VIO	Lamp Column 7A
J108-16/J109-16	YEL-GRY	Lamp Column 8A

Power Driver Board Continued...

J110-1/J111-1	BLU-BRN	Solenoid Drive 17
J110-2/J111-2	BLU-RED	Solenoid Drive 18
J110-3/J111-3	BLU-ORG	Solenoid Drive 19
J110-4/J111-4	BLU-YEL	Solenoid Drive 20
J110-5/J111-5	BLU-GRN	Solenoid Drive 21
J110-6/J111-6	BLU-BLK	Solenoid Drive 22
J110-7/J111-7	BLU-VIO	Solenoid Drive 23
J110-8/J111-8	BLU-GRY	Solenoid Drive 24
J110-9/J111-9	BRN-WHT	Solenoid Drive 41-Logic
J110-10/J111-10	ORG-WHT	Solenoid Drive 42-Logic
J110-11/J111-11	YEL-WHT	Solenoid Drive 43-Logic
J110-12/J111-12	BLU-WHT	Solenoid Drive 44-Logic
J110-13/J111-13	VIO-BRN	Solenoid Drive 1
J110-14/J111-14	VIO-RED	Solenoid Drive 2
J110-15/J111-15	VIO-ORG	Solenoid Drive 3
J110-16/J111-16	VIO-YEL	Solenoid Drive 4
J110-17/J111-17	VIO-GRN	Solenoid Drive 5
J110-18/J111-18	VIO-BLU	Solenoid Drive 6
J110-19/J111-19	VIO-BLK	Solenoid Drive 7
J110-20/J111-20	VIO-GRY	Solenoid Drive 8
J110-21/J111-21	VIO-WHT	Solenoid Drive 45-Logic
J110-22/J111-22	GRY-WHT	Solenoid Drive 46-Logic
J110-23/J111-23	BLK-WHT	Solenoid Drive 47-Logic
J110-24/J111-24	WHT	Solenoid Drive 48-Logic

J112-1	YEL-GRN	Solenoid Drive 33-Lower Right Flipper Power
J112-2	ORG-GRN	Solenoid Drive 34-Lower Right Flipper Hold
J112-3	YEL-BLU	Solenoid Drive 35-Lower Left Flipper Power
J112-4	ORG-BLU	Solenoid Drive 36-Lower Left Flipper Hold
J112-5	YEL-VIO	Solenoid Drive 37-Upper Right Flipper Power
J112-6	ORG-VIO	Solenoid Drive 38-Upper Right Flipper Hold
J112-7	YEL-GRY	Solenoid Drive 39-Upper Left Flipper Power
J112-8	ORG-GRY	Solenoid Drive 40-Upper Left Flipper Hold
J112-9	BLK-BRN	Solenoid Drive 25
J112-10	BLK-RED	Solenoid Drive 26
J112-11	BRN-BLK	Solenoid Drive 9
J112-12	BRN-RED	Solenoid Drive 10
J112-13	BRN-ORG	Solenoid Drive 11
J112-14	BRN-YEL	Solenoid Drive 12
J112-15	BRN-GRN	Solenoid Drive 13
J112-16	BRN-BLU	Solenoid Drive 14
J112-17	BRN-VIO	Solenoid Drive 15
J112-18	BRN-GRY	Solenoid Drive 16
J112-19	BLK-ORG	Solenoid Drive 27
J112-20	BLK-YEL	Solenoid Drive 28

J113-1	BLK-BRN	Direct Switch 17-Slam
J113-2	BLK-RED	Direct Switch 18-Interlock
J113-3	BLK-ORG	Direct Switch 19-Plumb Bob
J113-4	BLK-YEL	Direct Switch 20-Not Used
J113-5	GRY-YEL	+12V DC
J113-6	BLK-GRN	Direct Switch 21-Right Flipper Button
J113-7	BLK-BLU	Direct Switch 22-Left Flipper Button
J113-8	BLK-VIO	Direct Switch 23-Right Feature Button
J113-9	BLK-GRY	Direct Switch 24 Left Feature Button
J113-10	BLK	Ground

Power Driver Board Continued...

J114-1	ORG-BRN	Direct Switch 1-Left Coin Slot
J114-2	ORG-RED	Direct Switch 2-Center Coin Slot
J114-3	ORG-BLK	Direct Switch 3-Right Coin Slot
J114-4	ORG-YEL	Direct Switch 4-4th Coin Slot
J114-5	ORG-GRN	Direct Switch 5-Not Used
J114-6	ORG-BLU	Direct Switch 6-Not Used
J114-7	GRY-YEL	+12V DC
J114-8	ORG-VIO	Direct Switch 7-Not Used
J114-9	ORG-GRY	Direct Switch 8-Not Used
J114-10	GRY-BLK	Direct Switch 9-Diagnostic 'Escape' Button
J114-11	GRY-ORG	Direct Switch 10-Diagnostic 'Down' Button
J114-12	GRY-RED	Direct Switch 11-Diagnostic 'Up' Button
J114-13	GRY-BRN	Direct Switch 12-Diagnostic 'Enter' Button
J114-14	BLK	Ground

J115-1/J116-1	GRN-BRN	Switch Column 1
J115-2/J116-2	GRN-RED	Switch Column 2
J115-3/J116-3	GRN-ORG	Switch Column 3
J115-4/J116-4	GRN-WHT	Switch Column 4
J115-5/J116-5	GRN-BLK	Switch Column 5
J115-6/J116-6	GRN-BLU	Switch Column 6
J115-7/J116-7	GRN-VIO	Switch Column 7
J115-8/J116-8	GRN-GRY	Switch Column 8
J115-9/J116-9	BLK-GRN	Direct Switch 13-Lower Right Flipper E.O.S.
J115-10/J116-10	BLK-BLU	Direct Switch 14-Lower Left Flipper E.O.S.
J115-11/J116-11	GRY-YEL	+12V
J115-12/J116-12	WHT-BRN	Switch Row 1
J115-13/J116-13	WHT-RED	Switch Row 2
J115-14/J116-14	WHT-ORG	Switch Row 3
J115-15/J116-15	WHT-YEL	Switch Row 4
J115-16/J116-16	WHT-GRN	Switch Row 5
J115-17/J116-17	WHT-BLU	Switch Row 6
J115-18/J116-18	WHT-VIO	Switch Row 7
J115-19/J116-19	WHT-GRY	Switch Row 8
J115-20/J116-20	BLK-VIO	Direct Switch 15-Upper Right Flipper E.O.S. (not used)
J115-21/J116-21	BLK-GRY	Direct Switch 16-Upper Left Flipper E.O.S. (not used)
J115-22/J116-22	BLK	Ground

J117-1		+12V
J117-2		+12V
J117-3	WHT	Drive 48-Logic-Ticket Driver
J117-4	GRY-VIO	+12V Coin Door Power
J117-5	GRY-GRN	Direct Switch 25-Ticket Notch
J117-6	GRY-BLU	Direct Switch 26-Ticket Low
J117-7	BLUE	Coin Counter
J117-8	BLK	Ground

NOTES

LAMP MATRIX A

COLUMN / ROW	1A YEL-BRN J108-9 Q5	2A YEL-RED J108-10 Q9	3A YEL-ORG J108-11 Q13	4A YEL-BLK J108-12 Q17	5A YEL-GRN J108-13 Q21	6A YEL-BLU J108-14 Q25	7A YEL-VIO J108-15 Q29	8A YEL-GRY J108-16 Q33
1A BRN-BLK J108-1 Q3	NOT USED 11A	TICKETS LOW 21A	NOT USED 31A	RIGHT POPPER ARROW 41A	MULTIBALL 51A	SHOOTER LANE 9 (TOP) 61A	LEFT SIDE 1 (BOTTOM) 71A	RIGHT SLINGSHOT (BOTTOM) 81A
2A BRN-RED J108-2 Q7	NOT USED 12A	NOT USED 22A	NOT USED 32A	EXTRA BALL 42A	CAPTURE 2 52A	UNDER RIGHT RAMP (LOW) 62A	LEFT SIDE 2 72A	RIGHT SLINGSHOT (SAUCER) 82A
3A BRN-ORG J108-3 Q11	START BUTTON 13A	LAUNCH BUTTON 23A	NOT USED 33A	MARTIAN ATTACK 43A	CAPTURE 1 53A	UNDER RIGHT RAMP (HIGH) 63A	LEFT SIDE 3 73A	RIGHT RETURN LANE (RIGHT) 83A
4A BRN-YEL J108-4 Q15	NOT USED 14A	COIN DOOR ILLUMINA. 24A	NOT USED 34A	STROKE OF LUCK 44A	CAPTURE ZONE ACTIVE 54A	UPPER RT. CORNER (LOW) 64A	LEFT SIDE 4 (TOP) 74A	RIGHT RETURN LANE (LEFT) 84A
5A BRN-GRN J108-5 Q19	RIGHT TOP LANE 15A	MOTHERSHIP MULTIBALL (RIGHT) 25A	LEFT DRAIN TO TROUGH 35A	LEFT SIDE SPOTLIGHT 45A	SHOOT AGAIN 55A	RIGHT ARCH (RIGHT) 65A	UNDER LEFT RAMP (BOTTOM) 75A	LEFT RETURN LANE (RIGHT) 85A
6A BRN-BLU J108-6 Q23	LEFT TOP LANE 16A	MOTHERSHIP MULTIBALL (LEFT) 26A	RIGHT DRAIN TO TROUGH 36A	CENTER ARROW 46A	BEHIND CENTER TARGETS 56A	RIGHT ARCH (LEFT) 66A	UNDER LEFT RAMP (TOP) 76A	LEFT RETURN LANE (LEFT) 86A
7A BRN-VIO J108-7 Q27	MARTIAN TARGET 4 (CENTER) 17A	LEFT RETURN LANE 27A	RIGHT RETURN LANE 37A	RIGHT MARTIAN (HIGH) 47A	UPPER RT. CORNER (MIDDLE) 57A	LEFT ARCH (RIGHT) 67A	BETWEEN L/B JETS 77A	LEFT SLINGSHOT (SAUCER) 87A
8A BRN-GRY J108-8 Q31	CENTER LOOP ARROW 18A	LEFT OUTLANE 28A	RIGHT OUTLANE 38A	RIGHT MARTIAN (LOW) 48A	NOT USED 58A	LEFT ARCH (LEFT) 68A	UPPER LEFT CORNER 78A	LEFT SLINGSHOT (BOTTOM) 88A

LAMP MATRIX B

COLUMN / ROW	1B YEL-BRN J107-10 Q6	2B YEL-RED J107-11 Q10	3B YEL-ORG J107-12 Q14	4B YEL-BLK J107-13 Q18	5B YEL-GRN J107-14 Q22	6B YEL-BLU J107-15 Q26	7B YEL-VIO J107-16 Q30	8B YEL-GRY J107-17 Q34
1B RED-BRN J107-1 Q4	SECRET WEAPON 11B	BIG-O-BEAM 21B	MARS KNEADS WOMEN 31B	MARTIAN HARRY HOUR 41B	RIGHT LOOP ARROW 51B	MARTIAN TARGET 5 (RT. TOP) 61B	BOTTOM JET BUMPER 71B	SHOOTER LANE 1 (BOTTOM) 81B
2B RED-BLK J107-2 Q8	TOWER STRUGGLE 12B	RIGHT SAUCER BEAM (LEFT) 22B	RIGHT SAUCER BEAM (RT) 32B	ALIEN ABDUCTION 42B	RIGHT LOOP CIRCLE 52B	MARTIAN TARGET 6 (RT. MID.) 62B	NOT USED 72B	SHOOTER LANE 2 82B
3B RED-ORG J107-3 Q12	CENTER SAUCER BEAM (LEFT) 13B	WEAPONS 23B	SAUCER RIM 9 (RIGHT) 33B	LEFT SAUCER BEAM (LEFT) 43B	RIGHT RAMP ARROW 53B	MARTIAN TARGET 7 (RT. BOT.) 63B	LEFT JET BUMPER 73B	SHOOTER LANE 3 83B
4B RED-YEL J107-4 Q16	QUESTION MARK 14B	SAUCER 24B	SAUCER RIM 8 34B	SAUCER RIM 1 (LEFT) 44B	RIGHT RAMP CIRCLE 54B	MARTIAN TARGET 3 (LEFT TOP) 64B	LEFT OF LEFT TOP LANE 74B	SHOOTER LANE 4 84B
5B RED-GRN J107-5 Q20	CENTER SAUCER BEAM (RT.) 15B	FUEL 25B	SAUCER RIM 7 35B	SAUCER RIM 2 45B	LEFT LOOP ARROW 55B	MARTIAN TARGET 2 (LEFT MID.) 65B	BETWEEN L/R TOP LANES 75B	SHOOTER LANE 5 85B
6B RED-BLU J107-6 Q24	DRIVE-IN DEMOLITION 16B	LEFT SAUCER BEAM (RT.) 26B	SAUCER RIM 6 36B	SAUCER RIM 3 46B	LEFT RAMP ARROW 56B	MARTIAN TARGET 1 (LEFT BOT.) 66B	RIGHT OF RIGHT TOP LANE 76B	SHOOTER LANE 6 86B
7B RED-VIO J107-7 Q28	PARIS IN PERIL 17B	CENTER SAUCER BEAM (CNTR) 27B	SAUCER RIM 5 37B	SAUCER RIM 4 47B	LEFT LOOP CIRCLE 57B	RIGHT MARTIAN EYE 67B	TOP OF CENTER LOOP 77B	SHOOTER LANE 7 87B
8B RED-GRY J107-8 Q32	LEFT SLINGSHOT SPOTLIGHT 18B	RIGHT SLINGSHOT SPOTLIGHT 28B	NOT USED 38B	NOT USED 48B	LEFT RAMP CIRCLE 58B	LEFT MARTIAN EYE 68B	UPPER RT. CORNER (HIGH) 78B	SHOOTER LANE 8 88B

SWITCH MATRIX

COLUMN / ROW	1 GRN-BRN J116-1 U45-18	2 GRN-RED J116-2 U45-17	3 GRN-ORG J116-3 U45-16	4 GRN-WHT J116-4 U45-15	5 GRN-BLK J116-5 U45-14	6 GRN-BLU J116-6 U45-13	7 GRN-VIO J116-7 U45-12	8 GRN-GRY J116-8 U45-11
1 WHT-BRN J116-12 U51-7	RIGHT RAMP ENTRANCE 11	NOT USED 21	CENTER LOOP REED (BOTTOM) 31	TROUGH JAM 41	RIGHT LOCKUP 1 51	LEFT SLINGSHOT 61	MARTIAN TARGET 3 (LEFT TOP) 71	NOT USED 81
2 WHT-RED J116-13 U51-5	LEFT RAMP EXIT 12	NOT USED 22	CENTER LOOP REED (TOP) 32	TROUGH BALL 1 42	LEFT RAMP ENTRANCE 52	RIGHT SLINGSHOT 62	MARTIAN TARGET 2 (LEFT MID.) 72	NOT USED 82
3 WHT-ORG J116-14 U51-9	START BUTTON 13	LAUNCH BUTTON 23	CENTER TARGET 4 33	TROUGH BALL 2 43	NOT USED 53	LEFT JET BUMPER 63	MARTIAN TARGET 1 (LEFT BOT.) 73	NOT USED 83
4 WHT-YEL J116-15 U51-11	NOT USED 14	NOT USED 24	CENTER TARGET 3 34	TROUGH BALL 3 44	NOT USED 54	RIGHT JET BUMPER 64	CENTER LOOP ROLLOVER 74	NOT USED 84
5 WHT-GRN J116-16 U57-7	DROP TARGET DOWN 15	LEFT LOOP (LOW) 25	CENTER TARGET 2 35	TROUGH BALL 4 45	NOT USED 55	BOTTOM JET BUMPER 65	CENTER DEFLECTOR PANEL 75	MARTIAN TARGET 7 (RT. BOT.) 85
6 WHT-BLU J116-17 U57-5	LEFT OUTLANE 16	LEFT RETURN LANE 26	CENTER TARGET 1 36	RIGHT POPPER 46	NOT USED 56	NOT USED 66	RIGHT TOP LANE 76	MARTIAN TARGET 6 (RT. MID.) 86
7 WHT-VIO J116-18 U57-9	RIGHT RETURN LANE 17	RIGHT OUTLANE 27	MARTIAN TARGET 4 (CENTER) 37	JET EXIT 47	NOT USED 57	RIGHT LOOP (LOW) 67	LEFT TOP LANE 77	MARTIAN TARGET 5 (RT. TOP) 87
8 WHT-GRY J116-19 U57-11	SHOOTER LANE 18	RIGHT RAMP EXIT 28	UP/DOWN RAMP UP 38	NOT USED 48	NOT USED 58	RIGHT LOOP (HIGH) 68	LEFT LOOP (HIGH) 78	NOT USED 88

= OPTO, TYPICALLY CLOSED

COIN SLOT SWITCHES

DIRECT GROUND	ORN-BRN J114-1 U43-7	ORG-RED J114-2 U43-5	ORG-BLK J114-3 U43-9	ORG-YEL J114-4 U43-11	ORG-GRN J114-5 U48-7	ORG-BLU J114-6 U48-5	ORG-VIO J114-8 U48-9	ORG-GRY J114-9 U48-11
BLACK J114-14	LEFT COIN SLOT D1	CENTER COIN SLOT D2	RIGHT COIN SLOT D3	4TH COIN OPTION D4	NOT USED D5	NOT USED D6	NOT USED D7	NOT USED D8

DIAGNOSTIC SWITCHES

DIRECT GROUND	GRY-BLK J114-10 U50-7	GRY-ORG J114-11 U50-5	GRY-RED J114-12 U50-9	GRY-BRN J114-13 U50-11
BLACK J114-14	'ESCAPE' BUTTON D9	'DOWN' BUTTON D10	'UP' BUTTON D11	'ENTER' BUTTON D12

PLAYFIELD FLIPPER SWITCHES (E.O.S.)

DIRECT GROUND	BLK-GRN J115-9 U56-7	BLK-BLU J115-10 U56-5	BLK-VIO J115-20 U56-9	BLK-GRY J115-21 U56-11
BLACK J115-22	LOWER RIGHT FLIPPER E.O.S. D13	LOWER LEFT FLIPPER E.O.S. D14	NOT USED D15	NOT USED D16

CABINET FLIPPER SWITCHES

DIRECT GROUND	BLK-BRN J113-1 U49-7	BLK-RED J113-2 U49-5	BLK-ORG J113-3 U49-9	BLK-YEL J113-4 U49-11	BLK-GRN J113-6 U55-7	BLK-BLU J113-7 U55-5	BLK-VIO J113-8 U55-9	BLK-GRY J113-9 U55-11
BLACK J113-10	SLAM TILT D17	COIN DOOR CLOSED D18	PLUMB BOB TILT D19	NOT USED D20	RIGHT FLIPPER BUTTON D21	LEFT FLIPPER BUTTON D22	RIGHT ACTION BUTTON D23	LEFT ACTION BUTTON D24

WARNINGS & NOTICES

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